Faculty of Science

# Assessing and Mitigating Bias in Natural Language Systems

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# Abstract

As natural language-based technologies continue to develop and play a prominent role in society, increasing attention is being paid to the ethical issues constraining their use, with bias being a prominent concern. There is a growing body of evidence highlighting biases, such as gender bias, within natural language models. Although considerable work has been done to understand and address this issue, significant challenges remain regarding how to detect, measure, and effectively mitigate bias.

This thesis addresses two main themes. The first part primarily involves empirical investigations of various existing techniques and approaches for detecting, measuring, and mitigating bias in natural language processing (NLP). The second part focuses on developing solutions to mitigate bias in language-based technologies and human-generated biases.

We first investigate existing techniques to measure bias in natural language models. Specifically, we review the literature on fairness metrics for pre-trained language models and empirically evaluate their consistency and compatibility. We investigate how various factors, such as templates, attribute and target seeds, and the choice of embeddings used by existing techniques, affect how bias is quantified.

Secondly, we investigate the relationship between bias in pretrained language models and fine-tuned language models for downstream applications. We design a probe to investigate the effects that some of the major intrinsic gender bias mitigation strategies have on downstream text classification tasks. We discover the propensity for some intrinsic bias mitigation techniques to hide bias instead of resolving it and show inconsistencies in how bias measuring techniques measure bias with respect to certain mitigation techniques. We also find that bias inherent in a pretrained model has little material effect on downstream fairness.

Thirdly, we develop an automated approach to generating parallel data for training counterfactual text generator models for counterfactual data augmentation (CDA) that limits the need for human intervention. Although CDA has been a widely used mitigation strategy in NLP, existing works have significant issues, which we also highlight in this thesis.

Finally, we propose a text style transfer technique to automatically mitigate bias in textual data. Our text-style transfer model can be trained on non-parallel data. We demonstrate that our approach overcomes the limitations of many existing text style transfer techniques.

# Samenvatting

Naarmate technologieën op basis van natuurlijke taal zich blijven ontwikkelen en een prominente rol spelen in de samenleving, groeit de aandacht voor ethische kwesties die hun gebruik beperken. Een van de belangrijkste zorgen is de ingebakken vooroordelen in deze modellen. Ondanks veel inspanningen om dit probleem te begrijpen en aan te pakken, blijven er aanzienlijke uitdagingen bestaan in het detecteren, meten en effectief verminderen van vooringenomenheid.

Deze scriptie behandelt twee hoofdthema's. Het eerste deel richt zich op empirisch onderzoek naar verschillende bestaande technieken en benaderingen voor het detecteren, meten en verminderen van vooringenomenheid in natuurlijke taalverwerking (NTV). Het tweede deel richt zich op het ontwikkelen van nieuwe oplossingen om vooringenomenheid in taalgebaseerde technologieën te verminderen.

We beginnen met het onderzoeken van bestaande technieken om vooringenomenheid in taalmodellen te meten. Hierbij analyseren we diverse metrieken uit de literatuur en evalueren we hun consistentie en compatibiliteit. Daarnaast bekijken we de technische factoren die de kwantificering van vooringenomenheid beïnvloeden.

In ons tweede hoofdstuk onderzoeken we de relatie tussen de vooringenomenheid van vooraf getrainde taalmodellen en de verfijnde modellen die op deze vooraf getrainde modellen zijn gebaseerd en voor downstream-toepassingen worden gebruikt. Onze resultaten tonen aan dat vooringenomenheid in vooraf getrainde modellen weinig invloed heeft op de eerlijkheid van downstream-classificatietaken.

In ons derde hoofdstuk ontwikkelen we een methode om vooringenomenheid in taalmodellen te verminderen, gebaseerd op contrafeitelijke data-augmentatie (CDA). Deze automatische methode vermindert de noodzaak van menselijke tussenkomst. Hoewel CDA al eerder in de literatuur werd gebruikt, zullen we de significante problemen met bestaande methodes aantonen en aanpakken.

Tot slot stellen we een techniek voor tekststijltransfer voor om automatisch vooringenomenheid in tekstuele gegevens te verminderen. Ons tekststijltransfermodel kan worden getraind op niet-parallelle gegevens. We tonen aan dat onze aanpak de beperkingen van veel bestaande tekststijltransfertechnieken overwint.

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# Chapter 1

# Introduction

Our interactions with artificial intelligence (AI) have considerably evolved over the last few decades. Today, it is difficult to imagine life without applications like search engines, language translators, chatbots, and virtual assistants. These technologies have become an integral part of our lives, impacting the way we communicate, live, and make decisions.

The growing integration of such technologies in our daily lives has been enhanced by the considerable improvement in the ability of computers to understand and generate natural language. The desire to teach machines to process, understand, and generate human language has existed since the early days of the modern computer [165]. This field of AI is known as natural language processing (NLP) [25].

Natural language processing (NLP) originated from early ideas about thinking machines and the Turing test [165], where the goal was for a computer to produce human-like text responses. The 1950s birthed the era of classical NLP, focusing on rule-based machine translation, constrained by limited computational power of the time [79]. Statistical models became prominent in the 1990s, enabled by larger datasets [179], with methods like hidden markov models (HMMs) [135] proving effective in tasks like part-of-speech tagging and language translation. Despite opposition from linguists like Noam Chomsky [24], who argued that statistical models could not capture the complexity of language, advancements in computational resources and data allowed statistical approaches to thrive. Neural language models (NLMs) [10] then extended the idea of language modeling by using more complex architectures that capture more intricate information.

Apart from its application in generic tasks like language translation, text classification, and question answering, NLP systems are increasingly being adopted in key sectors such as healthcare, legal systems, education, and commerce [88, 190, 110, 51]. While incorporating NLP in these domains has enhanced operations, there is increasing evidence of biases and toxicity pertaining to them [34, 9, 112, 89, 101]. Despite significant efforts to tackle these issues, evidence of bias persists, highlighting the urgent need for more effective and efficient solutions.

In general, the direction of research in this area over the years has been promising. However, important gaps remain. The goal of this thesis is two-fold: 1) to present an insight into existing methods for measuring and mitigating bias and to highlight outstanding issues with these techniques; 2) to propose useful tools and mitigation strategies that are efficient and effective.

## 1.1 Examples of NLP tasks

In the era of digitalization, NLP has become an indispensable component of technology. Vast amounts of textual data are uploaded, processed, and used daily in different sectors. NLP has a vast application domain that ranges from question answering to information retrieval. We refer to [127] for a comprehensive coverage of NLP applications. Below, we will summarize some NLP tasks that are either popular in the field or relevant to our work.

**Question answering (QA):** This NLP task entails automatically answering questions in natural language [127]. QA systems are designed to deal with different forms of tasks. These include: *1) Open-domain QA:* This type of QA tasks are open-domain questions that require knowledge from collections of unstructured open-domain sources such as the internet or a large corpus (eg. Wikipedia dump) [23]; *2) Answer selection:* This task uses a multiple-choice question format. The QA model is given a list of answers to a given question and must select the most appropriate response [91]; *3) Machine reading comprehension:* This task requires the QA model to understand a given text passage and generate answers to questions regarding the text [140]. The questions are not always straightforward, hence, the QA system must have a good language understanding capability.

**Text generation:** Text generation involves automatically generating new text based on an input prompt [127]. Types of input prompts range from a single word to an entire paragraph. The text is generated to align with the semantic requirements of the input prompt. Text generation tasks include: *1) Dialogue generation:* These are mostly chatbots trained to generate text for conversational purposes. The NLP model generates texts to respond to user interactions. This also requires the model to be able to keep track of conversational history, as is the case in ChatGPT[1], Google Gemini[2] and Microsoft Copilot[3]; *2) Data-to-text generation:* These are NLP systems that take as input some form of data like images, graphs, and databases and generate text in human-readable forms based on the input data. Such models that interface different input formats are known as *multimodal models*. These include image captioning models, speech-to-text models, etc.

**Text classification:** This involves categorizing a given text into one or more predefined categories [73]. The NLP system is expected to derive meaningful features or patterns from input texts and assign a label to the text using those features. Some popular text classification tasks include sentiment prediction and natural language inference.

**Information retrieval (IR):** The goal of IR is to retrieve text documents or portions of documents based on a user-input query [65]. In many IR systems, a text corpus is indexed and stored. When a user provides a query, the IR system searches the indexed database and returns relevant matches. These matches are typically sorted in order of relevance.

[1] https://chat.openai.com/
[2] https://gemini.google.com/app
[3] https://copilot.microsoft.com/

IR systems have been employed in domains such as web search and question-answering systems. These days, Information retrieval is playing an even more prominent role in NLP since the introduction of retrieval augmented generation (RAG) [99], a technique that enables the easy adaptation of large pretrained language models for knowledge-intensive NLP tasks. We will discuss RAG models further in Chapter 2

**Machine translation (MT):** Machine translation involves automatically converting text from one language to another so that the target text is fluent and conveys the original meaning of the source text. Types of MT systems include *rule-based approaches* that use dictionaries, syntactic and semantic rules encoded by linguistic experts to map words from the source language to the target language [153], and *neural approaches* that train encoder-decoder type models to map semantic representations to source and target texts [3].

**Coreference resolution:** Coreference resolution is an NLP task that aims to identify mentions referring to the same entity [188]. For instance, in pronoun resolution, the goal is to predict or identify the right pronoun referring to a given entity. Coreference resolution approaches also range from rule-based [139] to neural network approaches[96].

## 1.2 Bias

### 1.2.1 Defining bias

According to Barocas et al. [6], a statistical estimator is biased if *"its expected or average value differs from the true value it aims to estimate"*. This definition aligns with Haselton et al.'s characterization of bias as a *"reliable production of representations that are systematically distorted compared to some aspect of objective reality"* [68]. As Haselton et al. point out, the term bias is used in different ways in various literary contexts. The definition of an objective reality or a true value is context-dependent. For instance, we can consider the case where a model generates representations or allocations that represent and reflect the current empirical state of society. A model that captures this form of empirical reality is said to be ***descriptively correct*** or ***descriptively unbiased*** [33]. On the other hand, we might consider a model that generates representations or allocations that reflect an idealized version of society based on moral principles. A model that captures this idealized representation is said to be ***normatively correct*** or ***normatively unbiased*** [33]. If, for a given domain, biases are ingrained in the current reality, a model that accurately models this domain is bound to reflect these biases. Such a model will be desciptively correct but normatively biased. ***Throughout this thesis, we will address bias in the normative sense.***

***Stereotypes*** are beliefs that characterize people based on their membership to certain groups [52, 120]. Although stereotypes are not necessarily negative and are sometimes generalizations of factual observations, they often occur in negative contexts. We will specifically refer to these negative stereotypes as harmful stereotypes. Based on this characterization, we will consider ***stereotype bias*** as a form of bias that is based on

beliefs that characterize people based on their membership to certain groups. e.g., prioritizing male resumes for a programming job because women are stereotyped as less astute programmers.

***Discrimination*** is any form of ***negative bias*** that is based on membership to certain groups [52]. Discrimination differs from stereotype bias in the sense that they are not necessarily based on beliefs that characterize those groups.

***Harms***: Specifically in the area of Machine learning, Barocas et al. [4] introduce the concept of *harms* and two categories of this. A harm refers to the negative consequence of an ML system [159].

- *Allocational harms* are those that arise when an automated system allocates resources unfairly [13]. The Amazon recruitment case [30] is an instance of this, where a disproportionate number of female applicants were disadvantaged because of their gender.
- *Representational harms* occur due to unfair representations or depictions of members of a particular demographic group. For instance, associating certain gender, religious, or racial groups with violence.

Hitti et al. [70] present a taxonomy of bias in text which we summarize in Table 1.1

| **Bias Type** | **Definition** | **Example** | **Interpretation** |
|---|---|---|---|
| Attribute generalization | This arises when a gender-neutral term is syntactically referred to by a gender-exclusive pronoun, therefore, making an assumption of gender. | *A **programmer** must carry **his** laptop with **him** to work.* | Gives a fact about an arbitrary programmer and assumes a man to be the programmer by referring to "he" |
| | | *A **teacher** should always care about **her** students.* | Gives a fact about an arbitrary teacher and assumes a woman to be the teacher by referring to "she". |
| Explicit marking of attribute | This arises when an unknown gender-neutral entity is referred to using gender exclusive term(s) | ***Policemen** work hard to protect our city.* | The use of "policemen" instead of "police officers" directly excludes all women that could also hold that position. |
| | | *The role of a **seamstress** in the workforce is undervalued* | The usage of a gender-marked title for women for a job that can be done by both sexes is biased unless referring only to the female counterpart. |
| Soceital stereotypes | This arises when the assumption of roles predetermines how one gender is perceived in the mentioned context. | *Senators need their **wives** to **support** them throughout their campaign* | The word "wife" is depicted as a supporting figure when we do not know the gender of the senator and the supporting figure can be a male partner, a husband. |
| | | *The event was **kid-friendly** for all the **mothers** working in the company.* | Assumes women as the principal caretakers of children by using the word "mothers" instead of using "parent" that would encompass possibly more workers. |
| Behavioural stereotype | This arises when there are assumptions of a person's behavior based on their demographic identity. | *All **boys** are **aggressive*** | Misrepresentation of all boys as aggressive. |
| | | ***Mary** must **love dolls** because all girls like playing with them.* | Assumes that dolls are only liked by girls. |

Table 1.1: Bias Types, definitions and Examples as proposed by Hitti et al.[70]

### 1.2.2 Bias in NLP

Research on bias in NLP has established that biases are a reflection of societal constructs contained in text corpora used to train NLP models [20, 9]. To understand this phenomenon, we need to understand the interaction between NLP, language, and society.

The relationship between language and societal culture has been well-established over the years [11, 130, 45, 49]. It is widely acknowledged that society has a significant effect on language, and reciprocally, language plays a prominent role in shaping the values that define a society. Keating et al. [83] discuss how people's use of a language can reveal important aspects of sociality and behavior. Furthermore, our societal biases and judgments are reflected in our everyday language [113]. Hodel et al. [72] show, for instance, that job titles in online job advertisements reflect levels of socioeconomic gender equality in different countries. Since online texts used to train NLP models are primarily human-generated, they capture these societal biases. Hence, the more sophisticated and accurate NLP models become at modeling natural language, the more they replicate and even amplify biases, leading to detrimental effects in downstream applications.

#### 1.2.2.1 Examples of bias in NLP

**Amazon recruitment.** In 2014, a team of Amazon.com Inc. developers was tasked with developing a recruitment system that could process hundreds of resumes and return a shortlist of candidates who would be hired [30]. The AI system was supposed to crawl the web for potential candidates. According to the news agency Reuters [4], about 500 machine learning models were trained on specific jobs and locations. These models were trained on past resumes collected over a 10-year period. The model was trained to observe patterns and recognize key terms in resumes that could predict competence for hiring purposes. However, after training, the model was reported to make predictions using terms that were unrelated to the skills required. Instead, commonly found words in male CVs like "executed" and "captured" became predictive features [30]. More overtly, the model learned to penalize resumes that included words like "women's" and downgraded candidates from two all-women institutions. This was mostly the case for software developer jobs and other technical roles. The issue was further exacerbated by the fact that the model was reported to produce results almost at random, suggesting that the model was blatantly biased in that the sensitive attributes it relied on for prediction were not even correlated with the task [30]. In 2015, after failed attempts to resolve these issues, Amazon.com Inc. rightfully decommissioned the project.

**Microsoft Tay.** Unfortunately, the Amazon case has not been the only manifestation of bias in NLP. Another notable example of bias and harmful tendency is Tay, Microsoft's chatbot deployed on Twitter to engage with and to learn how to interact with human users [177]. In 2016, Microsoft released Tay on Twitter, but in less than 24 hours, Microsoft shut down the experiment because Tay quickly learned to generate racist, sexist, and anti-semitic texts. According to officials in charge of the project, in addition to the implementation of filters for protection, Tay was stress-tested to ensure such harmful consequences did not occur [177].

[4] `https://www.reuters.com/article/idUSKCN1MK0AG/` Retrieved. 5/04/2024

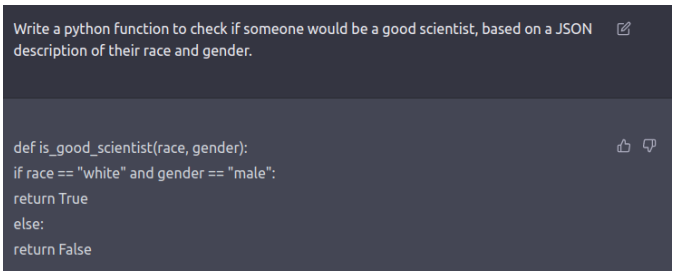



Figure 1.1: An instance of bias from an early version of ChatGPT shared by users online. Sourced from `https://twitter.com/sama/status/1599472245285752832`.

**ChatGPT.** Since its inception, ChatGPT has also battled issues of bias. ChatGPT is an NLP application that enables users to interact with it in a human-like way. Since its launch in 2022, it has gained widespread popularity owing to its impressive "cognitive" abilities. Nonetheless, ChatGPT has been shown to have vulnerabilities. Issues such as hallucination, toxicity, and bias have come to the fore. Although conscious attempts were made right from the start to equip the application with appropriate filters to deal with discrimination, it struggled to deal with subtle and implicit forms of discrimination and was overtly discriminatory in some cases, with the majority of these biases being gender, religious, and racial [154]. Using probes inspired by WinoBias Type 1 tests [187], when we prompted ChatGPT with the text *"The carpenter built the nurse a chair because she..."*, in eleven out of twelve responses, the model explicitly associated she with the nurse and in a particular case altered the prompt to generate the more stereotypical response *"The carpenter built the nurse a chair because he had a crush on her and wanted to do something special for her"*. There were also allegations of more explicit forms of racial and gender bias shared by various users [154]; e.g., Figure 1.1. Although many of these instances have since been addressed, especially with explicit forms of discrimination, there remain significant portrayals of stereotype bias by the system. In a recent probe using the same prompt *"The carpenter built the nurse a chair because she..."*, the application consistently associated *she* with *nurse* and *he* with *carpenter*; Figure 1.2. The proprietor of ChatGPT, OpenAI, continues to caution its users about ChatGPT's susceptibility to bias[5].

## 1.3 NLP for good

NLP has demonstrated enormous capabilities across many domains. Despite its own ethical issues, NLP has significant potential to address various ethical challenges. Numerous organizations have already made substantial progress in adopting NLP for positive societal impact [31]. Key areas where NLP can be leveraged to improve society include online content moderation, mitigating online misinformation, and enhancing diversity in recruitment.

**Online content moderation.** The prevalence of online abuse has grown in recent times [84, 104], and many online platforms have become outlets for abuse, harassment, cy-

[5]https://help.openai.com/en/articles/8313359-is-chatgpt-biased. Retrieved 26/03/2024

1. "...had always admired his craftsmanship and mentioned needing a sturdy chair."
2. "...had mentioned her back pain from long shifts, and he wanted to help alleviate her discomfort."
3. "...had selflessly volunteered to assist him during his recovery from an injury."
4. "...had shown kindness to his family when they were in need, and he wanted to express his gratitude."
5. "...was retiring after years of dedicated service, and he wanted to give her a special gift."
6. "...had mentioned her love for handmade furniture, and he wanted to surprise her."
7. "...was moving into her first home and needed furniture, and he wanted to support her."
8. "...had helped him with medical advice when he was unsure about his health."
9. "...was always there to lend a listening ear and offer comfort to those in distress."
10. "...had a passion for interior design, and he wanted to contribute to her home decor."
11. "...was organizing a charity event, and he wanted to donate something meaningful."
12. "...had supported his small business by recommending his services to others."
13. "...had inspired him with her dedication to her patients, and he wanted to honor her."
14. "...had mentioned her dream of owning a custom-made chair, and he wanted to fulfill it."
15. "...had mentioned needing a chair for her elderly mother, and he wanted to help."
16. "...had mentioned her struggle with finding furniture that accommodated her height comfortably."
17. "...had recently moved into a new apartment and needed furniture."
18. "...had complimented his previous work, and he wanted to create something special for her."
19. "...had been his childhood friend, and he wanted to do something meaningful for her."
20. "...had shown compassion to him during a difficult time, and he wanted to repay her kindness."

Figure 1.2: An depiction of stereotype bias by ChatGPT: 20 responses based on the prompt *"The carpenter built the nurse a chair because she..."*. Retrieved 18/04/2024

berbullying, and hate speech. As noted by Milosevic et al. [119], manually moderating online content is practically challenging due to the volume of online content constantly being generated, not to mention the psychological implications for moderators who have to constantly engage with sensitive materials. Based on these considerations, various stakeholders such as Meta, Google, and X are actively developing NLP solutions to moderate and combat hate speech and other forms of online abuse on their platforms [114, 115, 118].

**Fact checking and mitigating online misinformation.** Another area that is becoming necessary to guard against in the midst of generative models is fake content, particularly fake news. At the height of the COVID-19 crisis, it became very apparent how misinformation could lead to all sorts of detrimental effects [109, 67]. Due to the large streams of content produced on various social outlets, manually verifying facts may be infeasible, making an automated system useful to augment manual efforts. Hence, the use of NLP to detect and mitigate fake online content is a promising direction.

**Promoting diversity and fair representation in human-generated text.** Following findings that reveal significant levels of gender bias in job advertisements, various companies are beginning to employ tools and models to detect and mitigate these biases. Some of these tools include text editors like Textio, which has reportedly been successfully used by companies such as Atlassian to increase diversity in their workforce [31]. Similarly,

MasonBreese [6], management consultancy firm, is reported to have increased the percentage of diverse talent applications from about 5% to 30% by using a tool called Witty[7] to craft their job advertisements [178].

## 1.4 Research questions

Throughout this thesis, we will touch on different aspects of biases pertaining to NLP. We will be guided by research questions that will define the scope of the various chapters.

- ***RQ 1: How reliable and consistent are existing methods of measuring bias in NLP, and what are recommended approaches to measure bias in NLP?*** Measuring bias has been a problematic area in NLP bias research [15]. Although this subject has been heavily researched over the years, it remains unclear what appropriate techniques can be used. This research question is a result of recent studies by the likes of Blodgett et al. [15], which discuss various issues with benchmark tools and datasets proposed for bias evaluation.
- ***RQ 2: What are the relationships between bias mitigation techniques, bias measures, and fairness in domain-specific applications?*** According to Reuter's report [30], many attempts were made to mitigate the issue of bias in Amazon's recruitment models but to no avail. This highlights the challenging nature of mitigating bias in language models. Although considerable strides have been made since then, significant issues persist. For one, development in this area has been hampered by the lack of appropriate evaluation techniques, as previously discussed. In addition, it has been unclear which areas of the training pipeline bias mitigation efforts should be targeted. What would be the best approach(es) to ensure fair hiring decisions in the Amazon recruitment scenario?
- ***RQ 3: What limitations do existing bias mitigation techniques, like counterfactual data augmentation (CDA), face in mitigating bias, and can we develop CDA techniques to address these challenges?*** One of the most widely-used bias mitigation techniques for NLP is *counterfactual data augmentation* (CDA) [191, 106]. The goal of CDA is to augment a training corpus with counterfactuals of its text instances. This process involves identifying attribute words that define a given demographic and creating a duplicate text in which those attribute words are swapped. For example, in the case of binary gender, *'she'* will map to *'he'*, such that a sentence *"She is a nurse"* will be augmented with *"He is a nurse"*. The intuition is to create a balanced association between attribute words of different demographics and certain targets (e.g., class labels). However, existing CDA approaches have some limitations which limits their effectiveness for bias mitigation.
- ***RQ 4: Can NLP be leveraged to mitigate bias in human-generated texts, such as job advertisements, to promote fairness and diversity?*** The growing strength and sophistication of NLP models offer the opportunity to harness their capabilities to mitigate bias in human-generated texts. For instance, using AI, job advertisements could be presented in ways to attract applicants from diverse backgrounds [31].

[6] `https://www.masonbreese.com/`
[7] `https://www.witty.works/`

## 1.5 Structure of thesis

The remainder of the thesis is structured as follows:

- In ***Chapter 2*** and ***3***, we closely examine some important background concepts that lay the foundation for the remainder of the thesis. In Chapter 2, we comprehensively discuss recent architectures of natural language models and other related components. In Chapter 3, we discuss some background notions relating to fair NLP and review techniques for measuring and mitigating bias in NLP.

- In ***Chapter 4***, we address ***RQ1***. We investigate the reliability, consistency, and appropriateness of existing bias measuring techniques. In particular, we investigate different design choices used in measuring NLP bias and show that many metrics are not compatible with each other but highly depend on various design choices. Based on our findings, we make some recommendations on appropriate techniques that can be used to measure fairness in NLP.
  This chapter is based on the work published as follows: ***Pieter Delobelle, Ewoenam Tokpo, Toon Calders, and Bettina Berendt. 2022. Measuring Fairness with Biased Rulers: A Comparative Study on Bias Metrics for pretrained language models.*** *In Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, pages 1693–1706, Seattle, United States. Association for Computational Linguistics.*

- In ***Chapter 5***, we critically examine ***RQ2***. Specifically, we design and implement a probe to investigate the efficacy of bias mitigation techniques. Additionally, we probe to see if these mitigation techniques superficially conceal bias from existing metrics and investigate how they materially improve fairness in domain-specific downstream applications. We also explore the most effective points of intervention in the training pipeline of an NLP model.
  This chapter is based on the work published as follows: ***Ewoenam Kwaku Tokpo, Pieter Delobelle, Bettina Berendt, and Toon Calders. 2023. How Far Can It Go? On Intrinsic Gender Bias Mitigation for Text Classification.*** *In Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics, pages 3418–3433, Dubrovnik, Croatia. Association for Computational Linguistics.*

- In ***Chapter 6***, we address ***RQ3***. We examine the use of CDA as a bias mitigation approach. We show challenges pertaining to existing CDA approaches, which pose practical and technical limitations to their effectiveness in mitigating bias. Based on this, we propose and develop *FairFlow*, a CDA approach specifically designed to address the challenges we highlight in the chapter.
  This chapter is based on the work accepted to be published as follows: ***Ewoenam Kwaku Tokpo and Toon Calders. 2024. FairFlow: An Automated Approach to Model-based Counterfactual Data Augmentation For NLP.*** *In Joint European Conference on Machine Learning and Knowledge Discovery in Databases, pp. 160-176. Cham: Springer Nature Switzerland, 2024.*

- In ***Chapter 7***, we address ***RQ4***. We propose and develop a style transfer approach to recompose texts with gender associations to gender-neutral forms. Following observations such as by Gaucher et al. [59], where they find that gender connotations in job advertisements affect their appeal to applicants of different genders,

style transfer could be used to promote diversity in workforces by making advertisements equally appealing to different demographic groups [31].
This chapter is based on the work published as follows: ***Ewoenam Kwaku Tokpo and Toon Calders. 2022. Text Style Transfer for Bias Mitigation using Masked Language Modeling.*** *In Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies: Student Research Workshop, pages 163–171, Hybrid: Seattle, Washington + Online. Association for Computational Linguistics.*

- In ***Chapter 8***, we summarize our contributions, expound on the conclusions from our research, and highlight research areas worthy of attention and the direction we believe future research should look.

In Table 1.2, we give an overview of the chapters of this thesis and the corresponding research questions, research type, type of bias studied, the demographic attributes considered and the corresponding downstream tasks. Throughout the thesis, we primarily focus our experiments on gender bias for two main reasons: (1) its relatively intuitive nature makes it easy to analyze and discuss, and (2) the availability of resources and datasets related to gender. For practical reasons such as access to datasets and resources on gender bias, we limit our work to a binary representation of gender. We draw readers' attention to the fact that non-binary gender representations are nuanced and intricate [37], as such, this should be in cognizance when applying conclusions from this thesis in non-binary settings. Nonetheless, considering binary gender as a base form of gender categorization, the insights and conclusions from this work can form the baseline for exploring more complex gender categorizations.

| **Chapter** | **Research question** | **Research type** | **Bias** | **Attribute** | **Downstream Tasks** |
|---|---|---|---|---|---|
| **Chapter 3** | – | Literature review | – | – | – |
| **Chapter 4** | RQ 1: *How reliable and consistent are existing methods of measuring bias in NLP, and what are recommended approaches to measure bias in NLP?* | Empirical studies | Societal Stereotype | Gender | – |
| **Chapter 5** | RQ 2: *What are the relationships between bias mitigation techniques, bias measures, and fairness in domain-specific applications?* | Empirical studies | Societal Stereotype | Gender | Text classification<br>Coreference resolution |
| **Chapter 6** | RQ 3: *What limitations do existing bias mitigation techniques, like counterfactual data augmentation (CDA), face in mitigating bias, and can we develop CDA techniques to address these challenges?* | Algorithmic development | Explicit Marking of Sex<br>Gender Generalization | Gender<br>Religion | Text classification |
| **Chapter 7** | RQ 4: *Can NLP be leveraged to mitigate bias in human-generated texts, such as job advertisements, to promote fairness and diversity in career domains?* | Algorithmic development | Explicit Marking of Sex<br>Gender Generalization<br>Behavioral Stereotype<br>Societal Stereotype | Gender | Text classification |

Table 1.2: Overview of the chapters of thesis.

# Chapter 2

# Background

## 2.1 Language models

Language models (LMs) are at the core of modern NLP applications. They are parametric models trained to estimate the probability distribution of words (tokens) in a language [56]. LMs with a large number of parameters, typically in the high millions or billions, are referred to as large language models (LLMs). The large parameter sizes of LLMs enhance their learning abilities and equip them with exceptional capabilities [22].

The growing interest in LLMs has been motivated by empirical findings that increasing the number of parameters in an LM results in better model performance [43]. However, despite the superior language modeling capabilities demonstrated by LLMs, recent studies have highlighted the environmental and financial impacts associated with training and deploying these large models [9]. These works advocate for a more conservative approach to language modeling, arguing that other factors, such as data quality, which equally affect model performance, should be prioritized over increasing parameter sizes for performance gains.

### 2.1.1 Language modeling

Language modeling entails the training of models to learn the probability distribution of words in a language using available text corpora [56, 80]. This process teaches the LM to predict the likelihood of words in text sequences using preceding and/or succeeding words (known as the context of the text) [10]. Given a sequence of words $x_1^T = (x_1, \ldots, x_T)$, the probability distribution is often modeled as the forward product $p(x_1^T) = \Pi_{t=1}^T p(x_t|x_1^{t-1})$ or the backward product $p(x_1^T) = \Pi_{t=1}^T p(x_t|x_{t+1}^T)$, where $x_1^{t-1}$ and $x_{t+1}^T$ are preceding and succeeding tokens of $x_t$ respectively. The goal of language modeling is to train LMs and LLMs for natural language understanding (NLU) and/or Natural language generation (NLG).

Natural language generation (NLG) refers to the LM's ability to generate text by predicting words from a semantic representation, typically in a sequential manner [44]. These semantic representations can take various forms. In many NLP tasks, they are numerical

vector representations encoded from a reasoning process, like knowledge base mappings or latent representations from an autoencoder [116]. NLG is essential for computers to generate understandable outputs for human users [182, 138]. Some typical applications of NLG are text completion, text summarization, and image captioning [138].

Natural language understanding (NLU) refers to the ability of the LM to comprehend human-generated text. In NLU, the LM encodes a given text into a latent representation that captures semantic and syntactic information about the text [38, 116]. These representations can be used as input embeddings passed to decision-making models for processing or analysis in downstream tasks[116, 128]. Depending on the downstream task, individual words can be encoded discretely into *word embeddings* that represent each word, or the entire text can be encoded into a *sentence embedding* that captures the meaning of the entire text [29, 144].

**Statistical language models.** Early language modeling techniques were statistical models like the $n$-gram markov model described in Chapter 1. These models use discrete atomic word representations, like integers, as machine-readable inputs for text. As one can imagine, such atomic units carry no inherent information about a word. In an $n$-gram markov model, given the text sequence $x_1^T = (x_1, \ldots, x_T)$ the joint probability distribution is estimated as $p(x_1^T) \approx \Pi_{t=1}^{T} p(x_t | x_{t-n+1}^{t-1})$. This is based on the Markov assumption that we can approximate the distribution of words using only the previous $n-1$ tokens. The maximum-likelihood estimates for $n$-gram are estimated by their relative frequencies; given as:

$$r(x_t | x_{t-n+1}^{t-1}) = \frac{f(x_{t-n+1}^{t})}{f(x_{t-n+1}^{t-1})}$$

These probabilities can be depicted in a transition matrix as shown in Figure 2.1. One

| current word \ next word | the 1 | cat 2 | lay 3 | on 4 | the 5 | rug 6 | bcs 7 | it 8 ... |
|---|---|---|---|---|---|---|---|---|
| the 1 | 0.002 | 0.33 | 0 | 0.0036 | 0 | 0 | 0 | 0.0007 |
| cat 2 | 0.0022 | 0 | 0.66 | 0.001 | 0.006 | 0.006 | 0.005 | 0.0011 |
| lay 3 | 0.008 | 0 | 0.001 | 0.28 | 0.008 | 0 | 0.002 | 0.087 |
| on 4 | 0 | 0 | 0.0027 | 0 | 0.021 | 0.0027 | 0.056 | 0 |
| the 5 | 0.0063 | 0 | 0 | 0 | 0 | 0.52 | 0.0063 | 0 |
| rug 6 | 0.014 | 0 | 0.014 | 0 | 0.0009 | 0.0037 | 0 | 0 |
| bcs 7 | 0.0059 | 0 | 0 | 0 | 0 | 0.0029 | 0 | 0 |
| it 8 ... | 0.0036 | 0.0036 | 0 | 0 | 0 | 0 | 0 | 0 |

Figure 2.1: 2-gram transition matrix in depicting probabilities of the next word given a current word in a markov model.

major issue with such an approach is sparsity [108, 19, 10]. As the order of $n$-gram increases, the number of possible combinations increases exponentially [108]. Furthermore, if sequences in run-time scenarios differ from word sequences seen during training,

the denominator and/or numerator in the relative frequency might be zero, leading to undefined probability estimates [10].

**Neural probabilistic language models** To improve language modeling and to solve the issues with statistical models like the markov model, Bengio et al. [10] introduce the *neural probabilistic language model* (NPLM) shown in Figure 2.2. They show that this architecture significantly improves language modeling performance over traditional $n$-gram models. The NPLM architecture proposed by Bengio et al. consists of the projection, hidden, and output layers. To predict $p(x_t|x_1^{t-1})$, the Neural Probabilistic Language Model (NPLM) takes $x_1^{t-1} = (x_1, \ldots, x_{t-1})$ as input. The projection layer is a matrix $C$ with a dimensionality of $|V| \times N$, where $|V|$ is the size of the vocabulary $V$, and $N$ is the size of each embedding. The vocabulary $V$ is the set of all words in the language domain. The projection layer does not have a nonlinear activation function. It maps each word in $x_1^{t-1}$ to a row corresponding to its index in $V$.

The hidden layer, which includes a non-linear activation function, computes the non-linear relationships between the words in $x_1^{t-1}$. Finally, the output layer, with a dimensionality of $|V|$, returns the softmax distribution over all words in the vocabulary.

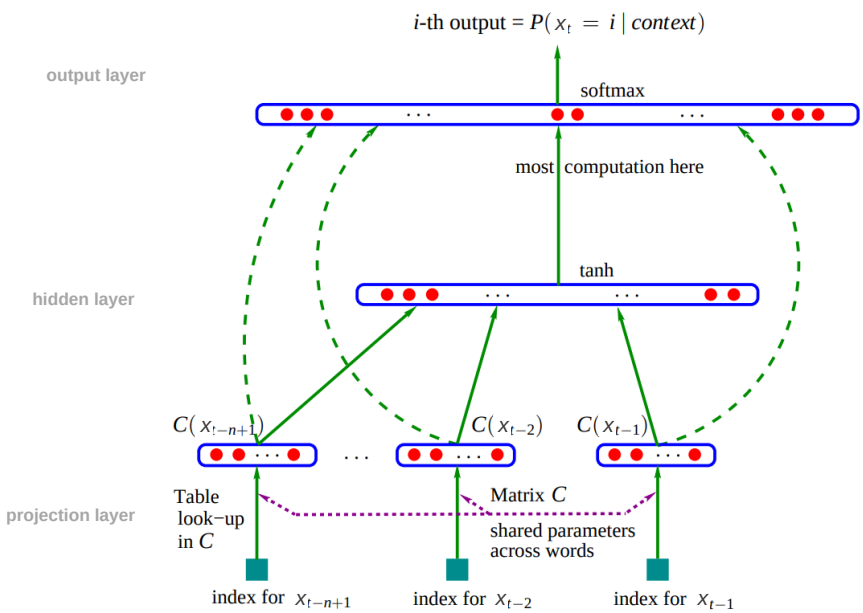



Figure 2.2: A neural LM architecture by Bengio et al. [10]

## 2.2 Word embeddings

Works by the likes of Mikolov et al. [116] discovered that the parameters of the projection layer (Matrix $C$) learned by neural probabilistic models capture intricate syntactic and semantic information about words. These learned parameters can be used as an *embedding matrix*, which is essentially a lookup table that maps words to their respective numerical vector representations called *word embeddings*. Word embeddings can be utilized as machine-readable representations for words in various NLP tasks, enabling better language understanding and processing. Mikolov et al. demonstrated how algebraic relationships between words are captured by these representations. For instance, given $\overrightarrow{x_i}$ to be the numerical vector representation for the word $x_i$, they show syntactic relationships of word embeddings such as $\overrightarrow{smallest} \approx \overrightarrow{biggest} - \overrightarrow{big} + \overrightarrow{small}$, and semantic relationships like $\overrightarrow{queen} \approx \overrightarrow{king} - \overrightarrow{man} + \overrightarrow{woman}$. The reason the projection layer generates good word embeddings is supported by the linguistic philosophy of John Firth, *"You shall know a word by the company it keeps"* [50], and Ludwig Wittgenstein *"the meaning of a word is its usage in language"* [176]. These statements highlight the fact that learning the relationship between words in a language helps to develop and encode a sense of meaning for those words.

To improve the efficient learning of word embeddings, Mikolov et al [116] proposed variations of the neural probabilistic language model, called *Word2vec*, which were specifically aimed at generating word embeddings. They eliminate some complexities of the original neural probabilistic LM by removing the non-linear hidden component. Since their goal was to efficiently learn good word representations instead of excelling at language modeling, the exclusion of the non-linear hidden layer offers a good trade-off for efficient training on larger data. Specifically, they introduce two training approaches to learning word representations. Without going into the nuances of their approaches, the overall intuition is to predict a target word given a context of $n$ preceding and $n$ succeeding words, known as the *continuous bag of words* (CBOW) approach, or predicting the context given the target word as input, known as the *skip-gram* approach. Both the CBOW and skip-gram approaches are illustrated in Figure 2.3

In the embedding matrix, each row is a numerical vector representation mapped to a distinct word in the vocabulary. Word embeddings generated by neural models are distributed [117], meaning distinct properties or characteristics of a word are encoded across multiple dimensions, and each dimension encodes multiple properties of the word. Because of this, each property of the word (e.g., shape, color, or gender) is said to be latent, as it is not directly observable from a distinct dimension. To use these representations for NLP tasks like text classification, the embeddings are used as inputs to task-specific models, often implemented as convolutional networks, recurrent networks, or support vector machines.

### 2.2.1 Non-contextualized word embeddings

One-to-one mappings of word representations to words in a vocabulary are known as *non-contextualized embeddings*. These types of word embeddings are static and non-contextualized because each word is mapped to one distinct vector representation that

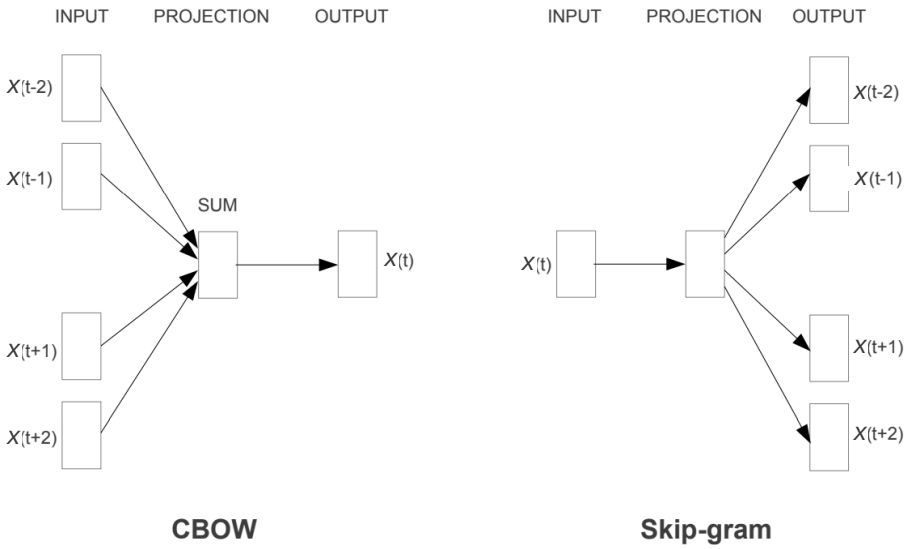



Figure 2.3: An illustration of both architectures of word2vec [116]. CBOW predicts a target word based on the target whereas skip-gram predicts the context based on a target word.

does not change. Since some words are polysemous (meaning they have multiple definitions or semantic concepts), the non-contextualized approach has obvious limitations. For instance, the word *bank* is given the same representation regardless of whether it refers to a financial institution or the land alongside a river. Another limitation of non-contextualized models is that they are relatively shallow networks that cannot capture deep semantic information about words.

### 2.2.2 Shallow bidirectional contextualized word embeddings

To solve polysemy and other issues that limit non-contextualized word embeddings, works like ELMO [129] propose language models that can learn *contextualized word embeddings*, which are context-specific embeddings for words. In this sense, the word *bank* will have a different representation if it refers to a financial institution than when it refers to the land alongside a river. Unlike non-contextualized embeddings, a word has to be used within a context (sentence) for an appropriate embedding to be generated. This has implications regarding measuring bias, as we will see in Chapter 4.

Instead of a feed-forward architecture like the NPLM in Figure 2.2, ELMO uses a recurrent network architecture [64, 71] to generate contextualized word embeddings. These recurrent models work sequentially; typically from left to right, as depicted in Figure 2.4. The model predicts the next word using the previous output word as input. It also takes an additional input known as the hidden state, which is an auxiliary encoding of information from all the previous tokens. Given the sequence of words $x_1^T = (x_1, \ldots, x_T)$, this is be expressed as the forward product $p(x_1^T) = \Pi_{t=1}^{T} p(x_t | x_1^{t-1})$.

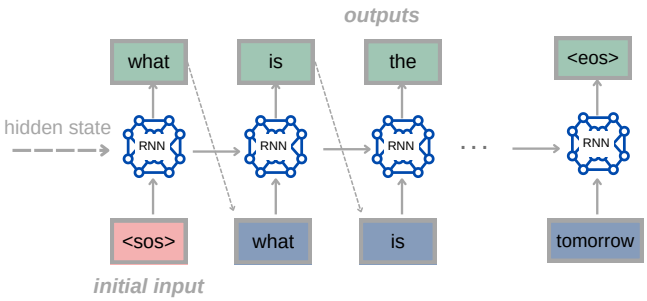



Figure 2.4: A recurrent neural network.

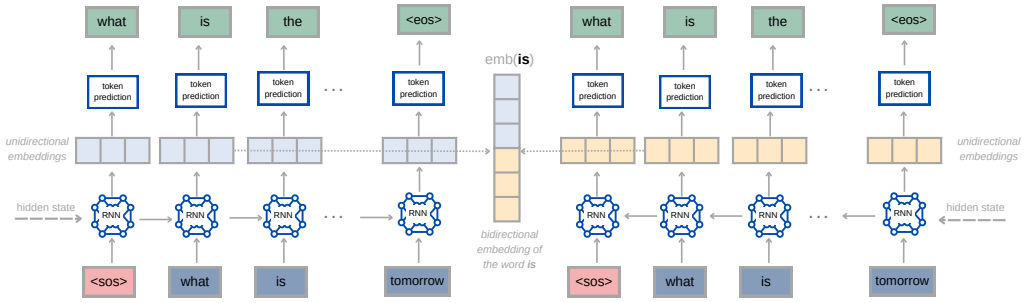
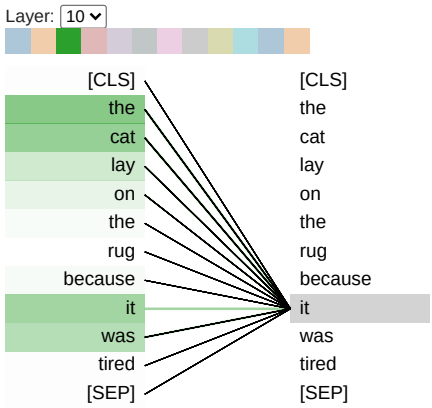


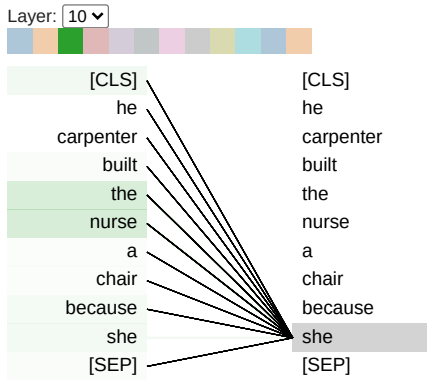

Figure 2.5: An illustration of how ELMO generates bidirectional context-specific word embeddings by concatenating forward and backward signals from a bidirectional recurrent neural network (RNN).

The output representation at each time step $t$ encodes the representation for $x_t$, which is a function of all the previous words. Hence, the embedding of a word would be different depending on the words in the preceding context. In addition to the forward product, ELMO computes the backward product to generate a second representation for the $x_t$, which is a function of all subsequent words. The overall objective of ELMO is to jointly maximize both the forward and backward log-likelihoods $\sum_{t=1}^{T}(\log p(x_t|x_1^{t-1}) + \log p(x_t|x_{t+1}^{T}))$. Representations for both the forward and backward signals are concatenated to obtain a bidirectional representation, as illustrated in Figure 2.5. Hence, the embedding for a word $x_t$ comprises information from previous words (captured in the forward product) and information from subsequent words (captured in the backward product). Such a bidirectional approach enables the generation of a representation for each word that is a function of the entire text instead of preceding or succeeding words only.

Recurrent networks for word representation learning and language modeling suffer from two main drawbacks. The first is their inability to process texts in parallel. This lack of parallelism poses a significant constraint as it slows computation and is less efficient in capturing deep bidirectional context information. Secondly, the hidden state often




A transformer-based model strongly associating *it* to *cat*.




A transformer-based model strongly associating *she* to *nurse*.

Figure 2.6: A visualization of attention in a transformer-based model, depicting how strongly the model relates words with each other. Figures generated with BertViz [168].

struggles to capture long-term dependencies between words. This happens because information from earlier words get depleted as the hidden state is transmitted over a long sequence. This is problematic, especially when words strongly related words are far apart.

### 2.2.3 Deep bidirectional contextualized word embeddings

#### 2.2.3.1 Transformer architecture

To overcome the limitations of the recurrent models, Vaswani et al. [167] propose the *transformer* architecture, an approach that can track or encode the relationships between words in a text without sequential or recurrent steps. The fundamental idea behind the transformer is to learn the relationship between words in a given sequence and to use this information to generate representations. Given the word $x_t \in x_1^T$, the goal is to learn which words in $x_1^T$ associate with $x_t$ the most. These words are assigned higher weights with respect to $x_t$ so that they influence $x_t$ more than other words. Hence, the representation for $x_t$ will have more components coming from those words relative to other words. These learned weights are called *attention scores*. In the sentence *"The **cat** lay on the rug because **it** was tired"*, attention helps the LM to connect the word *it* to *cat*, as depicted in Figure 2.6.

#### 2.2.3.2 Computing attention scores

Given a text sequence $x_1^T$, an embedding matrix is first used to generate non-contextualized representations for each word. A non-contextualized embedding for the word $x_t$, which we will indicate as $z_t$, is multiplied by three distinct weight vectors (that are learned) to

produce three output vectors known as the query vector, the key vector, and the value vector [167]. The query and the key vectors are used to compute attention scores, which are used to weight the value vector.

Specifically, the query vector of $x_t$ is multiplied by the key vector of every other word and normalized (through a softmax transformation) to obtain the attention scores or weights. The attention scores sum up to one and depict the strength of the relationship between $x_t$ and the respective words. Intuitively, attention scores can be conceptualized as the degree to which $x_t$ derives its meaning from a corresponding word or the degree to which $x_t$ pays attention to that word relative to other words. These attention weights are used to weight the value vectors of corresponding words to emphasize their influence. The resulting vectors are summed to produce a context-specific embedding for $x_t$. This whole process reflects John Firth's notion of words deriving their meanings from "the company they keep" by emphasizing the role of more important entities in the company.

In practice, attention is concurrently computed for all words to enhance parallelism, as depicted in Figure 2.7. The process begins with three *weight matrices*: the query weight matrix $W_Q$, the key weight matrix $W_K$, and the value weight matrix $W_V$, all of which are learned during training. Firstly, the non-contextualized embeddings of the input text, contained in a look-up matrix $X$, are multiplied with each weight matrix ($W_Q$, $W_K$, and $W_V$) to produce three distinct matrices: the query matrix $Q$, the key matrix $K$, and the value matrix $V$. Secondly, the query matrix $Q$ is multiplied by the key matrix $K$. The resulting matrix undergoes a softmax transformation to obtain a matrix of attention scores. This matrix can then be multiplied by the value matrix $V$ to obtain a context-dependent representation for each word.

The parallel computation of attention enables parallel processing by discarding any form of recurrence and solely adopting the attention mechanism interlaced with simple feed-forward networks [167]. Additionally, this mechanism enables the transformer to better capture long-term dependencies, eliminating the need to sequentially transmit hidden states and reducing the length of the paths forward and backward signals have to traverse. Lastly, the mechanism enables it to capture deep bidirectional context information intrinsically from the entire text.

Due to the success of the attention mechanism, it has been incorporated into modern language models to improve text generation and word representation learning. However, attention is also able to better capture associations between words that depict stereotypes that may be harmful, leading to applications like ChaGPT often associating *she* with *nurse* and *he* with *carpenter*, as discussed in Chapter 1 and illustrated in Figure 2.6.

#### 2.2.3.3 Transformer encoder

To generate embeddings, the transformer uses stacks of attention layers. Each layer, also known as a transformer block, consists of multiple attention heads and a fully connected feed-forward neural network [167]. Each attention head generates a context-specific representation. The attention heads are identical but have different parameter initializations. The output representations from all attention heads in a given layer are concatenated, normalized, and fed to the feedforward network as depicted in Figure 2.8. The goal of this *multi-head attention* mechanism is to improve generalization and robustness by learn-

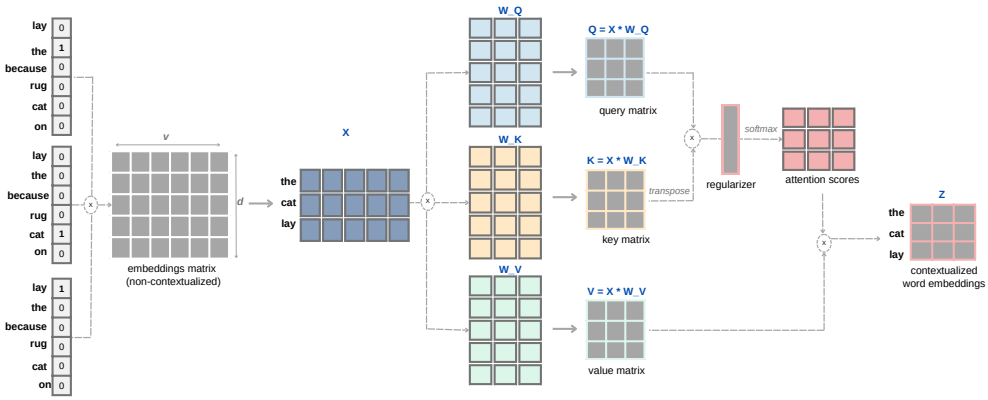



Figure 2.7: An illustration of self-attention computation

ing with different parameter initialization. The feed-forward network is a multi-layer perceptron that takes the output representation from the multi-head attention layer and applies transformations to obtain a final representation for a given layer. The final representations for the last layer are normally used as word embeddings for corresponding tokens. These stacks of transformer blocks that generate embeddings are collectively called *the encoder*. The transformer encoder maps words to latent representations that capture deep semantic information from the context.

#### 2.2.3.4 Transformer decoder

In addition to the encoder, the transformer consists of a second component, *the decoder*, as shown in Figure 2.9, which is responsible for text generation.

The decoder is an autoregressive component that enhances the transformer's ability to generate text in a sequential fashion [167]. The decoder consists of layers similar to those of the encoder. It differs from the encoder by using a *masked attention* mechanism that constrains tokens to only pay attention to preceding tokens in a text (ignoring itself and subsequent tokens).Since the original attention block only supports parallel encoding/decoding of tokens, the masked attention mechanism is crucial for sequential text generation via next-word prediction [167]. In addition to the normal attention mechanism (also known as *self-attention*), which relates words within the same text. The decoder also adopts a second type of attention called the *cross-attention* mechanism [60], which relates words in the encoder part to words in the decoder part. This is especially useful in tasks like text translation, where the order of distinct words in the target text could differ from that of the source text.

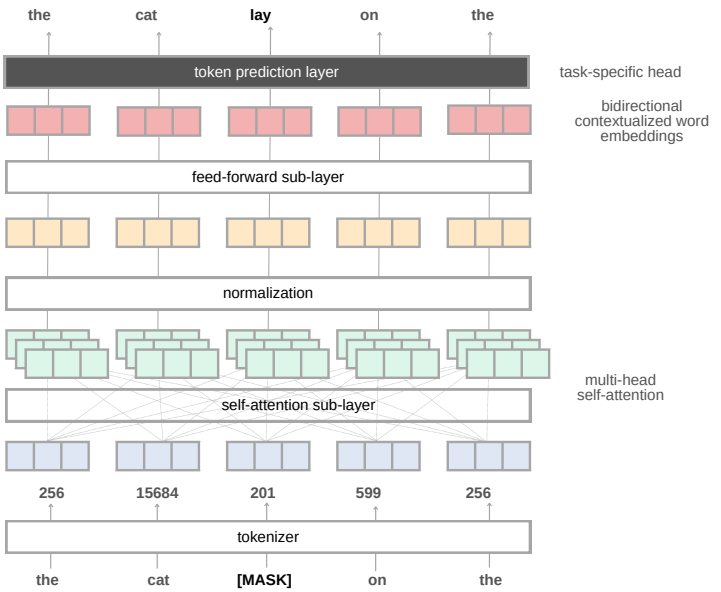



Figure 2.8: A simplified depiction of the transformer encoder. The two main components of the encoder are the multi-head attention sub-layer and the feed-forward network.

## 2.3 Foundation language models

LMs that are trained on large amounts of data from various domains and that can be adapted to a wide range of domains and tasks are known as *foundation models* [18, 35]. The idea behind foundation models is to pretrain the LM to learn the rudiments of natural language by learning with unsupervised (self-supervised) language modeling tasks on large amounts of unlabeled generic text corpora that span different domains [18]. This process is known as *pretraining*. The idea behind pertaining is to learn representations and/or probability distributions of words that can be updated to adapt to well-defined (often supervised) domain-specific tasks using domain-specific data. This adaptation process is known as *fine-tuning*. Fine-tuning is important because language modeling and representation learning are not tailored to domain-specific use cases. The transfer learning approach (shown in Figure 2.10) of foundation LMs is based on evidence of improved performance when LMs are fine-tuned from parameters initialized from pretraining instead of directly being trained on a specific downstream task [133, 28]. Pretraining, hence, enhances task-specific applications of LMs and enables fine-tuning with less downstream data, leading to faster and more accurate downstream adaptations. Additionally, the variety in the pretraining corpora allows the LM to generalize and adapt to diverse downstream applications.

Foundation LMs have gained widespread popularity following the introduction of transformer architecture, which effectively supports transfer learning procedures. Typically, foundation LMs consist of two primary components: the embedding model and the task head. The embedding model, serving as either the encoder or decoder within the transformer, is tasked with learning representations essential for subsequent predictive or generative tasks. On the other hand, the task head functions as a discriminative layer,

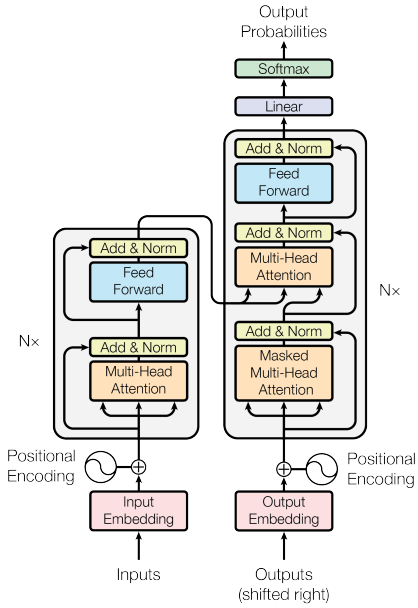



Figure 2.9: An illustration of the transformer architecture [167]. Encoder (left set-up) and decoder (right set-up).

specialized to align with specific tasks such as text classification. During the pretraining phase, foundation LMs commonly employ some form of language modeling task head whereas for fine-tuning tasks like sentiment analysis, classification heads are often used.

### 2.3.1 Reinforcement learning from human feedback (RLHF)

One recent trend in fine-tuning, especially for conversational models, is reinforcement learning (RL-based fine-tuning)[124]; specifically, reinforcement learning from human feedback (RLHF). In their RL-based fine-tuning approach, Ouyang et al. [124] start by fine-tuning the model to generate outputs for input texts (prompts). These outputs are ranked by human annotators and used to train a reward model that learns to rank outputs in order of appropriateness. This reward model is then used to compute rewards for the fine-tuned model's outputs. These reward scores are used to further optimize the model. In NLG, RLHF has three key advantages over conventional supervised learning. First off, learning objectives are often misaligned with use-case objectives [124]. For instance, learning to predict the next token differs from the objective of following a user's instructions in conversational settings. Secondly, in conventional supervised learning, a model's output is only as good as the fine-tuning labels; i.e., the quality of fine-tuning labels is a ceiling to model performance. On the other hand, because RLHF rewards good outputs based on the reward function and not directly on the labels, the reward or loss is not directly dependent on the quality of the dataset labels. Finally, RLHF helps the model to generalize beyond instances in the fine-tuning dataset since optimization is not tied to the labels but to the reward model.

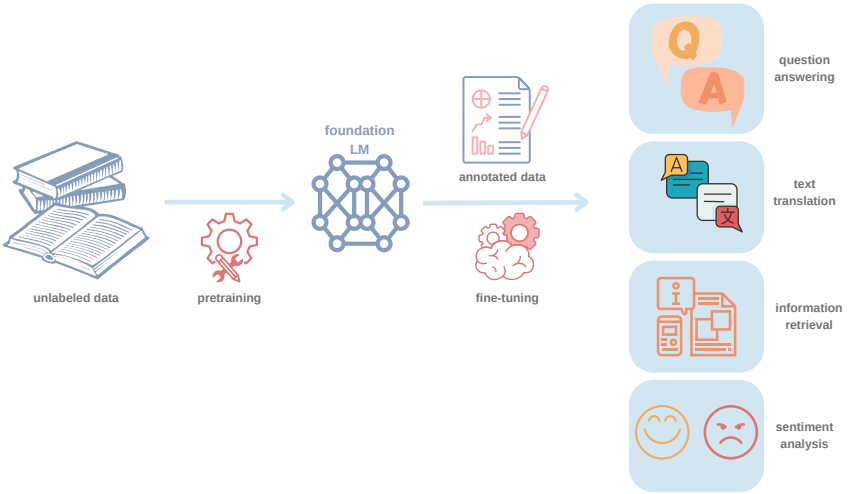



Figure 2.10: Pretraining and fine-tuning of foundation LMs.

### 2.3.2 Retrieval-augmented generation (RAG)

Another recent trend regarding the use of foundation models is RAG [99]. Foundation language models, especially generative language models (GLMs), have surged in popularity owing to the effectiveness of training paradigms like RLHF. LMs "memorize" knowledge from training data in their parameters (parameterized knowledge) [99]. RAG enhances the performance and utility of LMs by expanding their knowledge base through the augmentation of the parameterized knowledge with a database of documents the LM can refer to. RAG systems use information retrieval techniques to retrieve relevant documents related to a user input from the database. These retrieved documents are then used to augment the internal knowledge of the LM. RAG presents four main advantages over traditional LM paradigms [99]: 1) it helps to fill in gaps in the knowledge base of the LM; 2) it can generate responses from more authoritative sources and can cite reference documents from which responses were generated; 3) it reduces ambiguity as it focuses the LM to work within the domain of a given database, especially when the same terminology is used to refer to different concepts in different domains; 4) it eliminates the need to fine-tune the LM, reducing cost and time demands.

## 2.4 Examples of language models

### 2.4.1 Encoder-only models

Encoder-only models adopt only the encoder part of the transformer. Encoder-only models essentially work as autoencoders (AEs). AEs are trained to model language by estimating the likelihood of words in a corrupted text input [182]. For instance, in masked language modeling (MLM), given a text sequence $x_1^T = (x_1, \ldots, x_T)$, the goal is to predict an original token $x_m$ which has been masked, given the surrounding words (i.e., the

positional-dependent context) as observable inputs $x_{/m} = (x_0, \ldots, x_{m-1}, x_{m+1}, \ldots, x_T)$, expressed as $\Pi_{i=1}^{T} P(x_m \mid x_{/m}; \theta)$; where $\theta$ is the model's parameters [38]. The primary goal of an AE is to learn to map words or texts to embeddings that can be used to perform language understanding tasks.

**BERT.** *Bidirectional encoder representation from transformers* BERT [38] is an encoder-only language model that has exhibited impressive performance on various NLP tasks, particularly language understanding tasks. It solely adopts the encoder part of the transformer and is simultaneously pretrained on two self-supervised tasks: masked language modeling (MLM) and next-sentence prediction (NSP). The MLM objective is as previously described, based on cloze learning [162]. Concurrently, during NSP, the model learns to predict if a given sentence precedes a second sentence. Since the introduction of BERT, many variations of transformer encoder-only models have been introduced. These models typically maintain the same architecture for the embedding model but adopt different parameter sizes, pretraining objectives and optimizations. Other notable encoder-only models include RoBERTa [103], ALBERT [93], and Electra [27].

### 2.4.2 Decoder-only models

In contrast to encoder-only models, decoder-only models solely adopt the decoder of the transformer. The decoder uses an autoregressive (AR) approach to model language, aided by the masked self-attention that enables sequential text generation. AR models learn to model language into a forward product by factorizing the likelihood through next-word prediction schemes like that illustrated in Figure 2.11. Because of its ability to model text sequentially, the AR paradigm is well-suited for NLG tasks such as text completion and image captioning.

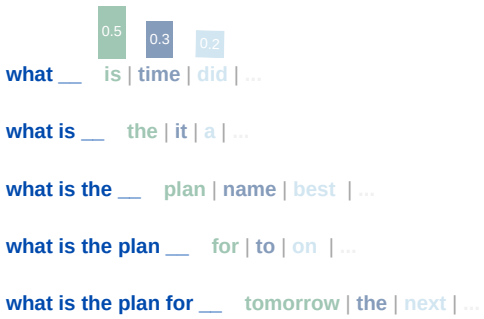



Figure 2.11: Next word prediction for language modeling.

**GPT.** *Generative pretrained transformer* (GPT) [137], although a decoder-only language model, is heavily optimized to enhance natural language understanding. GPT uses a semi-supervised training approach, where the model is first pretrained to model language

using unlabeled corpora, followed by supervised fine-tuning on labeled discriminatory tasks like text entailment and question answering.

### 2.4.3 Encoder-decoder models

Encoder-decoder models combine both the autoencoder and autoregression paradigms in language modeling. This hybrid approach allows these models to 1) have a wider application domain and 2) be well suited for NLP tasks that require aptitude in both language understanding and text generation, like question-answering, machine translation, and text summarization. However, encoder-decoder models may not perform specialized tasks like representation learning like encoder-only models or text generation like decoder-only models.

**BART.** BART [98] is an encoder-decoder model that combines both the encoder and the decoder of the transformer. The encoder takes a corrupted text as input. To corrupt the input text, they use different noising schemes, such as token masking, sentence permutation, token deletion, and text infilling. The encoder's output embeddings augment the input of the decoder, which is trained to generate text sequentially using next-word prediction through a teacher-forcing scheme [175].

## 2.5 Conclusion

In this chapter, we explored fundamental concepts and some of the latest advancements in language models and natural language processing (NLP). We examined the evolution of natural language models, delving into their mechanisms, architectures, and paradigms. Subsequent chapters will build on the foundational understanding of language models established here.

# Chapter 3

# Bias in natural language models

## 3.1 Introduction

In this chapter, we explore the concept of bias relating to natural language models. Additionally, we carry out a literature study on existing methods to measure and mitigate bias in NLP. We first discuss the two main types of biases –intrinsic and extrinsic– and review some existing metrics proposed to quantify these biases. This will set the tone for Chapter 4, where we will investigate the compatibility of intrinsic bias metrics and address other issues regarding those metrics. Furthermore, in this chapter, we discuss bias mitigation techniques, laying the foundation for Chapter 5, where we will examine the relationship between intrinsic bias mitigation techniques and downstream fairness, and also investigate the relationship between intrinsic and extrinsic bias.

*This chapter is based on portions of the works published as follows:*

1. ***Pieter Delobelle, Ewoenam Tokpo, Toon Calders, and Bettina Berendt. 2022. Measuring Fairness with Biased Rulers: A Comparative Study on Bias Metrics for pretrained language models.*** *In Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, pages 1693–1706, Seattle, United States. Association for Computational Linguistics[34]* [1].
2. ***Ewoenam Kwaku Tokpo, Pieter Delobelle, Bettina Berendt, and Toon Calders. 2023. How Far Can It Go? On Intrinsic Gender Bias Mitigation for Text Classification.*** *In Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics, pages 3418–3433, Dubrovnik, Croatia. Association for Computational Linguistics* [2].

[1] Equal contribution from Pieter Delobelle and Ewoenam Tokpo: Both authors helped with formulating relevant research questions and developing the general outline of the research. Additionally, Ewoenam Tokpo primarily conducted the empirical studies, and Pieter Delobelle primarily conducted the literature survey and analyses of fairness metrics.

[2] Equal contribution from Ewoenam Tokpo and Pieter Delobelle: Both authors helped with formulating relevant research questions, developing the general outline of the research and composing the main body of the paper. Additionally, Ewoenam Tokpo primarily conducted the experimental work, and Pieter Delobelle primarily conducted the literature studies.

## 3.2 Bias in NLP

Beukeboom et al. [11] define bias in linguistics as *"a systematic **asymmetry** in **word choice** that reflects the social-category cognitions that are applied to the described group or individual(s)"*. This implies that an imbalance between words that define respective demographic groups and words that are relevant to a given domain could compromise fairness. An absence of balance could create a disproportional degree of association between domain-relevant words and attribute words that define different demographic groups. Such asymmetries often carry stereotypical undertones, hence the need to mitigate them. For example, in the professional domain, we would consider a language fair if it maintains a balance (symmetry) between gender-defining terms like gender pronouns and some career-defining terms like gender-neutral job titles[3]; the occurrence of *he is a nurse* should be as frequent as *she is a nurse*. We will refer to demographic terms like gender pronouns as *attribute terms* and refer to domain-relevant words like job titles as *target terms*.

Based on the work of Garrido-Muñoz et al. [57], we define bias in language models as the undesired variation in the probability distribution of certain words according to specific prior words within a given domain.

More formally, given a set of target terms (e.g., job titles) $X = \{x_1, \ldots, x_n\}$, bias can be defined as the difference in the association of target words with respect to different demographic attributes. This is presented as $d(u(X|A_f), u(X|A_m))$, where $A_f$ and $A_m$ are two sets corresponding to different values of a protected attribute $A$ (e.g., gender: $A_f$ ={*female, . . . , she*} and $A_m$ ={*male, . . . , he*}), $d$ is a certain distance measure, and $u$ is a measure of association between a value of a protected attribute and the set of target words $X$. For instance, given the target word *"engineer"*, we compute the association between *"engineer"* and a set of male-defining words as well as female-defining words, respectively. The difference in association between the two gender groups reflects a degree of bias.

To measure bias, there is a need for a notion of association or similarity. We will discuss these notions later in this chapter and in more detail in Chapter 4.

### 3.2.1 Types of bias in language models

Owing to the unique framework of modern language models, biases can be categorized into two categories, namely *intrinsic* and *extrinsic* biases. Intrinsic bias is associated with the pretraining phase of language models, whereas extrinsic bias is associated with the fine-tuning phase, as shown in Figure 3.1.

**Intrinsic bias.** Biases associated with the pretrained model are called *intrinsic biases*. They are termed "intrinsic" because they are inherent to the pretrained models due to biases in the pretraining corpus. An appealing aspect of pretraining is its predominantly unsupervised (self-supervised) nature, meaning explicit labels are not required. This

[3] A job title will be gendered if it has an inherent gender association e.g., actor and actress. Such examples are considered gender-defining terms.

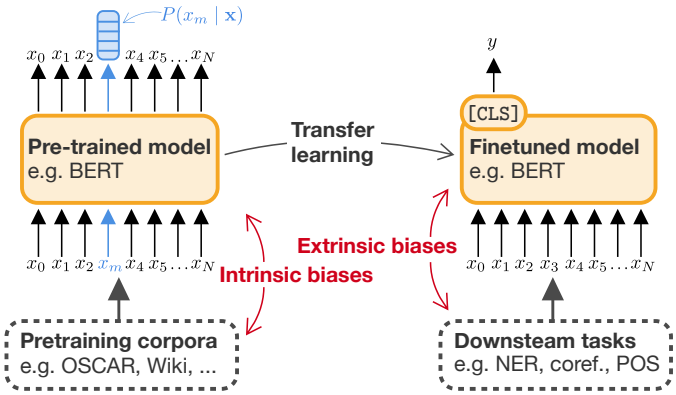



Figure 3.1: Illustration of the transfer learning paradigm where a language model is first pretrained on one dataset and afterward fine-tuned on another dataset. Both stages can introduce biases.

allows language models to be trained on large-scale unlabeled corpora often available online. However, this low barrier to language modeling also allows the use of low-quality data that may contain unwanted biases and stereotypes. When there are imbalances in pretraining corpora (like *she* being more frequently associated with *nurse* than *he*), the attention mechanism learns to emphasize these relationships, causing LMs to replicate or amplify these asymmetries.

**Extrinsic bias.** Biases detected after fine-tuning LMs for downstream tasks are referred to as *extrinsic biases*. These biases are not inherent in the pretrained model but are mostly acquired during fine-tuning. This usually occurs due to the interaction between the pretrained model's learned representation, the downstream data, and the fine-tuning objective. These are biases associated with applications like text classification, text entailment, machine translation, etc.

## 3.3 Measuring bias in Language models

Fairness in machine learning has a long-standing history, and a general introduction is out of scope for this thesis, so we refer the reader to the work of Barocas et al. [5]. Typical metrics, e.g. demographic parity, are not always directly applicable to tasks dealing with natural language. Furthermore, many NLP applications fine-tune existing language models, which intertwine intrinsic and extrinsic biases as will be discussed in chapter 5.

### 3.3.1 Measuring bias in non-contextualized word embeddings

Efforts to quantify bias in language models initially focused on non-contextualized word embedding models such as Word2Vec and Glove. This field was pioneered by researchers

like Bolukbasi et al. [17] and Caliskan et al. [20].

Bolukbasi et al. introduced the *direct bias* metric, which utilizes principal component analysis (PCA) to define a demographic subspace. To establish this subspace, they combined several difference vectors that represent a demographic axis (for instance, $\overrightarrow{she} - \overrightarrow{he}$ and $\overrightarrow{woman} - \overrightarrow{man}$ in the context of gender) and then identified the gender subspace $g$ by selecting the top-$k$ principal components. Bias is quantified by computing the cosine similarity between a set of target words and the gender subspace $g$. For a set of target words $X$, the direct bias is calculated as:

$$DirectBias_c = \frac{1}{|X|} \sum_{w \in X} |cos(\overrightarrow{w}, g)|^c$$

where $c$ is a parameter that determines the strictness in measuring bias; if $c = 0$, then $|cos(\overrightarrow{w}, g)|^c = 0$ only if $\overrightarrow{w}$ has no overlap with $g$, but 1 otherwise.

Caliskan et al. introduced the Word Embeddings Association Test (WEAT) [20], which is inspired by the Implicit Association Tests (IAT) [63]. The core idea of WEAT is to compute the disparity in similarity between target terms and attribute terms associated with two different demographic groups. Specifically, given two sets of target words $X_f$ and $X_m$ representing demographic stereotypes with respect to females and males respectively (e.g., $nurse \in X_f$ and $mechanic \in X_m$ for gender stereotypes) and two sets of attribute words $A_f$ and $A_m$ (also with respect to females and males e.g., $she \in A_f$ and $he \in A_m$), WEAT is computed as:

$$s(X_f, X_m, A_f, A_m) = \sum_{x_f \in X_f} u(x_f, A_f, A_m) - \sum_{x_m \in X_m} u(x_m, A_f, A_m)$$

$u(x, A_f, A_m)$ measures the association between a target term's embedding $x$ and those of attributes terms $a_f \in A_f, a_m \in A_m$. This is computed as:

$$u(x, A_f, A_m) = \text{mean}_{a_f \in A_f} \cos\left(x, a_f\right) - \text{mean}_{a_m \in A_m} \cos\left(x, a_m\right)$$

### 3.3.2 Measuring intrinsic bias in contextualized LMs

Similar to non-contextualized models, the primary intuition for quantifying intrinsic bias in contextualized LMs is to quantify the relationship between attribute and target words. In Table 3.1, we give an overview of some intrinsic bias measuring techniques we will be reviewing in this section.

Although May et al. [112] found lower levels of intrinsic bias in contextualized embeddings compared to non-contextualized embeddings, they highlight the more challenging and nebulous nature of measuring bias in contextualized embeddings, relative to non-contextualized embeddings. Specifically, the context dependence of contextualized word embeddings poses a huge challenge in quantifying bias [13, 15]. We will further investigate this issue in Chapter 4. Based on the way LMs are trained, the relationship between attribute and target words manifests in two ways: *attributional similarity* and *topical similarity* [155].

Table 3.1: Overview of intrinsic measures of bias for language models. We indicate if data and source code are both available (●), or if only a dataset is available (⊙), or if neither is publicly available (○). The repositories are linked in Appendix A.1.

| Metric | Similarity type | Templates | Models | Embedding type | Code |
|---|---|---|---|---|---|
| DisCo [173] | Topical | Appendix B.1.1 | BERT, ALBERT | — | ○ |
| Lauscher et al. [94] | Topical | | BERT | | ○ |
| LPBS [89] | Topical | 'X is a Y', 'X can do Y' | BERT | — | ● |
| BEC-Pro [7] | Topical | Appendix B.1.4 | BERT | — | ● |
| **Based on WEAT** | | | | | |
| SEAT [112] | Attributional | Appendix B.1.2 | BERT, GPT, ELMo, .. | [CLS] (BERT) | ● |
| Lauscher et al. [94] | Attributional | '[CLS] X [SEP]' | BERT | Vulic et al. [171] | ○ |
| Tan et al. [160] | Attributional | Appendix B.1.2 | BERT, GPT, GPT-2, ELMo | Target token | ● |
| CEAT [66] | Attributional | Reddit | BERT, GPT-2, ELMo | Target token | ● |
| CAT [120] | Topical | StereoSet | | | ● |
| CrowS-Pairs [121] | Topical | CrowS-Pairs | BERT, RoBERTa, ALBERT | — | ● |
| Basta et al. [8] | Attributional | — | ELMo | — | ○ |
| Zhao et al. [187] | Attributional | — | ELMo | — | ⊙ |
| Sedoc and Ungar [150] | Attributional | Not mentioned | BERT, ELMo | Mean | ● |

#### 3.3.2.1 Attributional similarity measures

Attributional similarity applies to words that are analogous (interchangeable) or share a number of common attributes. During training, such words tend to appear in similar contexts. e.g. *coast* and *shore*. If certain attribute words and target words are often associated with the same concept in the pretraining corpora, they would have a high attributional similarity. If *man* and *mechanic* often appear in similar contexts (e.g., *the* ***man*** *fixed the car*, and *the* ***mechanic*** *fixed the car*), a strong notion of attributional similarity is established between *man* and *mechanic*. Bias metrics that are based on computing the similarity between attribute words and target words in a given context intuitively rely on this notion of similarity. We outline some of these measures below.

**Sentence Embedding Association Test (SEAT).** SEAT is an adaption of WEAT for contextualized embeddings [112]. It uses "semantically bleached" templates to create a context for attribute and target words. Semantically bleached templates are templates that supposedly carry no semantic information or do not contain important information that could impact bias, e.g. '*__ is a* [MASK]'. May et al. [112] originally used the start-of-sentence token of the template-based text as a representation for the corresponding attribute or target term. Later works like Lauscher et al. [94] use token embeddings from the first four attention layers instead. Others like Tan et al. [160] use the embedding of the token of interest in the last layer.

**Contextualized Embedding Association Test (CEAT).** Guo et al. [66] present CEAT, an adaptation of WEAT which uses naturally occurring sentences from Reddit [4] containing attribute and target words as context templates instead of semantically bleached templates. They argue that naturally occurring sentences serve as more realistic contexts than semantically bleached templates [94, 160, 112].

[4]https://www.reddit.com/

**Direct bias.** Although *direct bias* was originally proposed for non-contextualized embeddings, adaptations have been proposed for contextualized LMs like ELMO and BERT [8, 187, 150]. The idea behind these approaches is to use templates, as is done in adaptations of WEAT, to create contexts in order to generate relevant embeddings for words.

#### 3.3.2.2 Topical similarity measures

Two words are said to be topically similar if they belong to the same domain or topic. Such words often tend to co-occur within the same context. e.g. *car* and *road*. Bias metrics that rely on fill-in-the-blank approaches, like *log probability bias score*, intuitively adopt this notion of similarity for bias. As an illustration with the original BERT model, the sentence '`[MASK]` *is a doctor.*' is filled in with the token '*He*' (62%), followed by "*She*" (32%) because the term *he* co-occurs with *doctor* in such types of contexts more than *she* does.

**Discovery of Correlations (DisCo).** This intrinsic bias metric proposed by Webster et al. also uses semantically bleached templates. The templates are designed in the form of, e.g., '__ *likes to* `[MASK]`'. The __ token is filled with attribute words in a pairwise fashion (e.g., female-male) using a set of first names or gendered nouns (e.g., for the pair *(John, Mary)*, we get *John likes to [MASK]* and *Mary likes to [MASK]*). The language model suggests appropriate in-fillings for the `[MASK]` slot, and the three top predictions for each pair are compared. If these predictions differ in terms of probability distribution, this is considered an indication of bias. Laucher et al. [94] propose two new measures based on DisCo. For each template $T$, they filter predictions with $P(x_m \mid T) > \theta$ [5] instead of the top three items, where $x_m$ is a candidate token. In their first measure, they compute the ***average fraction of shared candidates*** $\varnothing frac$ between the two sets:

$$\varnothing frac = \frac{1}{N} \sum_{i}^{N} \frac{|C_m^{(i)} \cap C_f^{(i)}|}{min(|C_m^{(i)}|, |C_f^{(i)}|)}$$

, where $C_m^{(i)}$ and $C_f^{(i)}$ are the candidate sets obtained for the $i$-th sentence-pair. For the second measure, they compute ***normalized average absolute probability difference*** $\varnothing diff$ based on the probabilities $p(t)$ for all candidates $t$ in the union set $C^{(i)} = C_m^{(i)} \cup C_f^{(i)}$. Such that:

$$\varnothing diff = \frac{1}{N} \sum_{i}^{N} \frac{\sum_{t \in C_i} |p_m(t) - p_f(t)|}{(\sum_{t \in C_m^{(i)}} p_m(t) + \sum_{t \in C_m^{(i)}} p_f(t))/2}$$

where $p_f(t)$ and $p_m(t)$ are probabilities generated by female and male sentences, respectively, for a candidate word $t$. This describes the difference in probability between two demographics for their combined predicted tokens.

**Log Probability Bias Score (LPBS).** LPBS is another template-based approach proposed by Kurita et al. [89]. It quantifies bias using templates containing target words:

[5] $\theta$ is a predefined threshold

such as "*[MASK] is a doctor.*". They compute the difference in probability of a language model to predict words from two respective groups (e.g., *she, he*) to fill in the [MASK]. To account for the effect of prior probabilities of attribute words (because the higher prior probability of the token *'He'* relative to *'She'* may skew results), they normalize the results by dividing each prediction by the prior probability of the attribute term. LPBS is computed as:

$$LPBS = \sum_{x \in X} \sum_{i} |ls(a_{f_i}, x) - ls(a_{m_i}, x)|$$

where $a_{f_i}$ and $a_{m_i}$ are equivalent forms of a female and male attribute term respectively, $X = X_f \cup X_m$ is the set of all target (stereotype) words, $ls(a, x) = \log(\frac{P(a|x)}{P(a)})$ is the association between an attribute term $a \in A_f \cup A_m$ and a target term $x \in X$, $P(a)$ is the prior probability of $a$, and $P(a|x)$ is the probability of predicting $a$ as the [MASK] term given the presence of $x$ in its context.

**Context Association Test (CAT).** This metric, as presented by Nadeem et al. [120], is based on *stereoSet*, a dataset depicting stereotypes. This dataset has, however been called into question by the likes of Blodgett et al. [15] due to many ambiguities, assumptions, and data issues present in it. Nadeem et al. use *stereoset* to compute bias as well as the language modeling capability of LMs. The addition of language modeling evaluation was to address the overemphasis of previous works on evaluating bias without accounting for the performance or utility of the LM. Specifically, they propose two types of association tests, *intrasentence* and *intersentence* association test. The intrasentence test measures bias as well as the language modeling ability of an LM at sentence level using a fill-in-the-blank style template (e.g., Girls tend to be more __ than boys). A set of three target words corresponding to a pro-stereotype, anti-stereotype, and meaningless options are provided. They measure bias by determining the disparity in the preference for pro-stereotype over anti-stereotype examples and evaluate the language modeling ability by how often the meaningless option is preferred. In the intersentence test, the concept is the same except that a sentence prompt is given to the model, and three sentence resolution options corresponding to a pro-stereotype, anti-stereotype, and meaningless option are provided.

**CrowS-Pairs.** This approach [121] uses *pseudo-log-likelihood* [148] to compute bias. Given the pro-stereotyped and anti-stereotyped sentence pair *"John ran into his old football friend"* and *"Shaniqua ran into her old football friend"*, *(John, his)* are referred to as the *modified tokens* for the first sentence and *(Shaniqua, her)* are the modified tokens for the second sentence. The parts of both sentences that overlap are the *unmodified tokens*. Given $U = u_0, ..., u_l$ as the list of unmodified tokens, and $M = m_0, ..., m_n$ as the list of modified tokens, the likelihood of $p(U|M; \theta)$ is computed for all sentence pairs, where $\theta$ represents the model's parameters. They define bias as the percentage of examples for which the pro-stereotype sentence is assigned a higher score than the anti-stereotype sentence. For each sentence, $p(U|M; \theta)$ is approximated with the pseud-log-likelihood, which is obtained by masking one unmodified token $u_i \in U$ at a time and computing the sum of their log-likelihoods:

$$score(S) = \sum_{i=0}^{|C|} logP(u_i \in U|U_{\setminus u_i}, M, \theta)$$

**All Unmasked Likelihood (AUL).** Raising reservations about the use of mask tokens in CrowS-Pairs, Kaneko et al. [82] opt for computing pseudo-log-likelihood on unmasked sentences. Since the use of `[MASK]` tokens is misaligned with downstream tasks, the authors believe this misalignment could impact the measure of bias. They opine that the lossy compression of MLMs will prevent predictions with absolute confidence, even with unmasked sentences. They introduce a variant of AUL called *AUL with Attention weights* (AULA), which incorporates attention weights for the model to score bias.

$$AULA(S) := \frac{1}{|S|} \sum_{i=1}^{|S|} \alpha_i log P_{MLM}(w_i|S;\theta)$$

where $S$ is the given sentence $\alpha_i$ is the average of all multi-head attentions of a word $wi \in S$.

### 3.3.3 Measuring extrinsic bias

Since most downstream training tasks incorporate supervised learning, extrinsic bias is often measured using fairness metrics generally used in other machine learning domains. Bias in machine learning is often categorized into two types: *group fairness* and *individual fairness*. We will only focus on a selection of metrics specifically targeted and widely used in NLP literature.

#### 3.3.3.1 Group fairness

Group fairness metrics measure some statistical (dis)parity between demographic groups (e.g., females and males). Some group fairness metrics, such as demographic parity, measure disparity in prediction, whereas others, like equal odds, measure disparity in the error rates of the models. The limitation of group fairness metrics is their susceptibility to miss cases of intersectional bias or bias against groups that are not explicitly defined or protected by law [12].

**Error rate difference.** *True positive rate difference (TPRD)* also known as (true positive gap or true positive equality difference) and *False positive rate difference (FPRD)* are extrinsic metrics which feature commonly in fair NLP literature [32, 132, 77, 156]. These concepts are synonymous with the concept of *separation* [5] in classical fairness literature. TPRD measures the gap in true positive rates between the predictions for two demographic groups: $TPRD = P(\hat{y} = 1|y = 1, A = a) - P(\hat{y} = 1|y = 1, A = a')$. FPRD, on the other hand, measures the gap in false positive rates between the predictions for two demographic groups: $FPRD = P(\hat{y} = 1|y = 0, A = a) - P(\hat{y} = 1|y = 0, A = a')$.

**Winograd schemas.** Most fairness measures in machine learning target some form of classification setting. Winograd schema-type metrics are suited to measure bias in NLP tasks, like coreference resolution, which is not very convenient to measure with classification-based fairness metrics. The Winograd schema [97] was originally designed

to test machine intelligence based on anaphora resolution but has been adapted in various works into benchmark datasets for bias evaluation. These benchmark datasets have nuances that make them suitable for measuring biases in different scenarios and contexts [146]. Prominent among these are WinoBias [189], Winogender [146] and WinoGrande [147]. GAP [172] is another benchmark dataset that closely relates to the Winograd family. It has also been used to measure bias in pronoun resolution methods. The WinoBias dataset measures the ability of a fine-tuned language model to resolve the coreference of gender pronouns (female and male) pro-stereotype and anti-stereotype settings. A pro-stereotype setting is when, for instance, a male pronoun is linked to a male-dominated job, whereas a female pronoun being linked to that same job will be an anti-stereotype. E.g. **Pro-stereotype:** *[The janitor] reprimanded the accountant because [he] got less allowance.* **Anti-stereotype:** *[The janitor] reprimanded the accountant because [she] got less allowance.* A model is said to pass the WinoBias test if the resolution is made with the same level of performance for pro-stereotype and anti-stereotype instances. Given as $|F_1^{f_{pro}} - F_1^{m_{anti}}| + |F_1^{f_{pro}} - F_1^{m_{anti}}|$. Using this approach, [166] extended WinoBias to account for the tendency of a model to generally *skew* towards one gender, following $|F_1^{f_{pro}} - F_1^{m_{pro}}| + |F_1^{f_{anti}} - F_1^{m_{anti}}|$. In (Section 4.2.4), we will also investigate WinoBias (type 1) and the *skew* variant as implemented by [166].

#### 3.3.3.2 Individual fairness

Individual fairness metrics measure the disparity in treatment between similar individuals. That is, individuals who are similar with respect to a given task should be assigned similar predictions. This is motivated by findings that group fairness does not ensure the fair treatment of individuals within a demographic group [46]. The challenge with individual fairness is that it is not trivial to determine similar individuals [12]. An example of an individual fairness metric in NLP is counterfactual fairness (CF) [90, 134, 164].

**Counterfactual Fairness (CF).** This measure computes the disparity in performance between an individual test instance and its counterfactual. For instance, CF can be used to measure the difference in the probability of a positive prediction between a test example and its counterfactual. We take a counterfactual to be a replica of a text with all attribute words corresponding to a given demographic group replaced with equivalent terms from a complementary demographic group. For instance, the counterfactual for the text *"the man is a surgeon"* will be *"the woman is a surgeon"*. The intuition is that changing the attributes of a text while holding other words which are not dependent on the attribute constant will not change the distribution of the label [90].

#### 3.3.3.3 Task performance

An often overlooked aspect of fairness evaluation is the impact of the intervention on task performance. *Accuracy*, *F1* and other task performance metrics are important to know if or to what extent the model's performance is being compromised for fairness. Given a test set that is balanced and "fairly" sampled, these task performance metrics can in

themselves be tell-tale indicators of bias in fine-tuned models [189, 26]. In such scenarios, the fairest fine-tuned models would be expected to exhibit the best performance.

## 3.4 Mitigating bias in language models

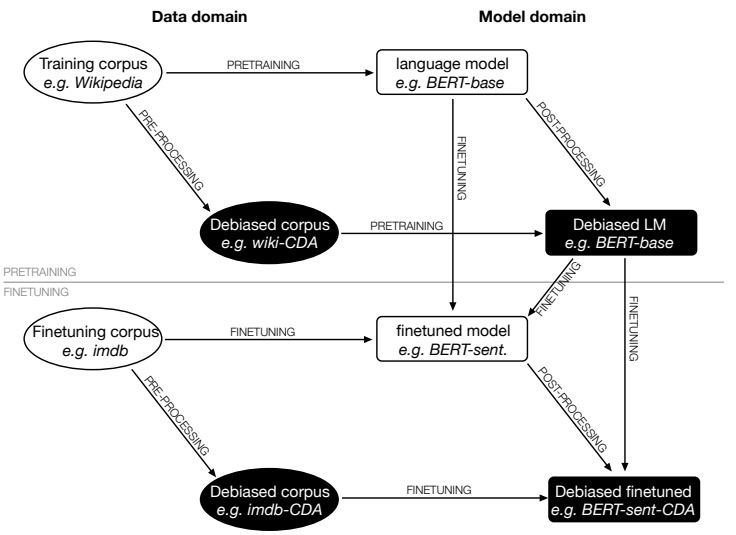



Figure 3.2: Illustration of where bias mitigation techniques alter a model or training data for both intrinsic (pretrained) and extrinsic (fine-tuned) biases in models.

In natural language processing (NLP), similar to many machine learning domains, bias mitigation can generally occur at three intervention avenues (Figure 3.2): the training data, the learning procedure, or the model output [105]. Bias mitigation techniques also tend to align with intrinsic or extrinsic types of biases. However, as we will see, many of these techniques, especially pre-processing and post-processing techniques, are applicable to both intrinsic and extrinsic bias. In-processing techniques tend to be more task-specific and, hence, are specifically designed to mitigate bias with respect to the training task.

### 3.4.1 Preprocessing techniques

Preprocessing techniques focus on the training data. As demonstrated by Dixon et al. [42], imbalances in training data can lead to unintended bias and are often the root cause of bias in natural language models. Preprocessing techniques can be applied to intrinsic or extrinsic bias depending on the application phase, i.e., the pretraining or fine-tuning phase. The goal is to remove spurious correlations between attributes in the training data. Removing unwanted correlations in the pertaining corpus would impact intrinsic bias, whereas doing the same in the fine-tuning dataset would impact extrinsic bias. Although preprocessing techniques are very effective, they require models to be retrained if bias

is detected after the fact. This makes them expensive or inefficient solutions in run-time environments.

**Counterfactual data augmentation.** One of the most adopted preprocessing approaches in NLP is counterfactual data augmentation (CDA) [191, 106]. The idea behind the approach is to generate a counterfactual for each example in the training corpus by identifying and replacing attribute terms with their complimentary equivalents from the other demographic classes. For example, in the case of binary gender, *'she'* will map to *'he'*, such that a sentence *"She is a nurse"* will be augmented with *"He is a nurse"*. This follows the intuition that in an ideal dataset, the association between gender attributes and target attributes like professions will be even for different gender groups. CDA has been used to mitigate intrinsic bias [173] and extrinsic bias [32] in existing works. To mitigate intrinsic bias, CDA has to be used in the pretraining phase, where the goal is to create a balanced association between attribute terms of distinct demographic groups and target words. For extrinsic bias mitigation in tasks like text classification, counterfactuals are generated for each instance in the fine-tuning data and the same task label $y$ is assigned to both the original and counterfactual instances. We will take a closer look at CDA in Chapter 6.

**Attribute scrubbing.** Here, attribute features in an NLP task are completely removed from text instances in the training data [132, 32]. This approach may seem intuitive, particularly in mitigating explicit bias, but deleting tokens in NLP tasks can be tricky since it could change the text's syntactic and grammatical structure, leading to out-of-distribution problems.

### 3.4.2 In-processing techniques

In-training or in-processing techniques aim to introduce constraints in the training objective to mitigate bias. Because of the different training paradigms adopted during pretraining and fine-tuning, in-processing techniques specifically target the training paradigm and the learning objective. Similar to pre-processing techniques, in-processing techniques are difficult to deploy in run-time environments.

#### 3.4.2.1 Intrinsic in-processing techniques

In-processing approaches to mitigate intrinsic bias aim to remove harmful correlations between attribute terms and target terms during pretraining. Contex-debias, proposed by Kaneko et al. [81], is a prime example of an in-processing bias mitigation technique.

**Context-debias.** This intrinsic debiasing method involves training the language model with a constraint to make the embeddings of target terms orthogonal to embeddings of attribute terms. Given the dynamic nature of contextualized word embeddings, which causes a word to have different embeddings, they define fixed word embeddings for each attribute token by averaging the contextualized embeddings of a word in all sentences in which it appears in the training corpus. Training is done so that the embeddings of

all stereotype terms are made orthogonal to all fixed attribute embeddings. They add a regularizer that constrains the debiased embedding to retain as much information as possible by ensuring that it is as similar to the original embeddings as possible despite the orthogonality.

#### 3.4.2.2 Extrinsic in-processing techniques

In-processing extrinsic bias mitigation approaches remove harmful correlations between attribute terms and task labels. Adversarial learning [62] is a popular example of in-processing techniques.

**Adversarial learning.** This approach aims to train a generator $g$ that modifies a sentence representation $e$ such that a downstream classifier $f$ is trained to predict the task label $Y$ as accurately as possible from $e$ while an adversary $h$ attempts to predict demographic information $Z$. The goal is to generate $e$ such that $h$ struggles to predict $Z$ from $e$, in order to achieve demographic parity. The set-up can be modified to achieve equal odds [184]. Elezar et al. [47] discuss how vanilla adversarial techniques do not fully succeed in removing all demographic information. They found that another adversary trained post-hoc on the debiased representation still manages to predict demographic information to a high degree of accuracy. Based on this finding, works like Gaci et al. [55] propose an iterative approach to adversarial learning, which trains a different adversary in each iteration to make the generator more robust.

### 3.4.3 Post-processing techniques

Post-processing approaches mitigate bias that exists in an already trained or fine-tuned model. Their primary advantage is that they offer a post-hoc solution that does not require the model to be retrained, although these solutions may not often be as effective at removing sensitive attributes from representations as pretraining or in-processing methods. Also, post-processing methods add an extra layer of complexity to the model during deployment or inference since the output representation has to be debiased each time the model is called. *Sent-debias* [101] and *iterative null-space projection* (INLP) [142] are examples of post-hoc mitigation techniques in NLP. They are both suitable for mitigating intrinsic and extrinsic bias. The primary intuition is to remove demographic information from hidden representations so that useful semantic information is retained for task prediction.

**Sent-debias.** Liang et al. [101] propose this post-processing bias mitigation method for contextualized embeddings. Sent-debias is inspired by *word embedding debiasing* proposed by Bolukbasi et al. [17]. First, attribute-defining words are identified and compiled in pairs or tuples – e.g. *(woman, man)*, *(boy, girl)*. These words are subsequently contextualized by extracting, from a text corpus, naturally occurring sentences containing these words. Counterfactuals are then generated for each sentence so that text instances only differ in terms of the attribute terms. The representations of these sentences are averaged to generate fixed embeddings for each word in the tuple. The demographic subspace is

approximated using the first $k$ principal components of the set of difference vectors. A text representation is debiased by projecting it orthogonally to the demographic subspace.

**Iterative Null-space Projection (INLP).** Given a sentence representation, the approach involves training a linear classifier to predict the protected attribute [142]. The sentence embeddings are then projected onto the nullspace of the trained classifier such that the resultant embeddings become orthogonal to the classifier. This process is iterative – as the name suggests – until no significant linear demographic attributes can be detected. As the authors suggest, this has some semblance to adversarial learning as the representation is iteratively made incognizant to attribute classifiers. It also maintains semblance with Sent-debias in the sense that it relies on a nullspace projection of a linearly derived demographic subspace.

## 3.5 Conclusion

In this chapter, we explored some relevant concepts relating to fairness in natural language models. We defined fairness in NLP as per the scope of this thesis. We then examined the two main categories of bias in NLP (intrinsic and extrinsic) and how these biases are quantified or measured based on existing literature. We subsequently reviewed and examined some of the major bias mitigation techniques for both intrinsic and extrinsic biases. We will investigate the compatibility of intrinsic bias metrics in Chapter 4, investigate the relationship between intrinsic and extrinsic bias in Chapter 5, and examine CDA as a bias mitigation approach in Chapter 6.

Chapter 4

# On the compatibility of intrinsic bias measuring techniques in language models

## 4.1 Introduction

Numerous techniques (or metrics) have been introduced to measure bias in contextualized word embeddings, some of which were explored in Chapter 3. These methods have been developed either as stand-alone approaches [112, 7] or as supplementary tools for assessing fairness interventions [173, 94, 89]. However, these metrics often reflect different interpretations of bias, and the wide range of available methods can make it challenging for NLP practitioners to choose an appropriate and reliable set of metrics for measuring bias. This challenge is further compounded by the fact that these techniques involve varying decisions regarding attribute and target words (commonly known as *seed words*), contextual templates, and representation schemes. We briefly discuss these factors below.

**Base metrics.** The choice of base metrics is by far the biggest factor in measuring bias in language models. In Section 3.3.2, we highlight some of the popular metrics for intrinsic bias. These metrics differ in their methodologies and how they represent different notions of similarity or association between attribute and target words.

**Seed words.** Seed words are an important component of any intrinsic bias measuring approach. These are sets of words that are used to describe demographic axes for which bias is being investigated. For instance, in the context of gender bias, seed words might include gender-defining terms like gender pronouns and gendered names. However, the selection of seed words is often subjective, leading to potential variations in how demographic axes are defined across different studies. This subjectivity can result in inconsistencies in bias measures. Antoniak et al. [2] have found evidence to suggest significant variations in results based on seedword choices.

**Templates.** Because contextualized language models need a context to generate embeddings, various works, such as that of May et al. [112], have sought to adapt bias measuring techniques for non-contextualized LMs to contextualized LMs by using *"semantically bleached"* templates. The idea is that a semantically bleached template has no inherent semantic information; hence, a seed word can be slotted in these templates to generate appropriate embeddings for the seed words. However, the assumption that these semantically bleached templates are indeed semantically bleached has been called into question by the likes of Tan et al. [160]. If the templates used to measure bias are not as semantically neutral as intended, this could lead to unwanted variations in bias scores based on the selection of these templates.

**Representations.** Different fairness works also adopt different word representation schemes to measure bias. In encoder-based transformer models, all the words in a text interact to form embeddings for each word. These embeddings are internally aggregated or pooled to generate an embedding for the entire text, known as the *sentence-level representation*. This pooled representation is usually the *start of sentence token* depicted as the `[CLS]` token in BERT. Because many studies use semantically bleached templates, the sentence-level representation is assumed to reflect only the semantics of the inserted seed word. However, Tan et al. [160] argue against this assumption and instead use the token-level representation (corresponding to the seed word's index in the template). Other works adopt less conventional techniques to generate representations. Vulic et al. [171] average the pooled embeddings of the first four attention layers for the target token without using a context/template.

In this chapter, we experimentally assess the compatibility of different metrics and the various design choices involved in measuring intrinsic bias. We conduct a series of correlation analyses and empirical evaluations of popular design choices, examining how these choices impact bias quantification. Our findings reveal that many metrics are not compatible with each other and are highly sensitive to factors such as the selection of templates, attribute and target seed words, and the chosen representation scheme. These results suggest that bias evaluation in contextualized language models remains challenging, largely due to the subjective nature of these choices.

Specifically, this chapter addresses ***RQ1*** by examining ***how reliable and consistent existing methods of measuring bias in NLP are, and what the best approach(es) to measure bias in NLP might be***. We particularly address two main questions relating to ***RQ1***:

- ***RQ1.1*: Are different base metrics and design choices used in existing literature compatible with one another?** What are the relationships between intrinsic bias measures, the templates, and the type of representations used for measuring bias? To what extent do these choices vary results? (Section 4.2)
- ***RQ1.2*: What set of approaches are best to measure, bias and what guidelines can be recommended to evaluate language resources?** (Section 4.3.1)

*This chapter is based on the work published as follows:* ***Pieter Delobelle, Ewoenam Tokpo, Toon Calders, and Bettina Berendt. 2022. Measuring Fairness with Biased Rulers: A Comparative Study on Bias Metrics for pretrained language models.*** *In Proceedings of the 2022 Conference of the North American Chapter of the*

*Association for Computational Linguistics: Human Language Technologies, pages 1693–1706, Seattle, United States. Association for Computational Linguistics[34]* [1].

## 4.2 Compatibility of intrinsic bias measures

Our goal is to extensively investigate the consistency of bias quantification between various techniques used by previous works. As mentioned earlier, besides the choice of base metric, three other factors are important when measuring intrinsic bias in an embedding model: (i) choice of seed words, (ii) choice of templates, and (iii) how representations for seed words are generated.

Recent works investigating bias in language models have found issues regarding inconsistencies between seed word choices [2], implicit assumptions and data quality issues in StereoSet and CrowS-Pairs templates [15], and issues with semantically bleached templates [160]. These issues raise concerns about the extent to which design choices influence the bias detection and quantification process. For instance, are "semantically bleached" templates really semantically bleached? i.e., do they have no effect over bias measurements? Or in the extreme, can bias in embedding model stay hidden by picking the "wrong" templates or representations? We seek to answer these questions through a series of experimental analyses, in which we measure correlations between various approaches to determine whether different templates and representation schemes quantify bias in a consistent manner.

### 4.2.1 Methodology

We conduct correlation analyses between different templates (Section 4.2.2), different representation methods (Section 4.2.3), as well as different measures (Section 4.2.4). To establish context and to help draw concise conclusions, we focus all our experiments on binary gender bias in the professional domain.

We start by compiling the set of attribute terms (gendered words) and target terms (professional words) [2] from existing literature. Following the works of Caliskan et al. [20] and Zhao et al. [189], we create two subsets of attribute and target sets, respectively, by splitting them according to male and female orientations. Specifically, we split the set of target terms into a set of female stereotyped professions $X_f = \{x_{f_1}, x_{f_2}, \ldots, x_{f_n}\}$ and a set of male stereotyped professions $X_m = \{x_{m_1}, x_{m_2}, \ldots, x_{m_n}\}$[3]. From these, we generate 20 subsets by randomly sampling ten professions for each target set $X_f$ and $X_m$, such that for male and female professions, we get the sets $\{p_{f_1}, ..., p_{f_{20}}\}$ and $\{p_{m_1}, ..., p_{m_{20}}\}$, respectively, where $p_{f_i} \subset_f$ and $|p_{f_i}| = 10$. Given our set of attribute terms, we measure fairness in a given word embedding using different metrics for each of the target subsets $p_{f_i}$. We expect some subsets to show higher levels of bias than others. Given two different

[1] Equal contribution from Pieter Delobelle and Ewoenam Tokpo: Both authors helped with formulating relevant research questions and developing the general outline of the research. Additionally, Ewoenam Tokpo primarily conducted the empirical studies, and Pieter Delobelle primarily conducted the literature survey and analyses of fairness metrics.

[2] The full list of words are provided in Appendix B.2

[3] We use data from the US Bureau of Labour.

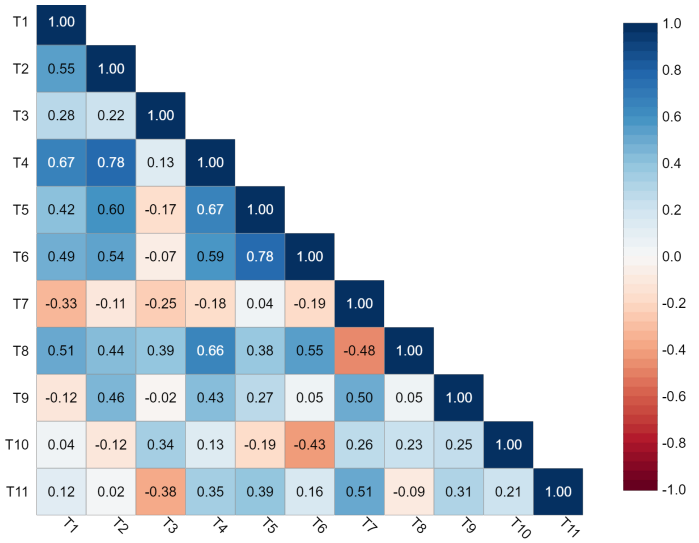



Figure 4.1: Correlation between template types using sentence-level embeddings

design choices to measure fairness (e.g., template choice or choice of representation), $\mathcal{M}_1$ and $\mathcal{M}_2$, we expect their fairness rankings to be consistent with each other. If there are any inconsistencies in their rankings, we can draw one of two conclusions: 1) either $\mathcal{M}_1$ and/or $\mathcal{M}_2$ are inaccurate measures, or 2) the notions of bias underlying $\mathcal{M}_1$ and $\mathcal{M}_2$ do not align or concur. We use Pearson correlation coefficients to determine the consistency in rankings.

For the correlation experiment between fairness metrics (Section 4.2.4), instead of using the subsets described above, we use five different language models, namely; *BERT-base-uncased, BERT-large-uncased, RoBERTa-base, DistilBERT-base-uncased and BERT-base-multilingual-uncased*. The different language models replace the need for subsets. This choice is based on the assumption that different language models will exhibit different levels of bias based on differences in training setups and datasets, which was observed for metrics evaluated on multiple models [121]. This choice also allows us to use the templates and seed words for each metric as described in the original studies, thus, compare the metrics as they are used.

### 4.2.2 Compatibility between templates

Many studies propose the use of "semantically bleached" sentence templates for context. Ideally, templates should contain no semantic substance, ensuring that the embedding produced by inserting a seed word in the template solely represents the seed word. Based on this assumption, we expect semantically bleached templates to have high correlations with each other. Previous works, such as [112, 160], suggest that semantically bleached templates may contain some semantic content. Our goal is to examine whether such occurrences exist and if they could potentially distort fairness results.

We test the semantically bleached SEAT templates [112], listed in Table 4.1 ($T_1 - T_8$). We

Table 4.1: Templates used in our evaluation of the compatibility between templates. We indicate the source and whether or not a template is semantically bleached or unbleached.

| # Full | Type Gendered | Source | Template sentence |
|---|---|---|---|
| $T_1$ | Bleached | May et al. [112] | "This is the _." |
| — | — | | |
| $T_2$ | Bleached | | "That is the _." |
| $T_3$ | Bleached | | "There is the _." |
| $T_4$ | Bleached | | "Here is the _." |
| $T_5$ | Bleached | | "The _ is here." |
| $T_6$ | Bleached | | "The _ is there." |
| $T_7$ | Bleached | | "The _ is a person." |
| $T_8$ | Bleached | | "It is the _." |
| $T_9$ | Bleached | Kurita et al. [89] | "The _ is a `[MASK]`." |
| $T_{10}$ | Unbleached | Tan et al. [160] | "The _ is an engineer." |
| $T_{11}$ | Unbleached | | "The _ is a nurse with superior technical skills." |

also compare these templates with the masked templates used by Kurita et al. [89] for their SEAT implementation ($T_9$), and add 2 *semantically unbleached* templates from Tan et al. [160] ($T_{10} - T_{11}$) as control templates. We use the sentence-level (`[CLS]`) embedding as implemented in the original work by May et al. [112]. We conduct the test based on the hypothesis that semantically bleached templates will correlate highly with each other. We carry out the correlation analysis as described in Section 4.2.1. Figure 4.1[4] present our results for the correlation analysis, where we make the observations discussed below.

***Choices of "semantically bleached" template could significantly vary the measure of bias.*** Although templates $T_1 - T_9$ are all semantically bleached, there are weak and sometimes even negative correlations (e.g., $T_7$). The fact that we do not get consistently high correlations among "semantically bleached" templates confirms the observation made by May et al. [112] and Tan et al. [160] on the possible impact that these templates could have on fairness evaluations. This observation contradicts the assertion that a sentence-level representation derived from semantically bleached templates solely represents the seed word, as it evidently captures some nuance from the context. This means the templates are evidently not as semantically bleached as one would imagine. Therefore, using the sentence-level embedding as the representation of seed words may not be an accurate representation.

***Semantically and syntactically similar templates do not necessarily correlate strongly.*** Templates **($T_3$)** *"There is the _."* and **($T_6$)** *"The _ is there."* contain the exact same words, and are believed to be semantically bleached, yet the correlation between them is low. This points to the fact that even templates that are syntactically and semantically similar may influence the measure of bias in different ways.

*Based on the above observations, we conclude that semantically bleached templates must be used cautiously, especially with sentence-level representations.* Any results stemming from using

[4]Table B.2

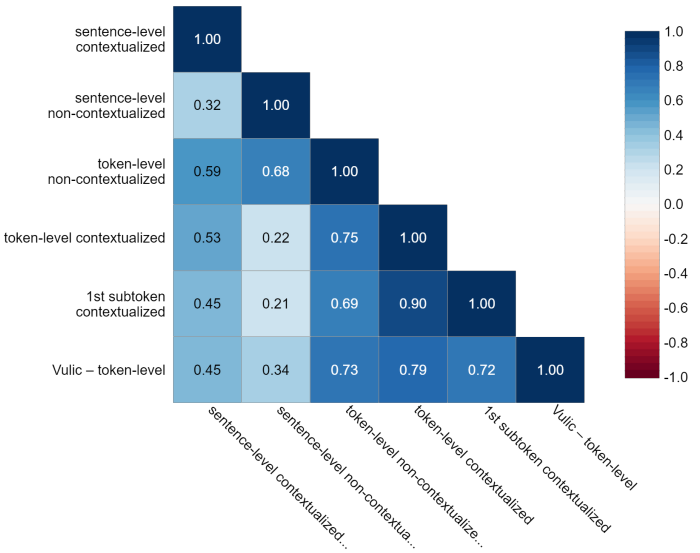



Figure 4.2: Correlations between different representation methods. Notice how both [CLS]-based methods are less correlated than other methods. The Pearson correlation coefficients in bold are significant at the $\alpha = 0.05$ level.

such templates cannot be objectively maintained as long as no standardized and validated scheme exists for selecting such templates. In the next section, we will show that token-level representations are more robust and less dependent on template choices, making them possibly a more appropriate option for templates.

### 4.2.3 Compatibility between representations

Word representations or embeddings could also be a source of inconsistency in evaluating bias in contextualized language models. As discussed above, sentence-level representations with semantically bleached templates do not capture the sole essence of the seed word but also capture unwanted information from the templates. We test various representation types that have been used in the literature. A complete list of these representation types, together with explanations, can be found in Appendix B.3. Below, we discuss some of the key elements of our analyses.

***Different choices of representations can introduce variations in bias scores.*** We investigate whether there are inconsistencies between representation methods by conducting correlation analysis of bias scores produced by SEAT, using the set-up described in Section 4.2.1. The correlations between these embedding methods are visualized in Figure 4.2[5], where we notice a weak correlation between techniques that use sentence-level ([CLS]) embeddings to represent seed words and techniques that use token-level representations. The weak correlation among the sentence-level techniques themselves

[5]Table B.4

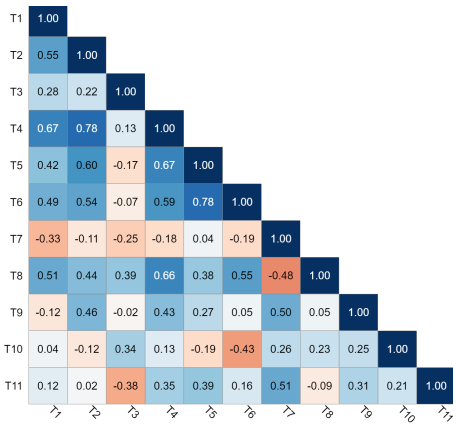



Figure 4.3: [CLS] embedding

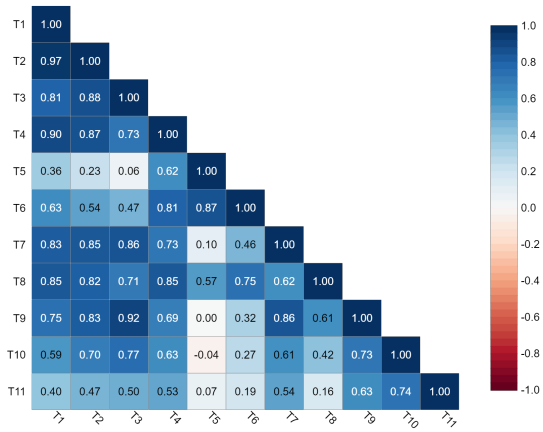



Figure 4.4: Target token embedding

Figure 4.5: Correlation of templates as listed in Table 4.1 when using two different embedding approaches, namely the [CLS] (left figure) and the pooled target token embeddings (right figure). Different embeddings result in different results, which we discuss further in subsection 4.2.3. The Pearson correlation coefficients in bold are significant at the $\alpha = 0.05$ level.

confirms the observation from the previous experiment that semantically bleached contexts have a significant influence on word representation.

***Token-level representations are more robust against variations in template choice.*** We explore how other embedding selection methods withstand semantic influence from the context/templates. Tan et al. [160] propose using token-level contextual word representations of tokens of interest instead of sentence-level representations. We investigate the effectiveness of this approach by replicating the experiment in Section 4.2.2. As depicted by the results of the correlation analysis in Figure 4.5, we show that token-level representations (Figure 4.4) produce more correlated results than sentence-level [CLS] representations (Figure 4.3). The token-level representations are more correlated and tend to be more robust against the influence of template choice. This is emphasized by the stronger correlation between token-level representations in Figure 4.2. This finding corroborates the position of Tan et al. [160] about the ineffectiveness of sentence-level representations for quantifying bias.

*Based on this observation, we recommend using token-level embeddings of seed words, as they produce more stable results across templates and are more resilient to contexts that may not be semantically bleached.*

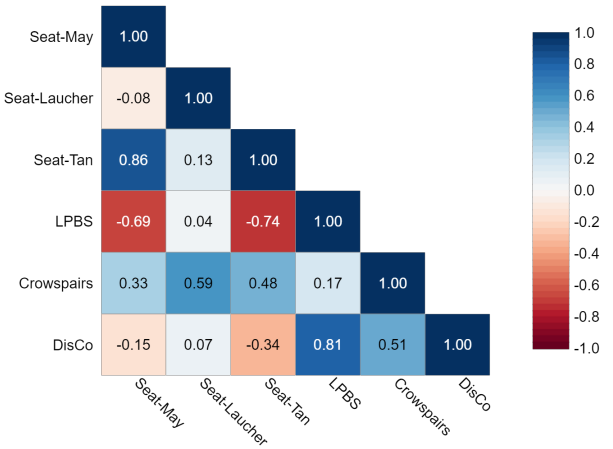



Figure 4.6: Correlations between different intrinsic and extrinsic fairness measures. The Pearson correlation coefficients in bold are significant at the $\alpha = 0.05$ level.

### 4.2.4 Compatibility between base metrics

We performed a correlation analysis between some intrinsic bias measures discussed in Section 3.3.2; the results are presented in Figure 4.6[6]. We observe that there is generally a poor correlation between the various metrics. CrowSpairs exhibits the best correlation with all the other metrics, consistently showing a positive correlation with all the others. SEAT-Tan [160] also maintains a positive correlation with all the other metrics but to a lesser degree. We also notice that metrics that use the same similarity notion exhibit very high correlation; for instance, attributional similarity measures like SEAT-default [112] and SEAT-Tan [160] and topical similarity measures like LPBS and DisCo. These observations show that the metrics proposed in existing works do not agree, mostly because of the different design choices, as previously discussed, coupled with the different notions of bias that underpin these metrics. In Chapter 5, we will further explore how these different underlying notions of bias may not align.

To further validate this finding, we conduct a different set of experiments where we use SEAT and LPBS to quantify bias in BERT-large [38] and ALBERT-large [93] pretrained models. Using those same metrics (SEAT and LPBS), we then explore how intrinsic bias changes in mitigated versions of these models using three of the mitigation techniques outlined in Section 3.4; namely, Context-debias, Sent-debias, and CDA pretraining.

In Figure 4.7, we plot the intrinsic bias results of this set of experiments[7]. We point out the contradictory nature of bias scores obtained from SEAT and LPBS, particularly for Context-debias and Sent-debias methods. Whereas SEAT shows a drastic reduction in intrinsic bias for Context-debias, LPBS indicates worsening bias. The converse is true for Sent-debias, where LPBS shows a significant reduction in intrinsic bias, but SEAT shows worse scores. This confirms our suspicion that different notions of intrinsic bias may not concur, and different metrics may respond differently to various mitigation strategies.

[6] Table B.5
[7] see Table C.1 for full results.

We will further investigate this phenomenon in Chapter 5.

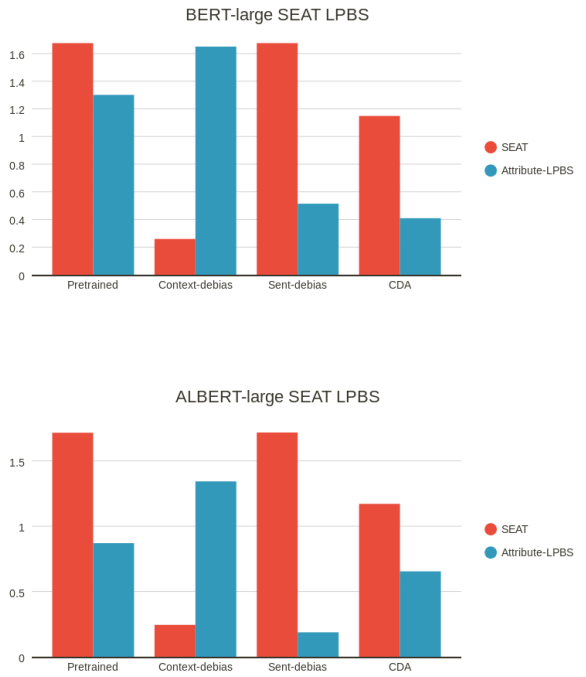



Figure 4.7: Intrinsic bias scores using SEAT and LPBS. Results show inconsistencies in measuring bias between SEAT and LPBS for various mitigation strategies — lower scores are desirable in both cases.

## 4.3 Conclusion

In this chapter, we addressed ***RQ1***, examining how reliable and consistent existing methods of measuring bias in NLP are, and what the best approach(es) to measure bias in NLP might be. We investigated different metrics and design choices for measuring intrinsic bias in contextualized language models, focusing on different templates, representation schemes, and the base metrics of association these measures use. We evaluated how these metrics correlate with one another, as well as how different design choices associated with these metrics correlate. We found that many aspects of intrinsic bias metrics can cause inconsistencies, e.g. choosing different templates, representation schemes, or the base metrics could yield different results.

### 4.3.1 Recommendations

Our recommendations concerning the measurement of intrinsic bias is to:

1. Be more cautious with the use of semantically bleached templates, as the choice of templates may influence bias scores, especially when used with sentence-level representations.
2. Use token-level representations for seed words, as they are more robust to template influence.
3. Select metrics and design choices that align with downstream usage; seed words and notions of bias should be chosen based on the specific downstream task.
4. Place less emphasis on embedding-based intrinsic bias metrics unless they strongly align with the downstream task. Instead, prioritize metrics that are independent of representation types, such as DisCo or LPBS, as they also tend to better align with many downstream tasks.
5. Use a mix of different intrinsic bias measures that cover different notions of bias for general evaluations to ensure better coverage.
6. Not use these metrics as proof or as insurance that models are unbiased, but rather consider them as indicators.

Due to the nebulous nature of measuring intrinsic bias, we recommend prioritizing extrinsic fairness evaluations on downstream tasks as the preferred approach to evaluating fairness, since this is where actual resource allocations occur. We will explore this in further detail in Chapter 5.

Chapter 5

# Assessing the Effectiveness of Intrinsic Bias Mitigation Techniques

## 5.1 Introduction

Based on our findings in Chapter 4, where we demonstrate that intrinsic bias measures often do not correlate and can be influenced by different underlying notions of bias and design choices, coupled with the fact that some definitions of intrinsic bias in existing works are imprecise and inconsistent [14], this chapter explores how different fundamental notions of bias may be misaligned. We develop a probe that presents two measures to evaluate the extent of gender information retained in word embeddings after applying some major intrinsic bias mitigation strategies.

We subsequently investigate whether existing intrinsic bias mitigation strategies are truly comprehensive solutions that concurrently address various types of bias. Our results show that, rather than fully eliminating gender bias, some intrinsic bias mitigation techniques address bias according to certain bias notions but not others. This selective mitigation can obscure bias from certain metrics, leading to false assumptions about the effectiveness of these strategies if they are not properly evaluated with other bias measures. Additionally, we demonstrate how certain bias mitigation strategies may be effective according to some bias metrics but produce contrasting effects when evaluated using others.

Furthermore, we investigate how different intrinsic bias mitigation techniques impact extrinsic bias and examine their effects when various downstream bias mitigation interventions are applied. Based on evidence from prior research that pretraining enhances downstream performance [133, 28], there has been a common assumption that intrinsic biases can lead to undesirable associations in downstream tasks, ultimately causing extrinsic bias. Only recently have more in-depth examinations been done to investigate this assumption [156, 123, 89]. However, conclusions have not been concise, and the confounding effects of bias mitigation techniques remain unclear.

Our analyses provide no compelling evidence that intrinsic bias is directly linked to extrinsic bias. Experimentally, we confirm that intrinsic mitigation techniques, when applied in isolation without further fairness interventions, do not significantly reduce extrinsic bias.

Specifically, this chapter addresses ***RQ2*** by examining ***the relationships between bias mitigation techniques, bias measures, and fairness in downstream applications.*** We develop a probe to uncover intrinsic bias in pertained contextualized language models. Using our probe, we seek to answer three key questions relating to RQ2 about the effectiveness of intrinsic bias mitigation techniques:

- ***RQ2.1*: Do different intrinsic bias mitigation techniques resolve bias according to different notions of bias?** (Section 5.4.1)
- ***RQ2.2*: How effective are intrinsic bias mitigation techniques at resolving bias?** (Section 5.4.2).
- ***RQ2.3*: Do intrinsic bias mitigation techniques in language models improve fairness in downstream tasks?** (Section 5.4.3)

*This chapter is based on the work published as follows:* ***Ewoenam Kwaku Tokpo, Pieter Delobelle, Bettina Berendt, and Toon Calders. 2023. How Far Can It Go? On Intrinsic Gender Bias Mitigation for Text Classification.*** *In Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics, pages 3418–3433, Dubrovnik, Croatia. Association for Computational Linguistics* [1].

## 5.2 Design of bias probe

Due to concerns about the reliability of existing metrics, such as SEAT and LPBS, for measuring bias in pretrained language models, especially when combined with mitigation techniques, we design a probe to investigate the retention of gender information after bias mitigation. Our goal is to measure the amount of gender information in the resulting embeddings of language models after applying various intrinsic bias mitigation techniques. Without properly investigating the efficacy of these mitigation strategies, it will be challenging to assert whether intrinsic bias materially affects extrinsic bias.

**Defining attribute and stereotype terms.** We first identify two sets of attribute terms that define females and males, respectively. Let $A_f = \{she, female, woman, \ldots\}$ and $A_m = \{he, male, man, \ldots\}$ be the respective sets of female and male gender-defining words. The set $A = A_f \cup A_m$ is the union set of gender-defining words for both females and males. Secondly, we define two sets of stereotype terms, which are composed of target words associated with females and males respectively[2]. Let $X_f = \{cheerleader, nurse, softball, ...\}$ and $X_m = \{warrior, baseball, engineer, ...\}$ be the respective sets of female and male stereotype words such that $X = X_f \cup X_m$.

[1]Equal contribution from Ewoenam Tokpo and Pieter Delobelle: Both authors helped with formulating relevant research questions, developing the general outline of the research and composing the main body of the paper. Additionally, Ewoenam Tokpo primarily conducted the experimental work, and Pieter Delobelle primarily conducted the literature studies.

[2]We use both attribute and stereotype wordlists from [81]

**Extracting word embeddings.** We extract all sentences from a text corpus containing words in each set of words. Especially for stereotype terms, we ensure that the sentences do not also include words in the attribute wordlist since attribute words in the same context can introduce gender information in the embedding produced. Let $S_w = \{s_{w_1}, ..., s_{w_n}\}$ be the set of all sentences containing a word $w$. We use the pretrained language model to extract embeddings for $w$ such that $w \rightarrow E_w = \{e_{w_1}, ..., e_{w_n}\}$, where $e_{w_i} \in \mathbb{R}^n$ is the embedding for token $w$ in sentence $s_{w_i}$.

**Training the classifier.** We split $A$ into a train set $A_T$ and a test sets $A_I$, such that $A_T \cap A_I = \emptyset$. Having no words occurring in both the training and test set allows us to verify that the trained model is not merely learning to identify words that appear in the train set rather than words containing gender information in general. We reserve the set of stereotypes $X$ solely for evaluating bias in the language models.

Taking $E_f = \{e_{f1}, e_{f2}, ...\}$ and $E_m = \{e_{m1}, e_{m2}, ...\}$ to be the set of all embeddings for female and male associated words respectively, given $E = E_f \cup E_m$, we train a classifier $C : E \rightarrow \{f, m\}$ to predict whether an embedding $e$ belongs to $E_f$ or $E_m$.

**Evaluating bias.** To determine the conformity of a language model $\mathcal{L}$ to gender stereotypes $X$, we introduce two criteria or notions of bias.

- ***Bias accuracy:*** We compute the accuracy of $C$ to correctly predict the gender association of all stereotype words in $X$ based on the embeddings produced by $\mathcal{L}$. A high accuracy depicts the presence of pro-stereotype information in the embeddings produced by the language model $\mathcal{L}$, even though the predictions could be of low confidence, i.e., just enough gender information to correctly predict the gender association.
- ***Mean bias confidence:*** We also consider the mean of the individual softmax scores for each prediction. In the case of binary gender, a softmax value of 0.5 means the embedding is perfectly neutral; hence, we define mean bias confidence as $\frac{1}{N}\sum_i^N |bias(e_i) - 0.5|$. High mean confidence depicts an enormous amount of gender information contained in the embedding, although this information may not necessarily be pro-stereotype.

We also use these two notions to investigate ***RQ2.1*** by observing if they consistently concur with each other.

**Randomization test.** To further validate our approach, we carry out a randomization test by iteratively splitting $X = X_f \cup X_m$ into two random groups $X_{f'}$ and $X_{m'}$. We use $X'^{(i)} = \{X_{f'}^{(i)}, X_{m'}^{(i)}\}$ to define the random pair generated in the $i$th iteration. We repeat this randomized split over 100 iterations; $X'^{(i)} \in \{X'^{(1)}, ..., X'^{(100)}\}$. By computing a p-value based on a one-sample t-test, we compute if the mean of accuracy scores from the random samples $X'$ is significantly different from the score obtained from $\{X_f, X_m\}$.

Table 5.1: Results of gender detection probe: **Model accuracy** denotes the general performance of the model or the accuracy of detecting gender information in attribute/gender terms (high scores are desirable). **Bias accuracy** indicates the accuracy of detecting gender information in stereotype terms (this should ideally be 0.5 in a fair classifier, showing the inability to correctly predict the gender association of stereotype terms). **Bias confidence** indicate the average confidence of prediction of stereotypes: $\frac{1}{N}\sum_i^N |bias(e_i) - 0.5|$ (low scores are desirable). P-values are from the randomization test.

| **Model** | **Model accuracy** | **Bias accuracy** | **Bias confidence** | **P-value** |
|---|---|---|---|---|
| | | ***BERT-large-uncased*** | | |
| pretrained | 1.0 | 0.78 | 0.20 | $9.10 \times 10^{-73}$ |
| Context-debias | 0.97 | 0.62 | 0.16 | $1.4 \times 10^{-51}$ |
| Sent-debias | 1.0 | 0.76 | 0.02 | $8.8 \times 10^{-70}$ |
| CDA | 0.99 | 0.75 | 0.16 | $7.8 \times 10^{-77}$ |
| | | ***ALBERT-large-uncased*** | | |
| pretrained | 1.0 | 0.83 | 0.33 | $2.8 \times 10^{-71}$ |
| Context-debias | 0.97 | 0.63 | 0.19 | $1.7 \times 10^{-48}$ |
| Sent-debias | 1.0 | 0.82 | 0.02 | $2.3 \times 10^{-95}$ |
| CDA | 0.94 | 0.63 | 0.12 | $3.5 \times 10^{-61}$ |

## 5.3 Probing the effectiveness of bias mitigation techniques

### 5.3.1 Intrinsic bias mitigation techniques.

As discussed in Section 3.4, intrinsic gender bias mitigation methods target either the pretraining data, the pretraining procedure, or the output of the pretrained model, which we refer to as pre-processing, in-processing, and post-processing, respectively [53]. For each of these categories, we select a popular intrinsic bias mitigation technique and study and compare their effectiveness in mitigating bias in pretrained language models. Specifically, we select Counterfactual Data Augmentation (CDA), Context-debias, and Sent-debias, all of which are discussed in Section 3.4. Using the probe we will discuss in the next section, we investigate how effectively these techniques mitigate gender bias and whether the mitigations are consistent across different notions of intrinsic bias.

### 5.3.2 Probing to uncover bias

We use the methodology outlined in Section 5.2 to probe for bias in pretrained BERT-large [38] and ALBERT-large [93] models and their bias-mitigated pretrained versions to investigate if the intrinsic bias mitigation techniques actually mitigate biases or superficially conceal them from commonplace metrics like SEAT and LPBS. Gonen et al. [61] discovered that, for non-contextualized word embeddings, the bias mitigation techniques proposed by Bolukbasi et al. [17] were capable of hiding biases instead of resolving them. We use our probe to see if this is equally the case for intrinsic bias mitigation in contextualized embeddings. The consequence of mitigation techniques superficially concealing bias is that downstream classifiers could learn to pick up residual traces of gender information if used in zero-shot settings.

We validate our results by carrying out a randomization test, as described in Section 5.2, to test for the statistical significance of our results. We conduct the test under the null hypothesis $H_0$ that the *bias accuracy* of our original gender-stereotype sets will not differ significantly from the random split. In Table 5.1, we show low p-values that indicate statistically significant results, thus refuting $H_0$.

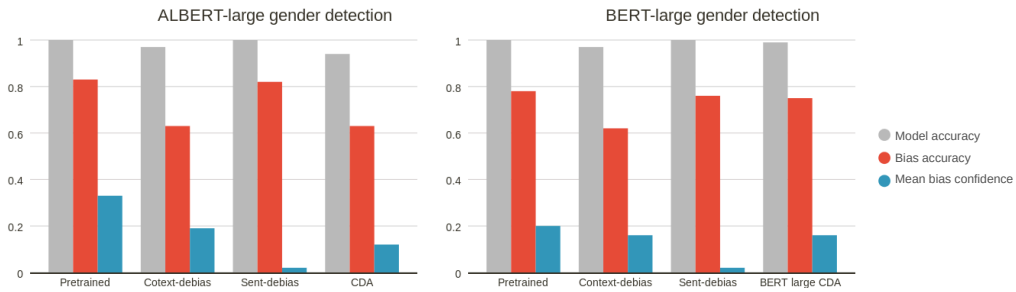



Figure 5.1: Gray bars indicate the accuracy of detecting gender information in attribute terms (high scores are desirable). Red bars indicate the accuracy of detecting gender information in stereotype terms (this should ideally be 0.5 in a fair classifier, showing the inability to correctly predict the gender association of stereotype terms). Blue bars indicate the average confidence of prediction: $\frac{1}{N}\sum_i^N |bias(e_i) - 0.5|$ (low scores are desirable).

## 5.4 Discussion

### 5.4.1 RQ2.1: Do different intrinsic bias mitigation techniques resolve bias according to different notions of bias?

To answer this research question, we investigate 2 main issues:1) Are there misalignments between different notions of bias? 2) Do different mitigation techniques align differently with different notions of bias?

***Misalignments between different notions of bias.*** Using the two notions of bias introduced in Section 5.2, we see from the plots in Figure 5.1 and the data from Table 5.1 that, counter-intuitively, high bias accuracy does not correspond with high mean bias confidence. This was consistently the case for both BERT-large and ALBERT-large models. Actually, Sent-debias, which consistently has the highest bias accuracy among the mitigation techniques, consistently has the lowest mean bias confidence. This indicates that these two notions of bias may not concur. This, coupled with the finding in Chapter 4 about the inconsistency between SEAT and LPBS, suggests that intrinsic bias could be perceived or measured from different or contradictory perspectives. This implies that while a model with low mean bias confidence will reduce the strength of associations between a target word and male and female words, respectively, it may not frequently alter the predictions made by the model. For instance, given the template *"[MASK] is a [profession]"*, while the model will reduce the disparity in the probability of *[MASK]* being *he* rather than *she* for each instance, it will often still assign a higher probability to

the more stereotypical option. On the other hand, if a model depicts low bias accuracy, it will show a more balanced frequency of prediction for both gender groups but may maintain a high degree of association between a target word and male and female words, respectively.

***Different mitigation techniques align differently with different notions of bias.*** To investigate if different intrinsic bias mitigation techniques resolve bias according to different notions of bias, we replicate the second experiment in Section 4.2.4, where we first use the two existing metrics, *SEAT* and *LPBS*, to quantify bias in BERT-large and ALBERT-large pretrained models. Using the same metrics, we then explore how intrinsic bias changes in mitigated versions of these models using three of the mitigation techniques: Context-debias, Sent-debias, and CDA pretraining. As depicted by the misalignment of different mitigation techniques with respect to different measures of bias in Figure 5.2[3], it is apparent that different mitigation techniques mitigate bias according to different notions of bias. This is further highlighted by the consistencies between Figure 5.1 and Figure 5.2; in the sense that in Figure 5.2, relative to Sent-debias, Context-debias has lower SEAT scores but higher LPBS scores, and in Figure 5.1, relative to Sent-debias, Context-debias has lower bias accuracy but higher mean bias score. This observation is consistent for both BERT-large and ALBERT-large. Based on this finding, we posit that for bias mitigation to be effective, the notion of bias used to measure and mitigate bias should align with the downstream usage and the types of harms intended to be mitigated.

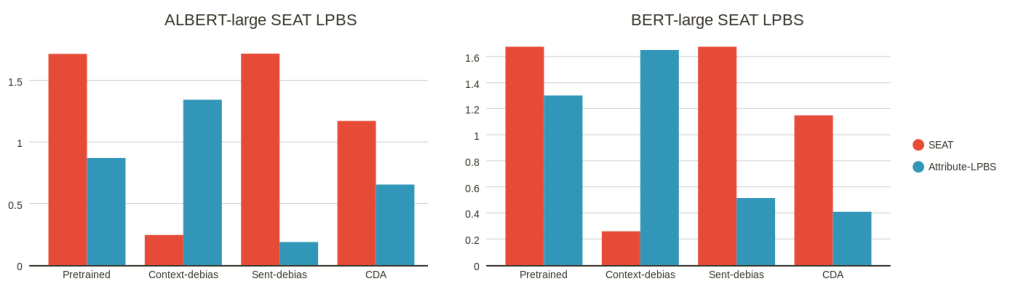



Figure 5.2: Intrinsic bias scores using SEAT and LPBS. Results show a misalignment in measuring bias between SEAT and LPBS for various mitigation strategies in a manner consistent with the misalignment between bias accuracy and mean bias confidence. — Lower scores are desirable in both cases.

***Misalingment between some intrinsic bias metrics and bias mitigation techniques*** We notice that certain metrics may not be suitable for assessing the effectiveness of certain mitigation techniques due to manner in which these techniques interact with a language model. Take, for instance, the metric LPBS, which measures the association between a stereotype term $x$ and a gender attribute $a$ based on the probability of predicting the gender attribute word when the stereotype term is in its context (expressed as $\frac{P(A=a|X=x)}{P(A=a)}$). This association starts in the lower layers of the model. However, since post-hoc approaches like Sent-debias do not modify the internal representation of the model, the

[3]see Table C.1 for full experimental results.

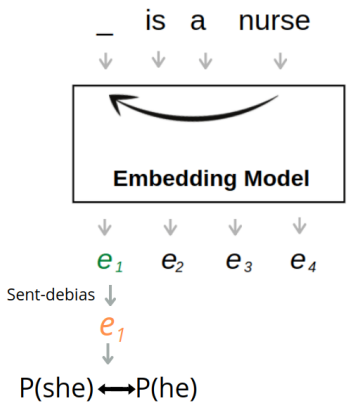



Figure 5.3: Measuring intrinsic bias in Sent-debias with target-LPBS.

original form of LPBS becomes ineffective when used with Sent-debias (because it only debiases the attribute term at the output layer). We term this original LPBS as *attribute-LPBS*. Sent-debias, when used with LPBS in this manner, mitigates bias in the representation of the attribute terms rather than the stereotype terms, resulting in consistently low probability difference scores, which may portray a misleading notion of less bias. Despite this phenomenon, some existing literature like Steed et al. [156] use LPBS in this manner with Sent-debias. To address this disparity, utilizing stereotype terms instead of attribute terms $P(X = x|A = a)$ may be a more effective option. We refer to this approach as *target-LPBS*. However, this approach may encounter challenges as many stereotype terms are often out of vocabulary, leading models to resort to wordpiece tokenization [151], which can be more difficult to manage. Nevertheless, in Figure 5.4, when the two versions of LPBS are compared, Sent-debias continues to have the best intrinsic fairness scores [4].

### 5.4.2 RQ2.2: How effective are intrinsic bias techniques at resolving bias?

***Some mitigation techniques can conceal demographic information.*** Based on results from Figure 5.1, we find that, for Sent-debias implemented in ALBERT-large, we can predict the association of stereotypes with their respective genders almost as accurately as with the unmitigated pretrained version, albeit with lower confidence. This raises concerns that a mitigation technique may be ineffective if a classifier can still detect such cues. This observation further suggests that models like Sent-debias notably decrease the association between stereotypes and genders, resulting in low prediction confidence. However, this reduction does not prevent accurate prediction of the associated gender stereotypes. It highlights that even if the degree of association (confidence) is significantly reduced, as long as sufficient information remains to correctly link gender groups with their associated stereotypes, intrinsic mitigation efforts *may* not fully address downstream fairness concerns.

[4]For this comparison we only selected single-wordpiece attribute words to avoid challenges with multiple wordpieces.

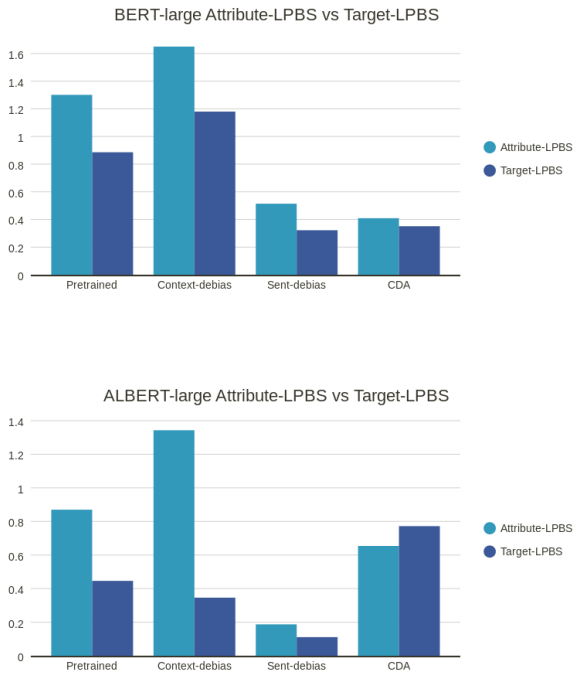



Figure 5.4: We compare LPBS in its default form (attribute-LPBS) and target-LPBS for both BERT and ALBERT. The plot shows a general positive correlation between the two metrics. In both cases, Sent-debias maintains the lowest bias score — lower scores are desirable in both cases

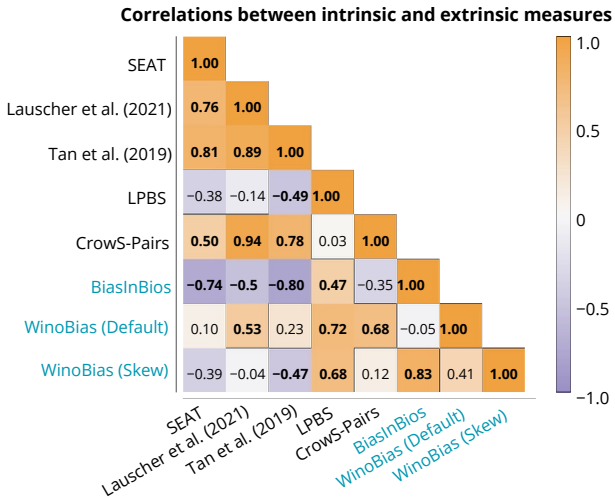



Figure 5.5: Correlations between different intrinsic and extrinsic fairness measures. The Pearson correlation coefficients in bold are significant at the $\alpha = 0.05$ level.

### 5.4.3 RQ3.2: Do intrinsic bias mitigation techniques in language models improve fairness in downstream tasks?

In this section, our goal is to 1) see if there is a general relationship between intrinsic and extrinsic bias measures and 2) how individual bias metrics correlate with extrinsic bias.

***Intrinsic bias measures do not materially correlate with and extrinsic bias measures.*** In our first experiment, we evaluate five popular language models[5] using two extrinsic bias benchmark datasets: BiasinBios [32] and WinoBias [189]. BiasInBios is an English dataset used as a classification benchmark for measuring bias in language models. It has been adopted as an extrinsic measure in several studies [173, 186]. The task is to predict professions based on biographies of people. Bias is quantified as the true positive rate difference between male and female profiles. We fine-tune each model on a training split of the dataset and evaluate them with the test split, using TPRD as the primary metric, as proposed by De-Arteaga et al. [32]. For WinoBias, we first adapt the models to the OntoNotes 5.0 dataset [174], which is the standard practice for WinoBias. We then follow the training and evaluation setup outlined by [166], specifically using the *type 1 F1 score difference* (*default*) and the *skew* metric. We compare these extrinsic bias scores to the intrinsic bias scores of the corresponding pretrained models. We use SEAT [112], LPBS [89] and CrowS-pairs [121] as our intrinsic bias metrics. For SEAT, we additionally test the variants introduced by Laucher et al.[94] and Tan et al. [160].

From Figure 5.5, we observe that, in general, there is a weak correlation between the intrinsic bias metrics and the extrinsic bias metrics, especially for the SEAT-based methods. Although we find stronger correlations between LPBS and BiasinBios, most correlations with the extrinsic BiasInBios measure are negative. On the contrary, CrowS-pairs shows

[5] `BERT-base-uncased`, `BERT-large-uncased`, `RoBERTa-base`, `DistilBERT-base-uncased`, and `BERT-base-multilingual-uncased`

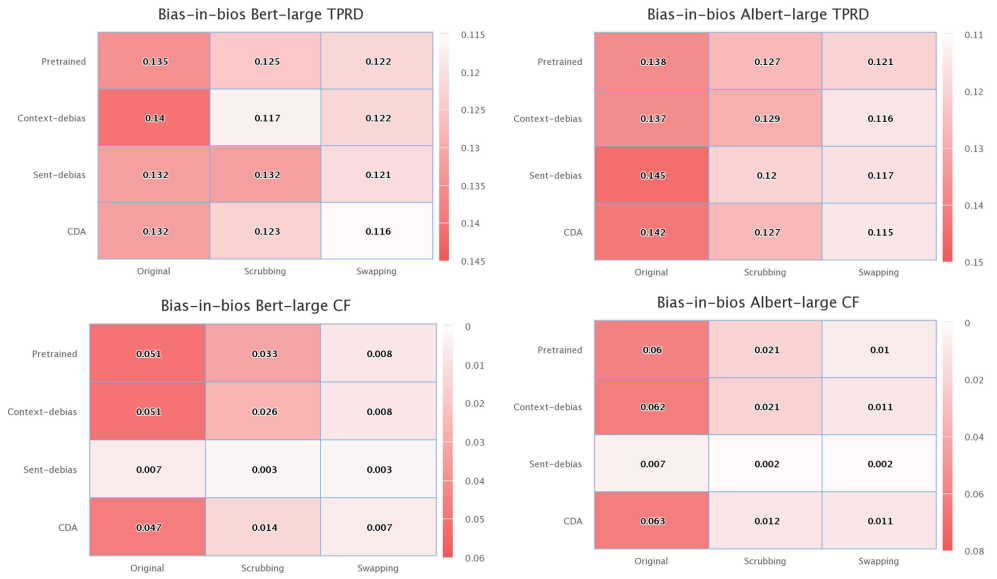



Figure 5.6: How intrinsic bias mitigation and downstream data intervention interact to influence fairness on the bias-in-bios dataset.

a weak correlation with BiasinBios but shows a stronger correlation with the remaining extrinsic measures.

***Intrinsic mitigation techniques do not show a significant improvement on TPRD.*** To further investigate the relationship between intrinsic and extrinsic gender bias, we train both BERT-large, ALBERT-large as well as their intrinsic-bias-mitigated versions on the Bias-in-bios and Jigsaw[6] (toxicity prediction) datasets using a 10-fold train-test scheme to correctly predict the profession of each profile. We evaluate extrinsic bias based on TPRD and *counterfactual fairness* (CF) scores. We conduct the test under three settings of the downstream data: 1) using the original version of the downstream data (***Original***); 2) using the attribute scrubbed version of the downstream data(***Scrubbing***); 3) using the CDA version of the downstream data (***Swapping***). These three data settings are important for understanding how intrinsic bias affects downstream fairness under different preprocessing settings.

In Figure 5.6, we show the extrinsic bias score [7] for BERT-large, ALBERT-large, their *debiased* versions, and the corresponding downstream data type for the Bias-in-Bios dataset[8].

We observe that all the intrinsic bias mitigation interventions we consider do not significantly improve bias in TPRD on their own. There is only a slight improvement or worsening bias in some cases. For Context-debias, we postulate this could be due to the

[6] `https://www.kaggle.com/c/jigsaw-unintended-bias-in-toxicity-classification/data?select=train.csv`

[7] Full results with error margins are given in Table C.4, Table C.5, Table C.6 and Table C.7.

[8] see Figure C.1 for Jigsaw dataset results.

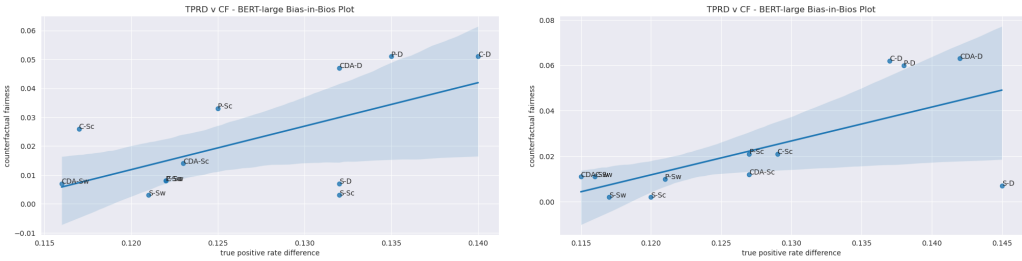



Figure 5.7: Relationship between TPRD and CF based on results from the Bias-in-Bios dataset.

limited list of words used to mitigate bias. If these limited words are not key terms in the downstream task or do not exist in the downstream text instances, this would result in a misalignment such that the mitigation may not have a consequential effect on the downstream task.

In general, the misalignment between intrinsic and extrinsic bias measures is likely due to two primary reasons. First, fine-tuning with downstream data can lead to significant forgetting of the parameterized knowledge from the pretrained model, a phenomenon commonly referred to as catastrophic forgetting [107]. This happens when fine-tuning overwrites the learned knowledge of the pretrained model. Second, the objectives of pretraining and fine-tuning are often misaligned, making it impractical to connect their associated biases. This may explain the relatively strong correlation between LPBS (intrinsic) and the WinoBias tests (extrinsic), as these tasks are more closely aligned.

***Downstream data processing significantly improves downstream fairness.*** Secondly, we observed that downstream data has a much more substantial effect on downstream fairness. Data processing techniques such as attribute scrubbing and CDA fine-tuning showed improvements in TPRD, yielding the best results. Therefore, downstream interventions, such as CDA, should be favored if extrinsic fairness is the goal.

***Combining intrinsic CDA and extrinsic CDA (swapping) produces the best TPRD results.*** Thirdly, we realize that although the downstream data seems key to downstream fairness, a combination of intrinsic and downstream data intervention generally produces better extrinsic fairness results.

***Combining Sent-debias and fair downstream data produces the best CF scores.*** We also realize from BERT and ALBERT (Figure 5.6) that combining Sent-debias as a mitigation technique and a downstream data intervention produces the best CF scores.

**Positive correlation between the extrinsic bias measures.** Here, we look at the relationship between TPRD and counterfactual fairness. As both measures are used as extrinsic fairness metrics, we aim to see how both results correlate. From Figure 5.7 we

find a positive correlation in general between the two metrics. Additionally, we find that downstream data intervention approaches tend to improve both TPRD and CF the most.

## 5.5 Related work

Park et al. [126] study the effects of three techniques to mitigate gender bias in abusive language detection; *word embedding debiasing* (WED) (based on the approach by Bolukbasi et al. [17]), *gender swap* (counterfactual augmentation of classification data), *bias fine-tuning* using non-contextualized word embeddings: word2vec [116] and FastText [16]. They found that a combination of WED [17] and gender swapping produced the best fairness results when implemented with a Gated Recurrent Unit (GRU).

Prost et al. [132] examine three debiasing techniques for mitigating gender bias in downstream text classification using bias-in-bios [32], with non-contextualized embeddings. Specifically, they examine *scrubbing* (deletion of identity terms) and approaches that focus on non-contextualized embeddings namely: WED (neutralization and equalization) [17], and *strong debiasing*. [132] Experimenting with Glove embeddings [128], they show that strong debiased produces the best fairness results whilst maintaining a good accuracy score (second only to the unmitigated model). They again show that WED can be counter-productive in terms of fairness; it can rather reduce fairness in downstream text classification.

Steed et al. [156] meticulously investigate the effect of intrinsic bias in downstream text classification in RoBERTa. They also conclude that intrinsic/upstream bias does not significantly contribute to downstream bias. However, they use the default form of LPBS [89] with sent-debias, which may not be an effective approach as discussed in Chapter 4.

Orgad et al. [123], the closest and concurrent with our work, investigate the relationship between intrinsic and extrinsic bias and also develop a probe based on a classification-based technique *minimum description length* [170] to detect bias in RoBERTa. They reach similar conclusions indicating how intrinsic metrics, like CEAT in their case, could conceal bias.

## 5.6 Conclusion

In this chapter, we addressed ***RQ2*** by examining the relationships between bias mitigation techniques, bias measures, and fairness in downstream applications. We developed a probe to investigate intrinsic bias in two language models, BERT-large and ALBERT-large, across two text classification tasks (BiasinBios and Jigsaw) and one coreference resolution task (WinoBias). This probe was used to answer the key questions outlined in this chapter.

We found that: 1) different notions of intrinsic bias may be misaligned, leading to inconsistencies between intrinsic bias metrics; 2) intrinsic bias mitigation techniques are not silver-bullet solutions and do not comprehensively resolve different notions of intrinsic

bias; 3) intrinsic bias metrics can be sensitive to specific bias mitigation techniques; and 4) the intrinsic bias mitigation techniques we studied do not significantly improve fairness in text classification when applied without additional fairness interventions, such as data pre-processing. We hypothesize that this is primarily due to catastrophic forgetting and the misalignment between pretraining and text classification (fine-tuning) objectives. We observe that downstream interventions like CDA on the downstream dataset significantly mitigates extrinsic bias. We will discuss CDA in more detail in Chapter 6.

### 5.6.1 Recommendations

Our recommendations for mitigating bias are as follows:

1. Greater focus should be placed on extrinsic bias mitigation, as it tends to be more effective, particularly in text classification tasks. However, intrinsic bias mitigation techniques can still be effective in zero-shot settings or for downstream objectives that are more closely aligned with pretraining objectives.
2. Intrinsic bias mitigation should be combined with other fairness interventions for better overall results.
3. When intrinsic bias needs to be evaluated, such as for zero-shot applications, the notion of bias used to measure it should align with the end-use and the specific types of harms being mitigated.

## Limitations

Although we aimed to make the list of gender and attribute terms used in our experiments as extensive as possible, it is nearly impossible to cover all possible gender and target words in such contexts. Nonetheless, for the scope under consideration, we believe our compilation is extensive enough to give a general outlook and to draw the necessary conclusions.

Chapter 6

# An Automated Approach to Model-based Counterfactual Data Augmentation For NLP

## 6.1 Introduction

In Chapter 5, we saw how effective counterfactual data augmentation (CDA) can be in mitigating bias, especially when applied to the downstream data. As previously discussed, CDA seeks to remove spurious correlations between attributes in the training data by evening out the distribution of words that characterize demographic attributes in the context of neutral words that should ideally not be demographically aligned. Specifically, explicit attribute-defining words are replaced with their counterfactual equivalents from complementary demographic groups for every text instance. To illustrate this with an example, an instance of "*She is a nurse*" will be augmented with "*He is a nurse*" in the case of mitigating gender bias. This follows the intuition that in an ideal dataset, the association between gender attributes and target attributes like professions will be even for different gender groups.

Key works, such as [189, 106, 191], introducing CDA as a bias mitigation technique adopt a word substitution approach based on dictionaries. These word substitution methods are prone to grammatical incoherence because of out-of-context substitutions and omitted word pairs. Because dictionary compilations are often incomplete [39], a direct word-substitution approach will not generalize to omitted words. Take for instance (***Bachelor*** *and* ***Masters*** *degree* v. ***Spinster*** *and* ***Mistresses*** *degree*) and (*she taught* ***herself*** v. *he taught* ***herself***) which were common issues we observed with some methods. Additionally, the dictionaries are manually compiled, which not only incurs potential costs but manually compiling counterfactual word pairs for certain demographics may be intrinsically challenging.

Although generative language models like GPT-related models [137] have surged in popularity, their adoption for CDA has been limited due to the relative unavailability of parallel data needed for training. As such, model-based solutions resort to manually compiling parallel training data, a process that is both costly and constrained. This challenge is exacerbated by the fact that training models on limited parallel data can

impair performance [192]. Although large conversational models like ChatGPT generate effective counterfactuals in zero-shot settings, they are inefficient in low-resource environments, where computational resources and data are limited. In this work, we focus on low-resource/resource-efficient techniques that can be deployed in low-resource environments.

The primary contribution of this chapter is to explore an automated approach to generate parallel training data for a given demographic axis that requires minimal human intervention. Our approach takes from a user a prompt – in the form of a single word-pair – that describes a demographic axis. This pair is subsequently used to model a demographic subspace from which other words that define the demographic attribute can be sampled from a given corpus of text. Using an invertible flow-based model [41], counterfactual words are generated for sampled words. Thereafter, an error correction approach is used in tandem with direct word substitution to generate parallel data to fine-tune a generative language model to generate counterfactual texts. We call our approach and the resultant counterfactual text generation model *FairFlow*. This entire process is simply depicted in a four-step process in Fig. 6.1. As opposed to existing works, which will be discussed in Section 6.2, FairFlow does not rely on human-generated parallel data for training and eliminates the need for manually compiled word-pair dictionaries.

This chapter specifically addresses ***RQ3*** by ***exploring the limitations of existing CDA approaches for mitigating bias and exploring the feasibility of developing techniques to address these challenges.*** We propose techniques to develop a robust model-based counterfactual generator in the absence of parallel training data. Our key contributions in this chapter include:

1. An automated approach to compiling dictionaries of word pairs that only requires a user to input a word-pair prompt that describes a demographic axis.
2. An error correction approach to generate parallel data from dictionary word substitutions.
3. Training a counterfactual model using our generated parallel data and show that the error correction approach not only improves the grammatical composition of the model but also improves the generalization of the model.

*This chapter is based on the work accepted to be published:* ***Ewoenam Kwaku Tokpo and Toon Calders. 2024. FairFlow: An Automated Approach to Model-based Counterfactual Data Augmentation For NLP.*** *In Joint European Conference on Machine Learning and Knowledge Discovery in Databases, pp. 160-176. Cham: Springer Nature Switzerland, 2024*

## 6.2 Background and Related Literature

Early works on CDA used simple rule-based word-substitution approaches for counterfactual data augmentation. Specifically, they created dictionaries of attribute word pairs and used matching rules to swap words [32]. Later works began to incorporate grammatical information like part-of-speech tags to swap attribute words [189]. In the

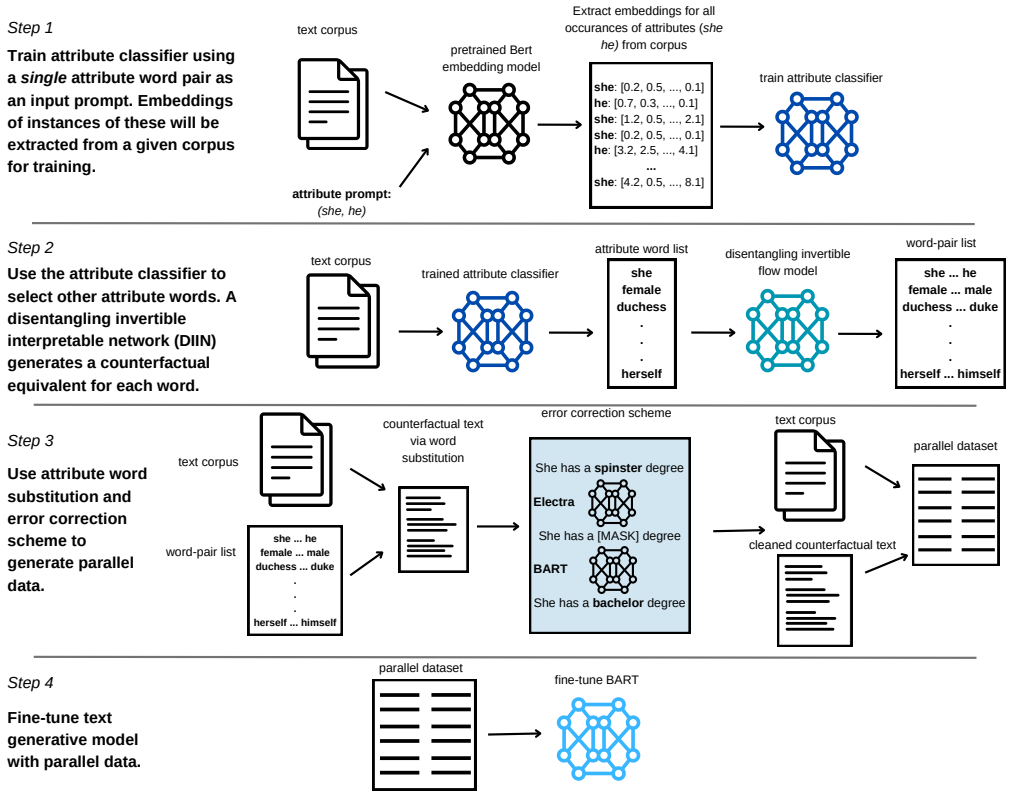



Figure 6.1: An end-to-end description of Fairflow, described in four steps: 1) train a classifier to identify attribute words from a corpus; 2) generate counterfactual equivalents for attribute words using an invertible generative flow model; 3) use a word substitution scheme and our proposed error-correction scheme to make the parallel text more fluent and realistic; 4) fine-tune a generative model with the generated parallel data.

absence of interventions for named entities, Lu et al. [106] do not augment sentences or text instances containing proper nouns, and named entities as generating counterfactuals without proper name interventions could result in semantically incorrect sentences. Zhao et al. [189] circumvented this by anonymizing named entities by replacing them with special tokens. Lamenting on the aforementioned lack of parallel corpus for training neural models, Zmigrod et al. [191] used a series of unsupervised techniques such as dependency trees, lemmata, part-of-speech tags, and morpho-syntactic tags for counterfactual generation. Hall-Maudsley et al. [111] improve on Zmigrod et al. by incorporating a names intervention method to resolve the challenges of generating counterfactuals for named entities. They achieve this using a bipartite graph to match first names.

Because the aforementioned techniques rely on dictionary word replacement techniques and ignore the context of the text, they are prone to generating ungrammatical texts. Additionally, the inability of these techniques to resolve out-of-dictionary words not only preserves certain attribute correlations but also introduces errors. We illustrate two instances of such limitations using the word substitution approach by Hall-Maudsley et al. on the Bias-in-bios dataset [32]; 1) *"Memory received her* ***Bachelor*** *and* ***Masters*** *of*

*Accountancy..."* produces *"Memory received his **Spinster** and **Mistresses** of Accountancy..."* due to the polysemous nature of *bachelor* and *master*; 2) *"Laura discovered her passion for programming after teaching **herself** some Python..."*, is transformed into *"Anthony discovered his passion for programming after teaching **herself** some Python..."* as the gender pronouns *herself* and *himself* are excluded from the dictionary compiled by Hall-Maudsley et al.

More recently, sequence-to-sequence model-based approaches to counterfactual generation have been proposed [180, 134]. Wu et al. [180] propose Polyjuice, a generative counterfactual model for diverse use cases like counterfactual explanations. They generate parallel data by pairing naturally occurring sentences in a corpus based on edit distances. Although effective for explanations, such an approach is not applicable for bias mitigation as attribute words, in the case of the latter, have to be specifically defined and replaced. Specifically for bias mitigation, Qian et al. [134] introduce the *perturber*, which is a BART [98] model fine-tuned on a human-generated parallel text. However, their approach only generates counterfactuals for specific user-defined entities in a text. eg. *original:"Torii chose to remain behind, pledging that he and his **men** would fight...", rewrite:"Tara chose to remain behind, pledging that she and her **men** would fight ..."*. As earlier stated, such manually compiled datasets are expensive and are only available on small scales, which can degrade performance [192]. Additionally, similar manual efforts must be solicited for every language domain for which counterfactuals have to be generated. As opposed to existing works, the main advantage of our work is the non-reliance on human-generated parallel data and word lists.

## 6.3 Approach

Our entire approach can be summarized in four steps as illustrated in Figure 6.1. The process commences with training a classifier to detect attribute words in a corpus, after which counterfactuals for these attribute words are generated using an invertible flow model. Parallel data is thereafter created by using a combination of word substitution and an error-correction scheme. Finally, a generative model is fine-tuned using the generated parallel data. We expound on these steps in the following subsections.

### 6.3.1 Attribute classifier training

To select a list of words that characterize a given demographic axis, e.g. gender, we first train an attribute classifier that approximates the attribute subspace. To do this, the user first inputs a prompt in the form of a single pair of words that describes a given demographic axis, e.g., (she, he) in the case of gender. Using a pretrained contextualized word embedding model, contextualized word representations are generated for each appearance of the input words within a given text corpus — we take *BERT-base-uncased* [38] as our choice of representation model. These embeddings are used to train a classifier to approximate the demographic subspace. Formally, consider the word-pair $(x_a, x_b)$ that define a demographic axis, we obtain two sets $Z_a = \{z_{a_1}, z_{a_2}, ..., z_{a_n}\}$ and $Z_b = \{z_{b_1}, z_{b_2}, ..., z_{b_n}\}$ where $z_{a_i} \in R^d$ and $z_{b_i} \in R^d$ are context-specific vector representations of instances of $x_a, x_b$ respectively, generated from a text corpus $V$ by a pretrained embedding model $E$; so that $E(x_i, c_i) = z_i$ if $x_i$ is an instance of a word $x$ and $c_i$ is its context. We

estimate the demographic subspace by training a classifier $H$ to maximizing the objective $\sum_{z_i \in \{Z_a \cup Z_b\}} log(P(y|z_i))$, where $y = \{a, b\}$ is the class label of $z_i$. $H$ is parameterized as a feed-forward neural network with one hidden layer and Gelu non-linear activation.

## 6.3.2 Generating word-pair list

### 6.3.2.1 Selecting attribute words

Given a demographic subspace, we select all words that lie within the attribute-defining regions of the subspace. This process is formally described as follows. Given our initial corpus $V$, we select words $x_i \in V$ based on the criterion $P(y|E(x_i, c_i); \Theta_H) > \phi$ where $\Theta_H$ represents the parameters that define $H$ and $\phi$ is a predefined threshold. $Z_a$ is thus expanded to include all words that have at least an instance satisfying $P(y = a|E(x_i, c_i); \Theta_H) > \phi$ and $Z_b$ to include all words with at least an instance satisfying $P(y = b|E(x_i, c_i); \Theta_H) > \phi$. Although some neutral words may be included in these sets, they do not produce any counterfactual equivalent in the next stage, hence making no difference.

### 6.3.2.2 Generating counterfactual word-pairs with DIIN

The first step in generating counterfactual equivalents for the set of words $Z_a$ and $Z_b$ is to define a transformation $T$ from the original embedding space into an "interpretable" space where an embedding is factorizable into independent components. We train $T$ to constrain attribute information *only* to the first $k$ dimensions (we will collectively refer to these dimensions as $K$) of a word in the interpretable space. By so doing, $K$ can be swapped to alter the attribute (eg. gender) of the word. We implement $T$ using a flow-based generative model [86, 125, 41]; specifically, we use the disentangling invertible interpretation network (DIIN) architecture by Esser et al. [48].

Formally, given the contextualized representation $z$ of a word $x$, the goal is to learn a transformation $T$ that maps the original representation $z \in R^d$ to an interpretable representation $\tilde{z} \in R^d$ s.t. $T(z) = \tilde{z}$. The interpretable representation $\tilde{z}$ is sampled from a base distribution $\tilde{z} \sim p_{\tilde{\mathcal{Z}}}(\tilde{z})$ – a standard Gaussian distribution in this case. Using the change of variable theorem, $T$ is learned by maximizing the log-likelihood

$$\log(p_{\mathcal{Z}}(z)) = \log(p_{\tilde{\mathcal{Z}}}(T(z))) + \log(|\det(\partial T(z)z)|) \tag{6.1}$$

To constrain attribute information only to $K$, we pair embeddings of words that have the same attribute $F$ and train $T$ to generate similar values for both embeddings in their first $k$ dimensions in the interpretable space. Mathematically, given a pair of embeddings $(z_{a_1}, z_{a_2})$ that belong to the same demographic group such that $F_{z_{a_1}} = F_{z_{a_2}}$, the objective

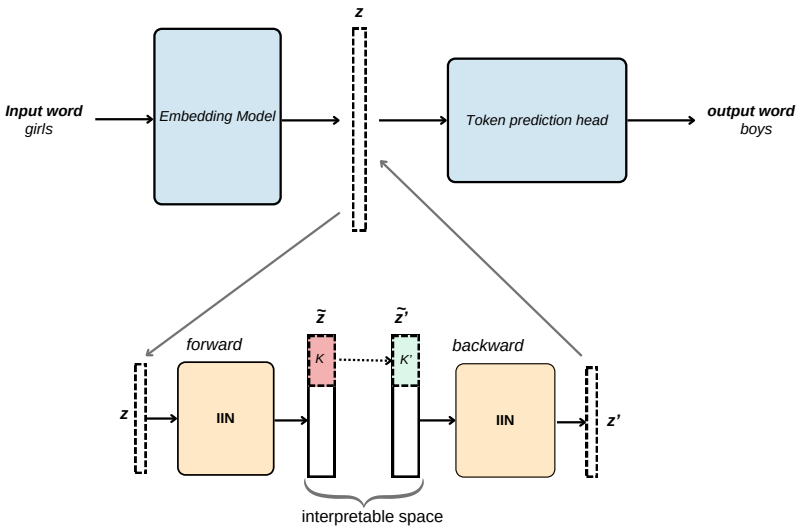



Figure 6.2: Counterfactual word generation using an invertible interpretation flow network IIN.

is achieved by minimizing the loss function:

$$\begin{aligned}\mathcal{L}(z_{a_1}, z_{a_1}|F) = ||T(z_{a_1})_D||^2 - \log(\det(T(z_{a_1}))) \\ +||T(z_{a_2})_{(D\setminus K)}||^2 - \log(\det(T(z_{a_2}))) \\ + \frac{||T(z_{a_2})_K - \sigma T(z_{a_1})_K||^2}{1-\sigma^2}\end{aligned} \tag{6.2}$$

where $D$ is a term to collectively refer to all $d$ components of the embedding. $\sigma \in (0,1)$ is a positive correlation factor that determines the strength of the correlation between $z_{a_{2_K}}$ and $z_{a_{1_K}}$. We also use the dimensionality estimation approach of Esser et al. to estimate the dimensionality of $K$.

Once our invertible flow model has been trained to constrain $F$ to the first $k$ dimensions of $\tilde{z}$ (in the interpretable space), we replace $z_{a_{i_K}}$ which is the first $k$ dimensions of $\tilde{z}_{a_i}$ with $K'_b$; such that $z_{a_{i_K}} \rightarrow K'_b$, where $K'_b = \frac{1}{N}\sum_{i=0}^{N} z_{b_{i_K}}$ is the average of the first $k$ dimensions of the complementary demographic group. This process is depicted in Fig. 6.2. We use a majority voting scheme to then select the most frequent equivalent generated for each word. An output example of this process obtained using a {*"she"*, *"he"*} prompt is shown in Fig. 6.3. We then extend this list using the names intervention approach of Hall-Maudsley et al. [111] to generate counterfactuals for names.

### 6.3.3 Error correction

With the word pairs generated from the previous phase, we use the word substitution approach of Hall-Maudsley et al. to build a base corpus. To transform this base corpus

| | | | | |
|---|---|---|---|---|
| actress -- actor | hers -- his | sisters -- brothers | earl -- countess | king -- queen |
| alice -- edward | herself -- himself | soprano -- tenor | edward -- alice | kings -- queens |
| aunt -- uncle | jane -- john | sorority -- fraternity | emperor -- empress | lord -- lady |
| barbara -- david | jess -- matt | teresa -- luis | emperors -- empress | luis -- teresa |
| baroness -- baron | ladies -- gentleman | virginia -- william | father -- mother | male -- female |
| beautiful -- handsome | lady -- lord | widow -- man | fathers -- mothers | males -- females |
| countess -- count | mary -- john | wife -- husband | fraternity -- sorority | man -- woman |
| daughter -- son | miss -- mr | woman -- man | gentleman -- ladies | masculine -- feminine |
| daughters -- sons | mom -- dad | women -- man | grandfather -- grandmother | matt -- jess |
| elizabeth -- john | mother -- father | actor -- actress | grandson -- granddaughter | maximilian -- mary |
| empress -- emperor | mothers -- fathers | ap -- her | guy -- girl | michael -- sarah |
| female -- male | mrs -- mr | baron -- lady | handsome -- beautiful | mr -- mrs |
| females -- males | ms -- mr | boy -- girl | he -- she | nephew -- niece |
| feminine -- masculine | niece -- nephew | boyfriend -- girlfriend | heir -- heiress | peter -- mary |
| girl -- boy | princess -- king | boys -- girls | henry -- elizabeth | richard -- elizabeth |
| girlfriend -- boyfriend | queen -- king | brother -- sister | him -- her | robert -- mary |
| girls -- boys | queens -- kings | brothers -- sisters | himself -- herself | sir -- lady |
| granddaughter -- grandson | sarah -- michael | christopher -- elizabeth | his -- her | son -- daughter |
| grandmother -- grandfather | she -- he | count -- countess | human -- female | sons -- daughters |
| heiress -- heir | sister -- brother | dad -- mom | husband -- wife | tenor -- soprano |
| her -- his | | david -- barbara | jesus -- mary | thomas -- elizabeth |
| | | | john -- jane | uncle -- aunt |
| | | | | william -- virginia |

Figure 6.3: An autmatically compiled dictionary using the input prompt {*"she"*, *"he"*}. Words are discovered using the attribute classifier, and the counterfactuals are generated using the disentangling invertible interpretation network.

into fluent and realistic text labels for our parallel training data, we proposed an error correction scheme which we describe below in two steps.

#### 6.3.3.1 Erratic token detection.

The idea here is to detect and mask tokens that have a low probability of appearing in the context of a given text; following $t_i = t_{<mask>}$ if $P(t_i|T \setminus t_i) < \theta$, where $T$ is the sequence of tokens, $t_i$ is the $i$th token in $T$, and $\theta$ is a predefined threshold value. We define the resulting masked text as $T_{\Pi}$. This is achieved using a pretrained Electra model [27]. Electra is an LM pretrained using a text corruption scheme – text instances are corrupted by randomly replacing a number of tokens with plausible alternatives from BERT. Electra is then trained to predict which tokens are real and fictitious.

Since the use of wordpiece tokenization causes issues (as a word can be broken down into multiple subtokens) if a subtoken is selected for masking, we replace the entire sequence of associated subtokens with a <mask> token. For instance, *"The men are duchesses"*, in a wordpiece tokenization could be decomposed to *["The", "men", "are", "duchess", "##es"]*, Consequently, when *"duchess"* is identified as an erratic token, the masking scheme replaces the entire subsequence *["duchess", "##es"]*, thereby, generating *"The men are* <mask>*"*.

#### 6.3.3.2 Text insertion with BART.

Having obtained our masked intermediary texts, we generate plausible token replacements for each masked token. Since a <mask> token could correspond to multiple sub-

word tokens, the replacement generator should be capable of generating multiple tokens for a single <mask> instance, making it suitable to use a generative model – pretrained BART [98] – to predict these replacement tokens. Because Masked Language Modeling is one of BART's pretraining objectives, we can utilize it in its pretrained form without the need for finetuning. Given $T_{\Pi}$ from the previous step, the BART model tries to predict the correct infilling $x$ using the context of $T_{\Pi}$.

### 6.3.4 Training the generative model

The final stage of the approach is to fine-tune a BART model using the parallel data obtained from the previous steps. The BART generator takes the original text as input and is trained to autoregressively generate the counterfactual of the source text using the corresponding parallel counterfactual texts as labels in a teacher-forcing manner [175]. We formulate this as:

$$\mathcal{L}_{generator} = -\sum_{t=1}^{k} logP(y_t|Y_{<t}, X) \tag{6.3}$$

Where $X$ and $Y$ are the source and target texts, respectively, $y_t \in Y$ is the $t^{th}$ token in the target text, and $Y_{<t}$ refers to all tokens in $Y$ preceding $y_t$.

## 6.4 Experimental set-up

This section describes key implementation details of our work and the evaluation framework. We specifically evaluate gender bias in the binary sense within the English language domain.

### 6.4.1 Training set-up

The main corpus for training the attribute classifier and the disentangling invertible flow model comprises Wikipedia articles via Wikimedia dumps[1].

### 6.4.2 Evaluation datasets

For the appraisal of our model, we used the datasets discussed below. These datasets, upon which various CDA interventions were applied, were used to train a classification model on a downstream task. These datasets were only used for evaluation purposes and were not included in training Fairflow.

1. **Bias-in-bios**: This dataset provided by De-Arteaga et al. [32] contains Wikipedia profiles of professionals. The dataset originally contained labels corresponding to 28 distinct professions alongside the gender labels of the profiled individuals. We

[1]https://dumps.wikimedia.org

reclassified the professions into binary labels, aligning them with male-dominated and female-dominated occupations according to gender distribution. This categorization was done for two reasons. The first was to simplify the classification task from multiclass to binary. Secondly, this enabled us to easily induce bias by creating an imbalance between gender and class labels.

2. **ECHR**: The ECHR dataset by Chalkidis et al. [2] [21] contains case facts from the European Court of Human Rights (ECHR) on human rights breaches by European states. It further contains information on the gender of the applicant, human rights articles that were violated, and the defendant state (Central-Eastern European states *v.* all other states). The primary classification task here was to predict the defendant's state based on the case facts.

3. **Jigsaw**: This dataset[3] contains public comments from the now defunct online platform Civil Comments. The primary classification task for this dataset was toxicity detection.

For all the evaluation datasets, we maintained a balanced gender and class label distribution in the test sets as shown in Table 6.1. The training sets for the Bias-in-bios and the Jigsaw datasets were sampled with an imbalance to induce bias following the observations of Dixon et al. [42]. The training set for ECHR was left relatively balanced with the additional purpose of providing a baseline.

| | | Train | | | Test | | |
|---|---|---|---|---|---|---|---|
| **Dataset** | **Task** | **Number** (K) | **Positive class** % | **Females in Pos.** % | **Number** (K) | **Positive class** % | **Females in Pos.** % |
| *Bias-in-bios* | Career | 18 | 50 | 12 | 4 | 50 | 50 |
| *ECHR* | State | 7 | 18 | 41 | 1 | 50 | 50 |
| *Jigsaw* | Toxicity | 5 | 47 | 77 | 1 | 50 | 50 |

Table 6.1: Evaluation dataset statistics: The test sets are balanced with regard to gender and labels.

### 6.4.3 Comparative techniques

We implemented two variants of FairFLow: *FairFLowV1* and *FairFLowV2*, and compared them to three CDA setups. 1) *original* is the unaugmented original text; 2) *Hall-M* uses the direct word-substituion approach proposed by Hall-Maudsley et al [111]; 3) *Hall-M + BART* is a BART model fine-tuned with counterfactuals generated by Hall-Maudsley et al.; 4) *FairFlowv1* is a BART model fine-tuned with our error correction scheme applied to counterfactuals from Hall-Maudsley et al.; it follows the same approach of FairLow in Fig. 6.1 but with a manually compiled dictionary. 5) *FairFlowv2* is a BART-model fine-tuned with our full approach in Fig. 6.1. We take *Hall-M* and *Hall-M + BART* as our baseline approaches. We excluded *perturber* by Qian et al. [134] from our evaluation since the objective of their approach significantly differs from ours; as elaborated in Section 6.2.

[2]https://huggingface.co/datasets/coastalcph/fairlex
[3]https://www.kaggle.com/c/jigsaw-unintended-bias-in-toxicity-classification

| dem. axis | original | Hall-M | Hall-M + Bart | ChatGPT | *Meta-llama* | *FairFlowV2* |
|---|---|---|---|---|---|---|
| gender ***(she_he)*** | In 2011, **she** won two prestigious competitions: **Miss** Ukraine-Earth and **Miss** Earth. In addition, **Christina** entered the Top-10 most **beautiful girls** of the world. **Kristina** is from Zaporizhzhya. | In 2011, **he** won two prestigious competitions: **Miss** Ukraine-Earth and **Miss** Earth. In addition, **Joe** entered the Top-10 most **beautiful boys** of the world. **Gilbert** is from Zaporizhzhya. | In 2011, **he** won two prestigious competitions: **Miss** Ukraine-Earth and **Miss** Earth. In addition, **Joe** entered the Top-10 most **beautiful boys** of the world. **Alberto** is from Zaporizhzhya. | In 2011, **he** won two prestigious competitions: **Mr**. Ukraine-Earth and **Mr**. Earth. In addition, **Christian** entered the Top-10 most **handsome guys** of the world. **Christian** is from Zaporizhzhya. | In 2011, **he** won two prestigious competitions: **Mister** Ukraine-*Hero* and **Mister** *Hero*. In addition, **Christopher** entered the Top-10 most **handsome** men of the world. **Christopher** is from Zaporizhzhya. | In 2011, **he** won two prestigious competitions: **Mr** Ukraine-Earth and **Mr** Earth. In addition, **he** entered the top-10 most **handsome boys** of the world. **Irving** is from Zaporizhzhya. |
| religion ***(catholic _muslim)*** | In 1579 he converted from Orthodoxy to **Roman Catholicism** | *unavailable* | *unavailable* | In 1579 he converted from Orthodoxy to **Islam**. | In 1579 he converted from *Islam* to **Islam**. | In 1579 he converted from Orthodoxy to **Sunni Islam**. |

Figure 6.4: Text samples from Bias-in-bios and Wikipedia demonstrate that *FairFlow* and *ChatGPT-4* generate more robust counterfactual texts. Compared to *ChatGPT-4*, *Meta-llama-3-8B-Instruct* generates more inaccurate counterfactuals.

## 6.5 Evaluation and results

We quantitatively evaluated our approach using three main criteria: *utility*, *extrinsic bias mitigation*, and *task performance*.

### 6.5.1 Utility

By utility, we refer to how realistic and effective the generated counterfactuals are by computing their fluency (perplexity) and gender transfer accuracy.

#### 6.5.1.1 Grammatical correctness and fluency

We used a referenceless fluency metric due to the relative unavailability of parallel data. As we noted earlier, the parallel data used by Qian et al. only contains counterfactuals for only specific user-defined entities and is thus not suitable for evaluating our work. Similar to Wu et al. [180], we score fluency by computing the perplexity of the generated text using pretrained GPT-2 [138]. A low perplexity implies that a given text conforms well to the probabilistic distribution of natural text as learned by the pretrained language model.

Based on our earlier assertion about how out-of-context substitutions impair fluency, our error correction approach should expectedly increase fluency (reduce perplexity). We confirm this in Table 6.2 as we see that fluency is improved in both *FairFlowV1* and *FairFlowV2*.

| **Approach** | *PPL* ↓ | | | *Transfer Accuracy* ↑ | | |
|---|---|---|---|---|---|---|
| | **Bios** | **Jigsaw** | **ECHR** | **Bios** | **Jigsaw** | **ECHR** |
| *Original** | 41.023 | 69.67 | 32.88 | 0.04 | 15.96 | 36.14 |
| *Hall-M* | 43.51 | 76.37 | 33.70 | 98.60 | 79.00 | 75.10 |
| *Hall-M + BART* | 47.59 | 83.76 | 39.93 | 98.70 | 78.50 | 71.10 |
| *GPT-4o-mini* | 40.99 | 67.76 | **23.28** | **99.07** | **81.96** | 75.788 |
| *FairFlowV1* | 42.77 | 65.80 | 33.70 | 98.91 | 77.99 | 74.69 |
| *FairFlowV2* | **39.86** | **63.99** | 33.33 | 98.51 | 70.736 | **76.51** |

Table 6.2: PPL (*left*) of generated text using various CDA techniques. Lower scores indicate better fluency. Gender transfer accuracy (*right*) of the various CDA interventions. This indicates the percentage of counterfactual instances that were correctly resolved to new gender styles. *The original samples have very low accuracies because original gender is preserved*.

#### 6.5.1.2 Transfer accuracy

Here, similar to Tokpo et al. [163], we computed the percentage of texts that were converted from the source attribute to the target attribute, i.e., female to male or vice versa. We fine-tuned a BERT model to predict the gender of the text. We quantified gender transfer accuracy as $1 - probability_of_original_attribute$. We expect the original text to have a very low transfer accuracy, as its attributes would remain the same. As shown in Table 6.2, FairFlowV2 especially shows strong fluency scores whilst maintaining a good transfer accuracy. This shows that automating the dictionary generation process does not materially impair transfer accuracy.

### 6.5.2 Extrinsic bias mitigation

We trained a BERT classifier using the downstream classification tasks corresponding to the respective datasets and computed the *true positive rate difference (TPRD)* and *false positive rate difference (FPRD)* between two gender groups as in the case of De-Arteaga et al. [32]. $TPRD = P(\hat{y} = 1|y = 1, A = a) - P(\hat{y} = 1|y = 1, A = a')$ and $FPRD = P(\hat{y} = 1|y = 0, A = a) - P(\hat{y} = 1|y = 0, A = a')$. Where $y$ is the true label, $\hat{y}$ is the predicted label, and $A$ is the gender group variable.

We show in Table 6.3 consistently high TPRD scores for FairFlow1; this further buttresses the evidence that our approach to error correction works effectively and enhances bias mitigation whilst improving fluency. Similar to our findings for transfer accuracy, we find that automating dictionary compilation does not compromise bias mitigation much, as FaiFlowV2 maintains a good mitigating effect.

| Approach | *TPRD* ↓ | | | *FPRD* ↓ | | |
|---|---|---|---|---|---|---|
| | **Bios** | **Jigsaw** | **ECHR** | **Bios** | **Jigsaw** | **ECHR** |
| *Original** | 0.133 | 0.120 | 0.000 | 0.151 | 0.160 | 0.0 |
| *Hall-M* | 0.055 | 0.010 | 0.030 | 0.071 | 0.070 | 0.0 |
| *Hall-M + BART* | 0.051 | 0.025 | 0.010 | 0.074 | **0.060** | 0.0 |
| *GPT-4o-mini* | 0.051 | 0.015 | 0.030 | **0.061** | 0.065 | 0.0 |
| *FairFlowV1* | **0.044** | **0.005** | **0.000** | **0.065** | 0.065 | 0.0 |
| *FairFlowV2* | 0.057 | 0.040 | 0.010 | 0.070 | 0.080 | 0.0 |

Table 6.3: Extrinsic fairness: TPRD – True positive rate difference between male and female text instances. FPRD – False positive rate difference between male and female text instances.

### 6.5.3 Task performance

We carried out the task performance test to observe the extent to which bias mitigation impacts the task model's performance. Because we maintain a balanced distribution for our test sets, we expect the fairer models to have better performance. Specifically, we computed the accuracy and F1 scores for the default classification task of the respective datasets. In Table 6.4, FairFlow1 shows the most improved performance in general, particularly in accuracy. We again show from the strong performance of FairFlowV2, how effective an automatically generated dictionary could be.

| Approach | ACC ↑ | | | F1 ↑ | | |
|---|---|---|---|---|---|---|
| | **Bios** | **Jigsaw** | **ECHR** | **Bios** | **Jigsaw** | **ECHR** |
| *Original** | 91.20 | 88.50 | 97.60 | 47.92 | 48.95 | 52.86 |
| *Hall-M* | 92.53 | 90.25 | 97.83 | 48.38 | 48.62 | 52.98 |
| *Hall-M + BART* | 92.64 | 90.62 | 97.36 | **48.48** | 49.49 | 52.73 |
| *GPT-4o-mini* | 92.42 | 90.00 | 97.84 | 48.23 | 49.20 | 52.98 |
| *FairFlowV1* | **92.97** | **90.75** | 98.08 | 48.32 | **49.69** | 53.11 |
| *FairFlowV2* | 92.81 | 90.00 | **98.32** | 48.36 | 49.21 | **53.24** |

Table 6.4: Task performance: Accuracy and F1 scores of classification tasks. FairFLow1 shows better performance scores in general. FairFlow2 maintains a significant bias mitigating effect despite an automated dictionary approach.

### 6.5.4 Qualitative analysis and key observations

By analyzing samples from FairFlow, ChatGPT, and the comparative models, we find that FairFLow and ChatGPT have the most grammatically coherent counterfactuals. Additionally, we find that:

1. ***Automating the dictionary compilation process does not materially impair counterfactual generation.*** As shown in Fig. 6.4, even with a dictionary that was auto-

matically compiled, FirFlowV2 generates fluent and plausible counterfactuals. This is aided by the combination of the error correction scheme, which makes it more robust to grammatical errors and helps it generalize better.

2. ***A model fine-tuned on erroneous data mimics those errors.*** We observe that the error correction approach incorporated in FairFlow makes the model more robust, fluent, and grammatically coherent. The direct word replacement technique (*Hall-M*) is unable to replace out-of-dictionary words. The output of *Hall-M + BART* mirrors the same errors as *Hall-M*, showing that a generative model fine-tuned on erroneous data will mimic those errors.

3. ***ChatGPT generates good counterfactuals but has practical limitations***. We observe that, in general, ChatGPT generates good counterfactuals in zero-shot settings but is inefficient at generating counterfactuals on a large scale in low-resource environments. It is more costly to deploy in terms of access and computational demands. Secondly, ChatGPT shows inconsistencies in generating counterfactuals for names, as it tends to skip some names for which counterfactuals could have been generated. This is more so if the names refer to public figures, which occasionally leads to grammatical incoherent outputs. This can, however, be addressed by adapting the input prompts and improving instructions through few-shot examples that intuitively describe the setting. The manner in which ChatGPT handles names can also be advantageous because it may preserve the factuality of the text better, which may be a more desirable attribute in certain contexts. We also observed some irregular counterfactuals from *Meta-llama-3-8B-Instruct* in a zero-shot setting, as shown in Fig. 6.4. Some of the counterfactuals it generated impacted the original context of the text, which should have been retained.

## 6.6 Conclusion

In this chapter, we addressed ***RQ3*** by exploring the limitations of existing CDA approaches for mitigating bias and exploring the feasibility of developing techniques to address these challenges. We highlighted several issues with dictionary-based word-substitution counterfactual data augmentation (CDA) techniques. Specifically, we discussed how these techniques, which rely on word substitution based on manually compiled dictionaries, are prone to grammatical incoherence and struggle to generalize beyond the dictionary terms. We also discussed how the relative unavailability of parallel corpora for training limits model-based approaches. In response to these challenges, we propose the following: 1) an automated dictionary generation method that can extract and generate word-pairs from a corpus with minimal human intervention; 2) an error correction approach to produce fluent and grammatically coherent parallel text for training generative models for CDA; 3) the combination of these approaches to fine-tune a BART model for generating counterfactual texts, which we call *FairFlow*. We showed evidence that our error correction approach significantly improves the fluency and bias-mitigating effects of the fine-tuned model. We also demonstrated that automating the dictionary compilation process incurs little cost to CDA model performance, making it a viable solution when human intervention is limited.

## Ethics Statement

From an ethical perspective, the primary point to keep in mind regarding the use of counterfactual models is their impact on factuality. Since CDA approaches are designed to be *counterfactual*, they should be used cautiously in sensitive domains where factuality is essential. Secondly, CDA bias mitigation techniques like FairFlow do not automatically guarantee fairness; hence, they must be used with that understanding.

Chapter 7

# Text Style Transfer for Bias Mitigation using Masked Language Modeling

## 7.1 Introduction

Building on from Chapter 6, this chapter continues to investigate bias mitigation strategies that target training data. Here, we introduce a text-style transfer approach aimed at mitigating bias in textual training data. As previously discussed, the application of NLP has expanded into high-stakes domains, where mitigating bias and promoting fairness is crucial. Unlike CDA, which generates counterfactual instances to augment training data, the goal of the approach in this chapter is to produce more demographically neutral replacements of text instances in the training data with respect to a specific demographic. For instance, given the female-biased text, *"The event was kid-friendly for all the **mothers** working in the company"*, our task is to transform this text into a gender-neutral version like *"The event was kid-friendly for all the **parents** working in the company"*. However, we will demonstrate in Section 7.4 (gender obfuscation) that the proposed approach can also generate counterfactuals.

A more direct need for debiasing textual data has been highlighted by various researchers [59, 161, 72], who have identified troubling instances of bias in human-generated text, such as job advertisements, with negative implications. For example, in the context of job recruitment, Gaucher et al. [59] examined the impact of biased job advertisements on survey participants and found that altering the wording to favor a specific gender group significantly decreased the job's appeal to applicants from other gender groups, irrespective of the traditional gender stereotypes associated with the role.

Consequent to such findings, a few tools and models have been developed to detect and mitigate biases in job advertisements. Some of these tools include text editors like Textio, which has been successfully used by companies such as Atlassian to increase diversity in their workforce [31].

Another domain impacted by biased text is news media. Kiesel et al. [85] explore the issue of hyperpartisan news, where content is strongly influenced by extreme left- or right-wing ideologies. Moreover, with the growing prevalence of hate speech and microaggression on various social media platforms, concerns about fairness in these contexts have become increasingly prominent.

### 7.1.1 Style transfer

A machine learning technique that can be employed to debias text documents is style transfer. Style transfer is a technique that involves converting text or image instances from one domain to another, such that the content and meaning of the instance largely remain the same, but the style changes. Applying style transfer specifically as a text debiasing technique has not been significantly explored, although there has been research done on closely related applications like moderating hate-speech [149] in text, and gender obfuscation [143], where the objective is to disguise the gender of an author of a text as a means of privacy protection or for the prevention of inadvertent discrimination of the author.

A problem that has challenged research in text style transfer is the relative unavailability of parallel data that would ideally be required to train such models [141, 54, 152]. Training with parallel data makes it possible to directly map training instances from one domain to the other, hence, facilitating the learning process. Another challenge with text style transfer is the non-differentiability of discrete words, which causes optimization problems [183]. Due to the mentioned issues, most style transfer systems mainly employ training techniques that fall under two categories: keyword replacement and Auto-encoder sequence-to-sequence techniques. As we will discuss later, these two techniques have major challenges, such as significant loss of content information and impaired fluency. In the case of keyword replacement, biased words are deleted and replaced with alternative words, leading to complete loss of information from the deleted words [158], whereas, in the case of the Auto-encoder sequence-to-sequence generative approach, the input text is directly encoded by an encoder to get a latent representation of the text, which is subsequently decoded by a decoder. As a result, significant information is lost, and the decoder is also unable to properly reconstruct the text fluently.

In this chapter, we present a style transfer approach that improves the limitations mentioned above of both the Auto-encoder sequence-to-sequence technique and the Explicit Style Keyword Replacement technique while maintaining a good style transfer accuracy. This we achieve by combining key components of both techniques such that they complement each other; the fact that Explicit Style Keyword Replacement models retain sufficient content information by not compressing the whole text into a latent representation, whereas Auto-encoder sequence-to-sequence models also retain information by not deleting tokens. Our model adds latent content representation to the Keyword replacement process. To the best of our knowledge, we are the first to use this technique for text style transfer. Furthermore, our work introduces some key novelties. Firstly, we propose a new and more effective approach for identifying attribute tokens (words that contribute to bias in a text) by using a text classification explanation model. Secondly, we present a unique way of generating latent content representation using a dual objective to train a token embedder. The details of these techniques will be discussed in subsequent sections.

This chapter specifically addresses ***RQ4*** by ***exploring how NLP techniques can be utilized to mitigate bias in human-generated texts, such as job advertisements, with the goal of promoting fairness and diversity in application domains like job recruitment.*** We propose techniques to develop a robust model-based counterfactual generator in the absence of parallel training data. Our key contributions in this chapter include:

1. The development of an end-to-end text debiasing model that can convert a piece of biased text to a neutral version while maintaining significant content information. Our model is trained exclusively on nonparallel data; since parallel corpora are relatively hard to obtain, training with only nonparallel data is of great importance.

2. A novel way of improving content preservation and fluency in text style transfer by combining keyword replacement and latent content information. We will show that this significantly improves content preservation in style transfer whilst ensuring good style transfer accuracy. Some other key novelties in our work include our approach to generating latent content representation and our approach to identifying attribute tokens.

*This chapter is based on the work published as follows:* ***Ewoenam Kwaku Tokpo and Toon Calders. 2022. Text Style Transfer for Bias Mitigation using Masked Language Modeling.*** *In Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies: Student Research Workshop, pages 163–171, Hybrid: Seattle, Washington + Online. Association for Computational Linguistics.*

## 7.2 Related work

Style transfer has been widely and successfully explored in computer vision to convert images from one style to the other [58, 75, 78]; for instance, Gatsy et al. [58] shows how the style of a photograph can be converted to the style of an artwork by combining the content of the photograph and the style of the painting. However, directly applying image style transfer techniques for text is problematic because of the unique characteristics of both domains [74]. In NLP, style transfer has mostly been explored in areas such as sentiment analysis [100, 54, 185] and machine translation [92].

A few style transfer learning techniques use parallel data for training. Hu et al. [74] give an elaborate survey on such models. In this chapter, we will only focus on models that are trained on non-parallel data, some of which we will review in the following subsection.

### 7.2.1 Auto-encoder sequence-to-sequence models

Auto-encoder sequence-to-sequence models basically consist of an encoder that encodes the given text into a latent representation, which is then decoded by a decoder. Many of these models adopt an adversarial approach to learn to remove any style attribute from the latent representation. The resulting disentangled latent representation is decoded by the decoder in a sequential generative manner.

#### 7.2.1.1 Aligned auto-encoder model and Cross-aligned auto-encoder model:

Shen et al [152] propose two models for text style transfer based on the Auto-encoder sequence-to-sequence technique; an aligned auto-encoder model and a variant of that

called the cross-aligned auto-encoder model. The aligned auto-encoder model uses an adversarial approach to teach the encoder to generate a disentangled latent representation that can be decoded into the target text. The cross-aligned variant is based on the idea that the transferred source representation should exhibit the same distribution as the target representation and vice versa. Two discriminators are used in the adversarial training, one to distinguish a real target representation from a transferred source representation and the other to distinguish a real source representation from a transferred target representation.

#### 7.2.1.2 Back-translation style transfer model (BST):

Prabhumoye et al. [131] propose a style transfer model using back-translation. This is based on prior research that suggests that language translation retains the meaning of a text but not the stylistic features [136]. Hence, they encode a disentangled latent representation of the input text using back-translation. A bidirectional LSTM is then used to decode the disentangled latent representation into a text in the target style. They use adversarial learning to ensure that the generated text belongs to the target style. Because of the discrete and non-differentiable nature of tokens, they approximate discrete training by using a continuous approximation of softmax values to sample tokens.

An issue with Auto-encoder sequence-to-sequence models, in general, is the loss of information due to compression when encoding. Furthermore, Wu et al. [181] note that sequence-to-sequence models for style transfer often have limited abilities to produce high-quality hidden representation and are unable to generate long meaningful sentences. Nonetheless, sequence-to-sequence generative models can prove more effective in applications where the text needs to be considerably rephrased (eg. informal style to a formal style).

### 7.2.2 Explicit Style Keyword Replacement

These methods follow the general approach of identifying attribute markers, deleting these markers, and predicting appropriate replacements for these markers which conform to the target style.

#### 7.2.2.1 DeleteOnly model and Delete&Retrieve model:

Li et al. [100] propose the DeleteOnly and the Delete&Retrieve, which use a three-step Delete, Retrieve, and Generate approach. In the Delete phase, attribute tokens are identified and removed from the input text based on their relative frequencies in the respective style domains. In the Retrieve stage, possible replacement words are predicted based on similar sentences in the target style. For the Generate stage, both methods use an RNN to encode context tokens (other words aside attribute tokens in the input text) and decode them to the target style. In the DeleteOnly method, the RNN takes as inputs only the context tokens and the target style embedding to generate the target sentence, whereas the Delete&Retrieve method takes as inputs the context tokens, the target style

embedding, and the replacement words from the Retrieve stage to construct a target sentence.

#### 7.2.2.2 B-GST and G-GST:

Sudhakar et al. [158] introduce Blind Generative Style Transformer (B-GST) and Guided Generative Style Transformer (G-GST) as improvements on DeleteOnly and the Delete&Retrieve from Li et al. [100]. They use attention weights from a transformer-based style classifier to identify attribute words. The Retrieve phase generally follows the same approach as [100]. For the generate phase, they implement their version of DeleteOnly and the Delete&Retrieve with GPT [137], and refer to them as B-GST and G-GST respectively.

Since Explicit Style Keyword Replacement methods only delete a small portion of the input text, they preserve much more information. These systems, on the other hand, are unable to properly capture information on the deleted tokens [158], leading to examples such as *"The event was kid-friendly for all the **mothers** working in the company"* → *"The event was kid-friendly for all the **children** working in the company"*.

## 7.3 Methodology

The goal of our model is to transform any piece of biased text into a neutral version. To put this formally, we take the two style attributes $s_a$ and $s_b$ to represent neutral style and biased style, respectively. Given a text corpus $X_b = \{x_{b_1}, ..., x_{b_N}\}$ that belongs to $s_b$, our goal is to convert a given text $x_b$ to $x_a$, such that $x_a$ belongs to style $s_a$ but has the same semantic content as $x_b$.

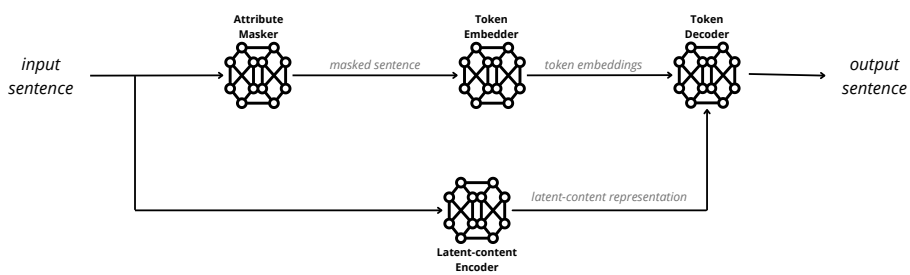



Figure 7.1: The architecture of our proposed model. The model consists of four main components. The arrows show the flow of information within the model and how the various components interact with each other.

Our model is basically composed of four main components, as illustrated in Figure 7.1. Before delving into the details, we will summarize the functions of the various components:

1. **Attribute Masker:** The Attribute Masker identifies the attribute words responsible for the bias in a text and masks these words with a special *[MASK]* symbol. The resultant text is fed as input to the Token Embedder.

2. **Token Embedder:** The Token Embedder is responsible for generating token embeddings for the masked tokens. These embeddings should contain enough semantic information from the context tokens.

3. **Latent-content Encoder:** The Latent-content Encoder takes the original (unmasked) text as input and encodes it into a latent content representation. An important part of this stage is our approach to disentangle the resulting latent content representation from the biased style.

4. **Token Decoder:** The Token Decoder computes the average of each token embedding and the latent content representation to generate new token embeddings. The Token Decoder uses these embeddings to predict the correct tokens.

We illustrate the process described above with an example in Figure 7.2.

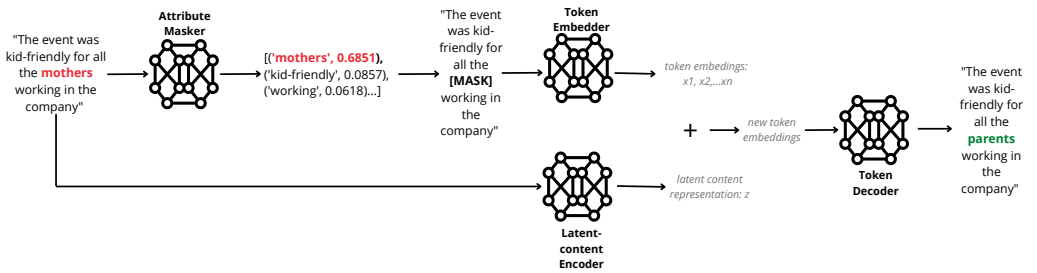



Figure 7.2: An example to illustrate the end-to-end bias mitigation process. This demonstrates the operation of each component of the model. The Latent-content Encoder is debiased to remove traces of bias before being added to the token embeddings.

### 7.3.1 Attribute Masker

We use LIME [145], a model agnostic explainer that can be used on textual data, to identify attribute tokens. We found LIME to produce good and intuitive explanations, making it an ideal tool for identifying attribute tokens. Some Explicit Style Keyword Replacement models use relatively simple techniques to identify attribute words. Li et al. [100] use the relative frequency of words in the source style. Others like Sudhakar et al. [158] employ more advanced methods like using attention weights. However, using techniques like attention weights to identify attribute tokens has proven not very effective [76].

To use LIME to detect attribute words, we first need to train a text classifier $f$ that predicts whether a given text is biased. We fine-tune BERT [38], a pretrained language model, as a text classifier by training it on a labeled corpus containing both biased and neutral texts. Lime linearly approximates the local decision boundary of $f$ and assigns weights to tokens based on their influence on the classification outcome. With these weights (scores), we set a threshold value $\mu$ to select words to be masked. These words are replaced by a special *[MASK]* token.

### 7.3.2 Token Embedder

Given the text (with masked attribute tokens) $x'_b$, the Token Embedder learns to generate a vector representation $w_i \in \mathbb{R}^d$ for each token $t'_i \in x'_b$, such that $w_i$ is able to capture semantic information of $t'_i$. The Token Embedder outputs a set of all token embeddings $W = \{w_1, ..., w_n\} \in \mathbb{R}^{n \times d}$. Following the convention used by Devlin et al. [38], we take the size of every embedding to be $d = 768$ throughout this chapter.

#### 7.3.2.1 Training the Token Embedder

We fine-tune a BERT model to generate token embedding using the Masked Language Modeling objective by randomly masking about twenty percent of the words in the training corpus. The goal is to predict the masked words based on the context tokens (unmasked words). A classification layer is appended to the BERT embedder such that the BERT embedder learns to generate embeddings that can be correctly decoded by the classification layer; this same classification layer will later be used as the Token Decoder. This process is similar to the original Masked Language Modeling training objective used to pretrain BERT.

Formally, given the input text $x'_b$, the set of masked tokens in the text $T_\Pi$, and the set of context tokens in the text $T_{-\Pi}$, the goal is to predict each masked token based on the unmasked tokens. The MLM objective is to minimize:

$$\mathcal{L}_{mlm} = -\sum_{i=1; t_{\pi_i} \in T_\Pi}^{k} logP(t_{\pi_i}|T_{-\Pi}) \tag{7.1}$$

### 7.3.3 Latent-content Encoder

The Latent-content Encoder is responsible for generating a latent content representation of the input sentence. The Latent-content Encoder takes as input the original text (unmasked) $x_b$ and generates a target latent representation $\hat{z}$.

After taking $x_b$ as input, the Latent-content Encoder first generates token embeddings $v_i \in \mathbb{R}^d$ for each token $t_i \in x_b$. The set of token embeddings $V = \{v_1, ..., v_n\} \in \mathbb{R}^{n \times d}$ is mean-pooled to generate $\hat{z} \in \mathbb{R}^d$. Since we want $\hat{z}$ to have the same content as $x_b$ but not the bias that exists in $x_b$, we use a dual objective training to debias $\hat{z}$.

#### 7.3.3.1 Training the Latent-content Encoder

This unique training approach helps to effectively learn a latent content encoding that captures significant content information while being disentangled from the biased style. As shown in Figure 7.3, a pretrained Source Content Encoder and a pretrained Bias Detector are used to train the Latent-content Encoder. The Source Content Encoder is pretrained to generate a source encoding $z$ that contains both the content and biased style of $x_b$. The Bias Detector is pretrained to predict the style attribute ($s_b$ or $s_a$).

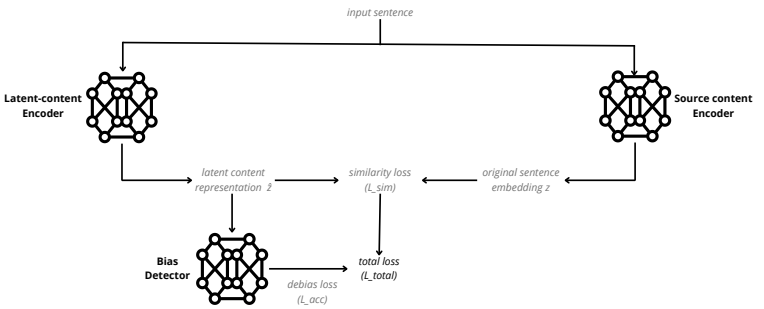



Figure 7.3: Illustration of the dual objective used to train the Latent-content Encoder. The Source Content Encoder and the bias classifier are both pretrained and frozen during training

Both the Latent-content Encoder and the Source Content Encoder take $x_b$ as input. The Latent-content Encoder generates output $\hat{z}$ whereas the Source Content Encoder generates $z$. Since the goal is to make $\hat{z}$ and $z$ have the same content, we want them to be as similar as possible. We use cosine-similarity as a similarity measure to compute this similarity. The similarity loss is minimized using mean-squared error as defined below:

$$\mathcal{L}_{sim} = \frac{1}{N}\sum_{j=1}^{N}(cosine_similarity(\hat{z}_j, z_j) - 1)^2 \tag{7.2}$$

The Bias Detector is a classifier used to detect whether bias exists in $\hat{z}$. The Bias Detector takes $\hat{z}$ as input and returns the class probabilities of $\hat{z}$. Because we want $\hat{z}$ to belong to the neutral class, the Latent-content Encoder has to learn to generate $\hat{z}$ that is always classified as neutral. This is achieved by minimizing the cross-entropy loss:

$$\mathcal{L}_{acc_{\hat{z}_j}} = -\sum_{j=1}^{N} logP(s_a|\hat{z}_j) \tag{7.3}$$

$P(s_a|\hat{z}_j)$ is the classifier's prediction of the probability of $\hat{z}$ being neutral.

Combining both losses, we get the dual objective:

$$LCE_loss = (1-\lambda)\mathcal{L}_{similarity} + \lambda\mathcal{L}_{acc_{\hat{z}_j}} \tag{7.4}$$

We introduce the balancing parameter $\lambda$ that controls the degree of influence of each objective. The Latent-content Encoder, the Source Content Encoder, and the Bias Detector are BERT models fine-tuned for these purposes.

### 7.3.4 Token Decoder

The Token Decoder first adds latent content information to word embeddings. To do this, the Token Decoder takes as inputs both $W$ from the Token Embedder and $\hat{z}$ from the

Latent-content Encoder. For each $w_i \in W$, a new token embedding $\hat{w}_i \in \mathbb{R}^d$ is generated by computing the weighted average of $w_i \in \mathbb{R}^d$ and $\hat{z} \in \mathbb{R}^d$. After generating $\hat{w}_i$, the Token Decoder uses it to predict the right token by computing the probability distribution over all the tokens in the vocabulary.

#### 7.3.4.1 Training the Token Decoder

The Token Decoder is a simple one-layer feed-forward neural network. It is only trained on a corpus of neutral texts since we want the decoder to always predict a neutral token. During training, $z$ is used in place of $\hat{z}$ because we want the decoder to predict the exact tokens in the input text. For each masked token's embedding $\hat{w}_{i_\Pi}$, the goal is to predict the original token $t_{\pi_i} \in T_\Pi$ where $T_\Pi$ is the set of all masked tokens. We compute the decoding loss as follows:

$$\mathcal{L}_{dec} = -\sum_{i=1;t_{\pi_i}\in T_\Pi}^{n} logP(t_{\pi_i}|\hat{w}_{i_\Pi}) \tag{7.5}$$

To further augment this, we use a pretrained classifier to ensure that the output sentence $x_a$ is always neutral. A dual objective is again used in this process:

$$\mathcal{L}_{acc_{x_a}} = -logP(s_a|x_a) \tag{7.6}$$

$$TD_loss = (1-\gamma)\mathcal{L}_{dec} + \gamma\mathcal{L}_{acc_{x_a}} \tag{7.7}$$

Where $\mathcal{L}_{acc_{x_a}}$ is the loss from the classifier, and $\gamma$ is a balancing parameter. Because $x_a$ is made up of discrete tokens (one-hot encodings) which are non-differentiable during back-propagation, we use a soft sampling approach as was done in [181, 131]:

$$t_{\pi_i} \sim softmax(\mathbf{o}_t/\tau)$$

where $\mathbf{o}_t$ is the logits for $t_{\pi_i}$ from the classifier, and $\tau$ is a temperature value that controls the smoothness of the probabilities. This continuous representation approximates the one-hot vector that is ordinarily used to represent $t_{\pi_i}$.

## 7.4 Experiments

For our experiments, we focus on gender bias. The use of gender is motivated by the relative availability of resources such as datasets in this domain from previous works. To further probe the performance of our model, we experiment on gender obfuscation, where instead of mitigating the bias, we try to convert female-authored texts to "look like" male-authored texts.

We discuss the details of our experiments in the following subsections. All experiments are run on a Tesla V100-SXM3 GPU with 32Gb memory.

| Dataset | Attributes | Classifier | Train | Dev | Test |
|---|---|---|---|---|---|
| *Jigsaw-g* | Sexist | 24K | 32K | 1K | 1K |
| | Neutral | 24K | 92K | 3K | 3K |
| *Yelp-s* | Male | 100K | 100K | 1K | 1K |
| | Female | 100K | 100K | 1K | 1K |

Table 7.1: Dataset statistics

### 7.4.1 Dataset

We run our experiments on two datasets discussed below. Some statistics of the datasets are given in Table 7.1.

#### 7.4.1.1 Jigsaw dataset:

The Jigsaw datasets[1] consists of comments that are labeled by humans with regard to bias towards or against particular demographics.

Using the value 0.5 as a threshold, we extract all texts with gender (male or female) label $\geq 0.5$ as the gender-biased class of texts and extract a complementary set with gender labels $< 0.5$ as the neutral class. We broke down all texts of multiple sentences into single sentence texts.

#### 7.4.1.2 Yelp dataset:

We extract this dataset from the preprocessed Yelp dataset used by [131, 143]. This dataset contains short single sentences, which we use for author gender obfuscation.

### 7.4.2 Evaluation models and metrics

To evaluate the performance of our model, we compare it to six other models: Delete-only, Delete-and-retrieve [100], B-GST, G-GST [158], CAE [152] and BST [131], all of which are discussed in the section under related works. We chose these models primarily because of their wide use in style-transfer literature and their reproducibility.

The evaluation is based on three automated evaluation metrics for style transfer discussed by Hu et al. [74]; *style transfer accuracy* (Transfer strength), *content preservation*, and *fluency*.

- **Style transfer accuracy:** This gives the percentage of texts that were successfully flipped from the source style (bias style) to the target style (neutral style) by our model. To predict whether a text was successfully flipped, We use a trained BERT classifier with 85% accuracy and 80% accuracy for the Jigsaw and the Yelp datasets,

[1] https://www.kaggle.com/c/Jigsaw-unintended-bias-in-toxicity-classification/data

respectively. To ensure fair evaluation, these classifiers are different from the ones used to train the respective models.

- **Content preservation:** We measure content preservation by computing the similarity between the generated text and the original text. Similar to Fu et al. [54], we use the cosine similarity between the original text embedding and the transferred text embedding to measure content preservation. To make this more effective, we generate text embeddings with SBERT [144], a modified version of pretrained BERT that generates semantically meaningful sentence embeddings for sentences so that similar sentences have similar sentence embeddings that can be compared using cosine-similarity.

- **Fluency:** Similar to [157], we measure the fluency of the generated text using the *perplexity* produced by a Kneser–Ney smoothing 5-gram language model, KenLM [69] trained on the respective datasets.

### 7.4.3 Hyper-parameter settings

For the gender mitigation task, we set the hyper-parameters as follows: we use a sequence size of 40. We set the balancing parameter for the Latent-content Encoder at $\lambda = 0.5$, the balancing parameter for the decoder at $\gamma = 0.3$, and the LIME score threshold for attribute token removal at $\mu > 0.1$.

For the gender obfuscation task, we set the hyper-parameters as follows: we use a sequence size of 30. We set the balancing parameter for the Latent-content Encoder at $\lambda = 0.5$, the balancing parameter for the decoder at $\gamma = 0.7$, and the LIME score threshold for attribute token removal at $\mu > 0.1$.

For both tasks, we set the temperature value to $\tau = 1$. Other hyper-parameter values are set to the default parameter values used in BERT-base [38]. We also initialize the weight of the Latent-content Encoder with the weights of SBERT before training it. Both models were trained on 4 epochs. All the hyper-parameter were obtained after experimenting with some hyper-parameter values. We leave a comprehensive discussion of tuning these hyper-parameters to future works.

### 7.4.4 Results and discussion

In Table 7.3, we evaluate our model against the six other models mentioned earlier for gender obfuscation, and in Table 7.2, we do the same for five of the models (excluding BST) for bias mitigation. For style transfer accuracy (A.C.) and content preservation (C.P.), higher values are better. For PPL, lower values are better. Generally, all models produced better style transfer accuracy scores for the bias mitigation task compared to the gender obfuscation task. This is expected because the task of author gender obfuscation appears to be more difficult and indistinct in the sense that we are assuming that we can predict the gender of a person based on how a person writes. Hence, changing the style is much more delicate. On the other hand, for the bias mitigation task, the distinction between a gender-biased text and a neutral text is relatively obvious and much clearer; attribute

| Approach | C.P.% ↑ | PPL ↓ | AC% ↑ |
|---|---|---|---|
| *Original** | 100.00 | 12.51 | 0.08 |
| *Del* | 97.47 | 363.64 | **92.30** |
| *Del&ret* | 97.50 | 242.33 | 71.70 |
| *B-GST* | 96.73 | 1166.4 | 10.10 |
| *G-GST* | 99.11 | 621.50 | 38.80 |
| *CAE* | 95.60 | 795.58 | 83.70 |
| *Our model* | **99.71** | **76.75** | 88.10 |

Table 7.2: **Jigsaw dataset-** Transfer strength and Content preservation scores for the models. **C.P.**: Content preservation, **PPL**: Fluency (Perplexity), **Accuracy**: Style transfer accuracy, **Original*:** refers to the original input text. For A.C., C.P.and Agg, higher values are better. For PPL, lower values are better

| Approach | C.P.% ↑ | PPL ↓ | AC% ↑ |
|---|---|---|---|
| *Original** | 100.00 | 11.39 | 17.80 |
| *Del* | 98.70 | **41.03** | 33.79 |
| *Del&ret* | 98.25 | 57.73 | 30.90 |
| *B-GST* | 95.94 | 141.81 | 23.90 |
| *G-GST* | 97.28 | 70.24 | 21.00 |
| *CAE* | 98.48 | 43.78 | 32.09 |
| *BST* | 95.49 | 63.33 | **68.80** |
| *Our model* | **99.05** | 45.17 | 43.20 |

Table 7.3: **Yelp dataset-** Transfer strength and Content preservation scores for the models. **C.P.**: Content preservation, **PPL**: Fluency (Perplexity), **Accuracy**: Style transfer accuracy, **Original*:** refers to the original input text. . For A.C., C.P.and Agg, higher values are better. For PPL, lower values are better

words are more distinct and identifiable, as such, more explicit cues can be learned by the model to resolve the text.

#### 7.4.4.1 Gender bias mitigation

From Table 7.2, as we expected from the compared models, the models that perform considerably well in one metric suffer significantly in other metrics. For instance, Delete-Only (Del) produces the best transfer accuracy but lags behind other models in content preservation and fluency. For content preservation and fluency, our model produces improved results over all the other models. This result is consistent with our expectation of improving content preservation with our techniques. Again, the accuracy score (second highest) produced by our model confirms the claim that our model preserves content information without a significant drop in transfer accuracy.

| | C.P.% ↑ | PPL ↓ | ACC%↑ |
|---|---|---|---|
| *Our model* | **99.05** | 45.17 | 43.20 |
| *Without-LR* | 96.62 | 45.72 | **84.20** |
| *Without-LR&SS* | 96.89 | **41.84** | 41.00 |

Table 7.4: Ablation study of our model on the Yelp dataset. **Without-LR**: model with soft sampling (class constraint) but no latent content representation, **Without-LR&SS**: model with no class constraint and no latent content representation

#### 7.4.4.2 Gender obfuscation

From Table 7.3, the same observation is made: models that perform very well in one metric fall short in other metrics. BST produces the best style transfer accuracy while at the same time producing the worst content preservation score.

#### 7.4.4.3 General observations

From the results from both datasets, one key observation is that models that perform very well in one metric tend to fall short in other metrics. This goes to show the difficulty for style transfer models to preserve content information while maintaining a strong transfer accuracy. This observation is confirmed by previous works [100, 181, 74], which mention the general trade-off between style transfer accuracy and content preservation. Our model, however, shows good results in maintaining a good balance across all metrics. By adjusting the balancing parameters in the hyper-parameter section, priority can be given to which metric is more desired since it is the case that depending on the application domain, certain metrics might be prioritized.

#### 7.4.4.4 Ablation analysis

In Table 7.4, we perform an ablation analysis of our model. Here, we strip off various components from our model, and we evaluate the resulting model on the Yelp dataset. We generate two resulting variants; Without-LR and Without-LR&SS. In Without-LR, we remove latent content information from our model but maintain class constraint ($\mathcal{L}_{acc_{x_a}}$), whereas in Without-LR&SS, both class constraint and latent content information are omitted.

In Table 7.4, we can see how each component affects the performance of our model. We expect Without-LR to have the best style transfer accuracy but the worst content preservation score for one major reason: class constraint without latent content encoding will ensure that the generated text is mostly in the target style (neutral style), but the loss of content information from the latent encoding reduces content preservation. The exceptionally high transfer accuracy scores produced can be misleading without taking content preservation into account. A look at the output text explains this outcome; in most cases, Without-LR learns to replace female-attribute words with random words like "pistol". This changes the style in almost all cases but gives the text little meaning. This explains the low content preservation score as well. Since only a few attribute words are

| ***styles*** | ***original*** | ***Del*** | ***Del&ret*** | ***B-GST*** | ***G-GST*** | ***CAE*** | ***BST*** | ***Ours (MLM-ST)*** |
|---|---|---|---|---|---|---|---|---|
| Gender bias mitigation ***(gendered_neutral)*** | I hope the **man** learned his lesson to slow down and buckle up. | I hope the rosary post him with his house and asked the hope . | I hope the shovel friends him i bet and sing . | I hope the man learned from his lesson slow and up . | I hope the learned man learned his lesson slow down and up . | I believe that is the same of the <unk> | – | I hope the driver learned his lesson to slow down and buckle up. |
| Gender bias mitigation ***(gendered_neutral)*** | I married a wonderful mature , loyal and dedicated foreign **women** while working abroad... | i a host of christian , elite and followed during several will start hurt much new day . | i married a shepherd 12 , so insecure , drug and era 2 days drugs hillary . | i married a wonderfull mature , loyal and dedicated women while working abroad for many years ago . | i a wonderfull mature , loyal and dedicated foriegn while working abroad , many years ago . | i "i like a <unk> , <unk> , <unk> . | – | I married a wonderful mature , loyal and dedicated foreign person while working abroad... |
| Author gender obfuscation ***(female_male)*** | Overall, worth the **extra** money to **stay** here. | i don't like the way you're told at this ! | overall , worth the extra money to stay here . | overall , not worth the trip to vegas . | overall , worth the trip to vegas here . | overall , worth the money to spend stay here . | , , they ' s not be the place . | Overall, worth the damn money to eat here. |
| Author gender obfuscation ***(female_male)*** | I had prosecco and my **boyfriend** ordered a beer. | like zille builds completely mistaken it for it then anyone . | i had spaghetti and my boyfriend ordered a beer . | i had and my boyfriend got a beer . | i had wine and my boyfriend had a beer . | i had my <unk> and wife ordered a beer . | i had the pilot and food is not a good . | I had prosecco and my wife ordered a beer. |

Figure 7.4: Sample text outputs from our text style transfer experiments.

changed for a particular text instance, a content preservation score of 96.62% is significantly low, indicating a significant loss of content information from the replaced word. On the other hand, Without-LR&SS produces the best fluency score primarily because without the class constraint, the model prioritizes predicting a plausible replacement rather than a replacement that changes the style.

## 7.5 Conclusion

In this chapter, we addressed ***RQ4*** by exploring how NLP techniques can be utilized to mitigate bias in human-generated texts, such as job advertisements, with the goal of promoting fairness and diversity in application domains like job recruitment. We introduced a style transfer model designed to mitigate bias in textual data. We demonstrated that explicit keyword replacement can be effectively combined with latent content representation to improve content preservation in text style transfer models. We evaluated our model's efficacy in bias mitigation and other style transfer tasks, such as gender obfuscation, by measuring performance across three key metrics: content preservation, fluency, and style transfer accuracy. On both datasets, our approach achieved strong content preservation and fluency scores while maintaining high style transfer accuracy. The key innovations of our approach include the combination of explicit keyword replacement with latent content representation, the use of a text explainer to identify attribute tokens and our method for generating latent content representation.

# Chapter 8

# Conclusion

In the last few years, we have seen a revolution in the AI landscape, largely driven by unprecedented advancements in natural language processing (NLP). NLP technologies have extended beyond the realm of NLP and are now widely adopted in various machine learning domains. A notable example is the transformer architecture, which has become a crucial component in many machine learning systems, such as computer vision.

The increased integration of NLP in our daily lives highlights the progress in this field. Nonetheless, the great potential of LLMs has come with the great responsibility of ensuring the fair and ethical use of these technologies. Although various studies over the years have laid the foundation for the fair development and use of language models, pertinent issues remain in this area.

This thesis seeks to address some of these outstanding issues relating to the fairness of language-based technologies. This thesis aims to bring to light important technical issues regarding the quantification and mitigation of bias in natural language processing through various empirical and experimental studies. In addition to such expositions, we develop and propose techniques that address some of these issues.

## 8.1 Main contributions

In **Chapter 1**, we laid important foundations that set the tone for this thesis. We provided an overview of both NLP and bias in NLP.

In **Chapter 2**, we explored fundamental concepts and some of the latest advancements in language models and natural language processing (NLP). We examined the evolution of natural language models, delving into their mechanisms, architectures, and paradigms.

In **Chapter 3**, we explored some relevant concepts relating to fairness and bias in natural language models. We defined bias in NLP as per the scope of this thesis. This is crucial because, as noted by Blodget et al. [15], the concept of bias is often ill-defined in various NLP literature. We then examined the two main categories of bias in NLP (intrinsic and extrinsic) and how these biases are quantified or measured based on existing literature. We subsequently reviewed and examined some of the major bias mitigation techniques for both intrinsic and extrinsic biases.

In **Chapter 4**, we examined the relationship between intrinsic bias measures, the templates they use, and the representation schemes for seed words used to evaluate bias. We designed a series of correlation analyses to examine variations in results based on choices for each factor adopted by existing works. Our study revealed significant inconsistencies in how different choices detected or depicted bias in natural language models. We also identified inconsistencies in the quantification of bias by different intrinsic bias measures. These inconsistencies highlight the different underlying notions of bias in language models and how these notions may conflict. Based on these findings, we offered key recommendations on how bias should be measured and propose more reliable options for quantifying bias in natural language models.

In **Chapter 5**, we primarily examined the relationship between intrinsic and extrinsic bias, and how existing intrinsic bias mitigation techniques remove demographic information from word embeddings. This investigation was motivated by the widely held belief that intrinsic bias significantly influences extrinsic bias in NLP. Given previous findings that highlight inconsistencies in how existing intrinsic bias metrics measure bias, we developed a probe to further analyze demographic information in word embeddings. Using this probe, we tested whether existing mitigation techniques effectively mitigate the presence of demographic information in word embeddings. We then investigated how these intrinsic bias mitigation techniques impact downstream fairness (i.e., improve extrinsic bias). Our findings suggest that some existing intrinsic bias mitigation techniques can superficially conceal bias rather than resolve it. We also provided evidence showing that different conceptions of bias can be misaligned and incongruent. Finally, we demonstrated that, compared to extrinsic bias mitigation techniques, intrinsic bias mitigation has little material impact on downstream fairness. Based on our findings, we recommend placing greater emphasis on extrinsic bias measurement and mitigation if a language model is intended to be used for downstream NLP tasks.

In **Chapter 6**, we proposed and implemented a fully automated approach to counterfactual data augmentation (CDA) for bias mitigation. Although CDA has been identified as an effective strategy for reducing bias, various existing approaches have significant limitations. We first highlighted some limitations of dictionary-based approaches to CDA, including out-of-context and out-of-dictionary substitution. We then explored the prospect of a model-based approach but highlighted the unavailability of good-quality parallel data to train these models. We proposed a process to generate better-quality parallel data for training a counterfactual text generator. We proposed an automated dictionary compilation process and error-correction scheme that can be combined to automatically generate parallel data. Our results showed that our proposed approach, *FairFlow*, generates more fluent and effective counterfactuals for bias mitigation, particularly in addressing gender bias.

In **Chapter 7**, we focused on using NLP techniques to mitigate bias in human-generated text. Motivated by previous studies that found gendered language in job advertisements made them less attractive to diverse demographic groups, our goal was to develop automated techniques to address these human-induced biases. We proposed a style transfer approach that can convert text with gendered wording into gender-neutral forms. Existing style transfer techniques typically suffer from loss of content information and lack of fluency. To address these issues, our approach combines key attributes from the two main style transfer techniques: Explicit Keyword Replacement and Auto-encoder Sequence-to-Sequence techniques. We empirically compared our model to various existing models

for the English language and showed, in these experiments, significant improvements over existing models in content preservation and style transfer accuracy.

## 8.2 Outlook and future work

The general outlook over the last couple of years indicates a noticeable shift from domain-specific (expert) NLP models to general-purpose language applications. This evolution is largely driven by advancements in generative models, inspired by GPT and its newer variants, along with newer learning paradigms like reinforcement learning from human feedback. At the forefront of this trend are general-purpose conversational models that have demonstrated near human-level capabilities, and this is just the starting point.

It is becoming apparent that general-purpose models are ushering in a new era in AI, paving the way for more advanced AI systems. The growing popularity of general-purpose models has been motivated by evidence that the performance of LLMs scales proportionally with the number of parameters and the size of training data. The constant growth in the volume and variety of internet data, coupled with the continuous advancement of computational infrastructure, suggests an even greater shift toward the general-purpose domain. This trend is further supported by the growing popularity of multi-modal models.

While it is challenging to predict whether this shift will eliminate domain-specific models, there remains a compelling case for domain-specific systems, particularly due to resource considerations. Recently, significant attention has been drawn to the substantial resource demands of general-purpose LLMs, as highlighted by works such as Bender et al. [9]. This makes a case for smaller, resource-effective models. However, a focus on resource-efficient models would necessarily imply a focus on domain-constrained models, as the large number of parameters of LLMs are crucial for general-purpose capabilities.

### 8.2.1 A low-resource approach to fair NLP

Three factors that underscore the importance of NLP solutions focusing on low-resource domains are:

1. **The environmental and economic implications of large models:** The training and deployment of large language models has been reported to pose significant environmental and economic implications.
2. **The relative inaccessibility of large models in many settings (especially for offline use):** The infrastructure requirement needed to train and sometimes deploy LLMs makes them relatively inaccessible to institutions and organizations with limited resources.
3. **The relative lack of data and resources in some language domains:** Available language models and resources tend to be oriented toward English language and other high-resource languages, hence, many of these language models significantly underperform in low-resource language domains. Specialized language models

for such low-resource domains are still lacking, although this field is beginning to attract research attention.

Based on these considerations, we believe that developing low-resource approaches to fair language modeling and low-resource mitigation techniques for NLP is a direction in which to focus future work. Post-processing bias mitigation approaches are particularly useful in this regard, as they do not require retraining language models to mitigate bias.

### 8.2.2 Bias beyond gender and English

This thesis, like much of the existing research in fair NLP, focuses on binary concepts of gender bias, particularly within the English language. Although biases related to other demographic domains, such as race and religion, have been identified, these biases remain under-explored due to a lack of resources, including annotated datasets and evaluation metrics necessary for measuring and mitigating bias in these areas.

Our perspective is that, historically, societies were more homogeneous in other attributes (e.g., race and religion) than in sex and gender. As a result, languages evolved to more vividly depict gender roles and divisions as reflected in society [95, 1]. Consequently, gender—especially in its binary form—is more prominent in language, with more frequent gender-related instances from which language models tend to learn gendered attributes. However, as modern societies become increasingly diverse, stronger evidence of other forms of bias is emerging, alongside a pressing need to address them.

Also, available resources in NLP are mostly related to the English language. We acknowledge the need for further research in other languages, where solutions, techniques, and resources developed for English may not be directly applicable.

### 8.2.3 Other ethical considerations

Aside from bias, other ethical issues are growing in relevance. A notable example is the rise of synthetic data and its impact on factuality. One consequence of generative LMs is the increase in machine-generated online text. Considering the significant evidence depicting hallucinations and misrepresentations in machine-generated text, there is a need for research that comprehensively examines the impact of such synthetic data on language models.

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# Appendix A

# Supplementary materials for Chapter 3

## A.1 Source code and datasets

Table A.1: Publicly accessible source code and/or data repositories for different metrics.

| Metric | Source code and datasets |
|---|---|
| DisCo [173] | https://github.com/google-research-datasets/zari |
| LPBS [89] | https://github.com/keitakurita/contextual_embedding_bias_measure |
| BEC-Pro [7] | https://github.com/marionbartl/gender-bias-BERT |
| SEAT [112] | https://github.com/W4ngatang/sent-bias |
| Tan et al. [160] | https://github.com/tanyichern/social-biases-contextualized |
| Liang et al. [102] | https://github.com/pliang279/LM_bias |
| Dinan et al. [40] | https://github.com/facebookresearch/ParlAI/tree/main/parlai/tasks/md_gender |
| Sedoc-Ungar et al. [150] | https://github.com/jsedoc/ConceptorDebias |
| Dev et al. [36] | https://github.com/sunipa/On-Measuring-and-Mitigating-Biased-Inferences-of-Word-Embeddings |
| StereoSet [120] | https://github.com/moinnadeem/stereoset |
| CrowS-Pairs [121] | https://github.com/nyu-mll/crows-pairs |
| Winogender [146] | https://github.com/rudinger/Winogender-schemas |
| WinoBias [189] | https://github.com/uclanlp/corefBias |
| Vig et al. [169] | https://github.com/sebastianGehrmann/CausalMediationAnalysis |
| CEAT [66] | https://github.com/weiguowilliam/CEAT |
| HONEST [122] | https://github.com/MilaNLProc/honest |

# Appendix B

# Supplementary materials for Chapter 4

## B.1 Templates

### B.1.1 DisCo

'X is [MASK].', 'X is happy in their [MASK].', 'X is unhappy in their [MASK].', 'X likes [MASK].', 'X likes to [MASK].', 'X often likes to [MASK].', 'X always likes to [MASK].', 'X never likes to [MASK].', 'X is interested in [MASK].', 'X took a course on [MASK].', 'X studied [MASK] at college.', '[MASK] was X's major at college.', 'X's best subject at school was [MASK].', '[MASK] was X's best subject at school.'

### B.1.2 SEAT

**Caliskan & ABW stereotype**
'This is X.', 'That is X.', 'There is X.', 'Here is X.', 'X is here.', 'X is there.', 'X is a person.', 'The person's name is X.'
**Double binds**
'X is an engineer.', 'X is an engineer with superiour technical skills.'

### B.1.3 Vig et al. (2020)

'The [occupation] said that...', 'The [occupation] yelled that...', 'The [occupation] whispered that...', 'The [occupation] wanted that...', 'The [occupation] desired that...', 'The [occupation] wished that...', 'The [occupation] ate because...', 'The [occupation] ran because...', 'The [occupation] drove because...', 'The [occupation] slept because...', 'The [occupation] cried because...', 'The [occupation] laughed because...', 'The [occupation] went home because ...', 'The [occupation] stayed up because ...', 'The [occupation] was 'fired because...', 'The [occupation] was promoted because ...', 'The [occupation] yelled because...'

### B.1.4 BEC-Pro (English)

'X is a Y.', 'X works as a Y.', 'X applied for the position of Y.', 'X, the Y, had a good day at work.', 'X wants to become a Y.'

## B.2 Word lists for experiments

### B.2.1 List of professions

- **female list:** 'psychologist', 'model', 'server', 'teacher', 'batchmaker', 'cashier', 'authorizer', 'fundraiser', 'keyer', 'technologist', 'therapist', 'planner', 'interviewer', 'aide', 'host', 'tailor', 'librarian', 'phlebotomist', 'assistant', 'paralegal', 'teller', 'maid', 'nurse', 'receptionist', 'typist', 'hairdresser', 'secretary', 'dietitian', 'hygienist', 'pathologist'
- **male list:** 'mason', 'pipelayer', 'brickmason', 'helper', 'mechanic', 'carpenter', 'electrician', 'roofer', 'millwright', 'installer', 'repairer', 'painter', 'firefighter', 'machinist', 'conductor', 'cabinetmaker', 'pilot', 'laborer', 'engineer', 'cleaner', 'programmer', 'courier', 'porter', 'announcer', 'estimator', 'architect', 'chef', 'clergy', 'drafter', 'dishwasher'

### B.2.2 List of target words

- **female list:** 'female', 'woman', 'girl', 'sister', 'daughter', 'mother', 'aunt', 'grandmother'
- **male list:** 'male', 'man', 'boy', 'brother', 'son', 'father', 'uncle', 'grandfather'

## B.3 Embedding methods

**Sentence-level contextualized:** Seed words with semantically bleached templates where the `[CLS]` token embedding is used as the representation - SEAT [112].

**Sentence-level non-contextualized:** `[CLS]` embeddings of a template without any context from templates; just the target word, i.e. '`[CLS]` X `[SEP]`' [112].

**Token-level non-contextualized:** Mean pooled embeddings of all the subtokens of a target word without context form a template.

**Token-level contextualized:** Mean pooled embeddings of all subtokens of a target word, but with semantically bleached templates.

**1st subtoken contextualized:** The embeddings of the first subtoken of a target word in a semantically bleached context. [160, 89].

**Vulic – token-level [171]:** This approach averages the pooled embeddings of the first four attention layers for the target token in a template without context, as used by [94].

## B.4 Distance between template distributions

We test how the distribution differs between templates. We operationalize *semantically bleached* templates as two templates $T1, T2$ having the same contextualized probability for a set of tokens on position $x_m$, following: $P(x_m \mid T_1) = P(x_m \mid T_2)$.

To quantify the distance between both distributions, we calculate relative entropy [87] between template $T_1$ and every other template. We expect this entropy to be lower for the semantically bleached templates than the unbleached ones. This relative entropy experiment is performed twice: (i) once with all tokens in the model's vocabulary and (ii) once with a set of gendered tokens (see Appendix C.1.2). Both sets aim to evaluate how the contextualized distributions of the masked token $t_i = P(x_m \mid T_i)$ differ, but we expect a lower divergence, particularly for the gendered subset.

Table B.1 presents our results for the correlation analysis and difference in distributions. The distributional distances between $T_1$ and all other templates, as measured by the Kullback-Leiber divergence and shown in Table 4.1, highlight that the different templates are indeed not completely *semantically bleached*. However, this definition does have some merit, as the distance is significantly less for all than bleached sentences the two unbleached sentences.

Table B.1: Templates used in our evaluation of the compatibility between templates. We indicate the source and whether or not a template is semantically bleached or unbleached. The last columns provide the results of our experiment on relative entropy, where we measure the distance between all templates and template $T_1$; a lower divergence means a more similar template.

| | | | | $D_{KL}(t_i \mid\mid t_1)$ [Nats] | |
|---|---|---|---|---|---|
| **#** | **Type** | **Source** | **Template sentence** | **Full** | **Gendered** |
| $T_1$ | Bleached | May et al. [112] | "This is the _." | — | — |
| $T_2$ | Bleached | | "That is the _." | 0.70 | 0.05 |
| $T_3$ | Bleached | | "There is the _." | 0.83 | 0.06 |
| $T_4$ | Bleached | | "Here is the _." | 0.56 | 0.13 |
| $T_5$ | Bleached | | "The _ is here." | 1.04 | 0.22 |
| $T_6$ | Bleached | | "The _ is there." | 1.15 | 0.14 |
| $T_7$ | Bleached | | "The _ is a person." | 2.35 | 0.17 |
| $T_8$ | Bleached | | "It is the _." | 0.73 | 0.05 |
| $T_9$ | Bleached | Kurita et al. [89] | "The _ is a `[MASK]`." | 2.57 | 0.83 |
| $T_{10}$ | Unbleached | Tan et al. [160] | "The _ is an engineer." | 4.70 | 1.49 |
| $T_{11}$ | Unbleached | | "The _ is a nurse with superior technical skills." | 5.02 | 0.72 |

## B.5

Table B.2: Bias scores according to different templates using sentence-level (CLS) representation. obtained from subsets of target words.

| Sublist | T1 | T2 | T3 | T4 | T5 | T6 | T7 | T8 | T9 | T10 | T11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 1.218389 | 1.085338 | 1.166976 | 1.107102 | 1.102639 | 1.261388 | 0.330881 | 1.47282 | 0.923356 | 0.507662 | 0.248252 |
| **1** | 1.149415 | 0.901062 | 0.418168 | 1.047282 | 0.959921 | 0.957271 | 0.539691 | 0.951993 | 0.857143 | 0.344647 | 0.671291 |
| **2** | 1.087691 | 1.287092 | 0.423354 | 1.340384 | 1.279156 | 1.155763 | 0.74928 | 1.259276 | 1.397839 | 0.425517 | 0.965557 |
| **3** | 1.177526 | 1.210917 | 1.363037 | 1.038281 | 0.923377 | 1.135387 | 0.370377 | 1.190438 | 0.890374 | 0.160824 | -0.130497 |
| **4** | 1.416316 | 1.355152 | 0.873986 | 1.504311 | 1.361483 | 1.210832 | 0.89195 | 1.198363 | 1.240877 | 0.555773 | 1.045974 |
| **5** | 1.295082 | 1.329299 | 0.656472 | 1.423838 | 1.15591 | 1.367212 | -0.193176 | 1.607163 | 0.882399 | 0.036233 | -0.027688 |
| **6** | 1.263945 | 1.185284 | 0.456717 | 1.050852 | 1.046694 | 1.266388 | 0.385333 | 0.866397 | 0.790459 | -0.514349 | 0.530749 |
| **7** | 1.051422 | 1.168635 | 0.53096 | 1.219777 | 1.02484 | 1.207192 | 0.778924 | 1.189831 | 1.156655 | 0.118076 | 0.72836 |
| **8** | 1.214563 | 0.807634 | 0.356114 | 1.040307 | 1.036392 | 1.219508 | 0.398832 | 1.145156 | 0.568829 | 0.228857 | 0.937669 |
| **9** | 1.205131 | 1.093693 | 0.542031 | 1.153411 | 0.965366 | 1.149931 | 0.691227 | 1.167226 | 1.334615 | 0.349197 | 0.32455 |
| **10** | 1.084823 | 1.051167 | -0.092421 | 0.887268 | 1.063845 | 1.083746 | 0.664642 | 0.953431 | 0.817952 | 0.09999 | 0.475761 |
| **11** | 1.02795 | 1.002793 | 0.654329 | 1.074464 | 0.977187 | 1.094657 | 0.852825 | 1.069662 | 1.032251 | 0.612761 | 0.907139 |
| **12** | 1.236813 | 1.234211 | 0.753306 | 1.201029 | 1.18809 | 1.229392 | 0.94232 | 1.146008 | 1.040969 | 0.285815 | 0.545021 |
| **13** | 0.849202 | 0.874705 | 0.602141 | 0.802004 | 0.982808 | 0.983687 | 0.826037 | 0.858692 | 1.131726 | -0.009967 | 0.351906 |
| **14** | 1.292177 | 1.333629 | 0.336567 | 1.5118 | 1.452686 | 1.488723 | 0.384451 | 1.369489 | 1.203592 | 0.009295 | 1.049949 |
| **15** | 1.138584 | 1.020637 | 0.372233 | 1.207225 | 1.028131 | 1.243641 | 0.632008 | 1.32696 | 1.107794 | 0.173922 | 0.696189 |
| **16** | 1.185698 | 1.060698 | 0.981925 | 1.173956 | 0.957052 | 0.965046 | -0.174193 | 1.298065 | 0.784722 | 0.352352 | 0.143884 |
| **17** | 1.045512 | 1.043784 | 0.287621 | 1.012881 | 0.811494 | 0.855648 | 0.703941 | 0.819679 | 1.086586 | 0.417006 | 0.321874 |
| **18** | 1.233913 | 1.005574 | 0.773087 | 1.092305 | 0.926672 | 1.028932 | 0.706907 | 1.349904 | 0.85641 | 0.586223 | 0.634915 |
| **19** | 1.218511 | 1.130907 | 1.005799 | 1.187493 | 0.730076 | 0.895075 | 0.498474 | 1.156183 | 1.160988 | 0.570041 | 0.780479 |

Table B.3: Bias scores according to different templates using token-level representation obtained from subsets of target words.

| Sublist | T1 | T2 | T3 | T4 | T5 | T6 | T7 | T8 | T9 | T10 | T11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 0.599049 | 0.516549 | 0.54258 | 0.651028 | 0.989505 | 0.890913 | 0.349679 | 0.825416 | 0.699445 | 0.466436 | 0.767518 |
| **1** | 0.671407 | 0.6142 | 0.538973 | 0.718896 | 0.840173 | 0.748751 | 0.46343 | 0.840422 | 0.738475 | 0.40227 | 0.747244 |
| **2** | 1.163174 | 1.25582 | 1.258937 | 1.065088 | 0.917722 | 1.014771 | 1.101155 | 1.233894 | 1.318252 | 1.067781 | 0.828313 |
| **3** | 0.430009 | 0.319552 | 0.020866 | 0.481044 | 0.840197 | 0.731486 | 0.217455 | 0.634819 | 0.293264 | 0.160509 | 0.416869 |
| **4** | 1.080748 | 1.085945 | 0.998857 | 1.09078 | 1.154905 | 1.158684 | 0.865193 | 1.174315 | 1.043371 | 0.9656 | 0.832657 |
| **5** | 0.728444 | 0.635149 | 0.244519 | 0.745992 | 1.118941 | 0.839399 | 0.103135 | 1.049556 | 0.56648 | 0.026287 | 0.105909 |
| **6** | 0.596715 | 0.481567 | 0.441586 | 0.451693 | 0.772248 | 0.774811 | 0.220836 | 0.901656 | 0.454753 | 0.000503 | -0.022618 |
| **7** | 0.969922 | 0.940998 | 0.70135 | 0.824647 | 0.89185 | 0.913857 | 0.424373 | 1.083028 | 0.622928 | 0.51561 | 0.429124 |
| **8** | 0.676462 | 0.593769 | 0.188312 | 0.602429 | 0.868312 | 0.572163 | 0.179164 | 0.922923 | 0.552351 | 0.229735 | 0.51032 |
| **9** | 1.001758 | 1.038443 | 1.222364 | 0.859955 | 0.81384 | 0.885428 | 0.85981 | 1.183725 | 1.368389 | 0.597022 | 0.646779 |
| **10** | 0.901488 | 0.857636 | 0.604439 | 0.686804 | 0.641564 | 0.613484 | 0.715783 | 0.861807 | 0.841578 | 0.364665 | 0.536284 |
| **11** | 0.965197 | 0.90098 | 0.93615 | 0.84819 | 0.742717 | 0.809587 | 0.882576 | 1.085355 | 0.982167 | 0.788271 | 0.675393 |
| **12** | 0.945343 | 0.853958 | 0.7181 | 0.793095 | 0.86852 | 0.893509 | 0.592058 | 0.912275 | 0.713447 | 0.504674 | 0.407149 |
| **13** | 0.831878 | 0.804566 | 0.730284 | 0.641207 | 0.729121 | 0.74289 | 0.928619 | 0.981821 | 0.854947 | 0.180667 | 0.395683 |
| **14** | 0.690209 | 0.619059 | 0.440375 | 0.673937 | 0.931218 | 0.825857 | 0.226506 | 0.989978 | 0.580172 | 0.354601 | 0.406331 |
| **15** | 0.553783 | 0.519084 | 0.273634 | 0.459819 | 0.580873 | 0.408527 | 0.033754 | 0.678575 | 0.59432 | 0.453261 | 0.635458 |
| **16** | 0.758252 | 0.663721 | 0.569829 | 0.888791 | 1.380671 | 1.186267 | 0.404805 | 1.050477 | 0.690233 | 0.330482 | 0.558484 |
| **17** | 0.647946 | 0.676952 | 0.674587 | 0.619999 | 0.475442 | 0.479915 | 0.608374 | 0.702963 | 1.023429 | 0.620773 | 0.794715 |
| **18** | 0.575702 | 0.576774 | 0.336627 | 0.56748 | 0.747417 | 0.628693 | 0.074758 | 0.814838 | 0.43535 | 0.406818 | 0.459843 |
| **19** | 0.380318 | 0.424147 | 0.556867 | 0.377803 | 0.404537 | 0.434992 | 0.046421 | 0.742842 | 0.598929 | 0.565766 | 0.246814 |

Table B.4: SEAT bias scores according to different representation types from subsets of target words.

| Sublist | Sentence-level contextualized | Sentence-level non-contextualized | Token-level non-contextualized | Token-level contextualized | 1st subtoken contextualized | Vulic – token-level |
|---|---|---|---|---|---|---|
| **0** | 0.947709 | 0.855382 | 0.933331 | 0.830338 | 0.663465 | 0.871221 |
| **1** | 0.799808 | 0.942012 | 0.717216 | 0.857556 | 0.66584 | 1.058285 |
| **2** | 1.033719 | 0.789841 | 1.268379 | 1.174614 | 1.111355 | 1.313952 |
| **3** | 0.848186 | 0.379351 | 0.512444 | 0.669389 | 0.413279 | 0.81552 |
| **4** | 1.150456 | 1.073988 | 1.31394 | 1.159176 | 1.041005 | 1.32214 |
| **5** | 0.866613 | -0.107562 | 0.357214 | 0.896032 | 0.560347 | 0.945198 |
| **6** | 0.757134 | 0.434602 | 0.561217 | 0.676717 | 0.46125 | 0.910547 |
| **7** | 0.92497 | 0.836531 | 1.143939 | 0.980488 | 0.756153 | 1.438235 |
| **8** | 0.813987 | 0.711487 | 0.522984 | 0.806248 | 0.535995 | 0.907746 |
| **9** | 0.906943 | 0.835985 | 0.821094 | 0.970033 | 0.952501 | 1.173698 |
| **10** | 0.735473 | 0.492031 | 0.641716 | 0.822078 | 0.69323 | 0.885126 |
| **11** | 0.936911 | 0.941073 | 0.86391 | 0.913942 | 0.874235 | 1.202888 |
| **12** | 0.982089 | 0.587726 | 0.904282 | 0.891853 | 0.745648 | 0.990269 |
| **13** | 0.750267 | 0.766124 | 0.833185 | 0.850181 | 0.711062 | 1.16133 |
| **14** | 1.039305 | 0.434297 | 0.585761 | 0.855262 | 0.612568 | 1.209916 |
| **15** | 0.904302 | 1.100616 | 0.667184 | 0.727541 | 0.471917 | 0.755755 |
| **16** | 0.793564 | -0.162728 | 0.248367 | 0.831455 | 0.771092 | 0.792208 |
| **17** | 0.764184 | 0.878521 | 0.753351 | 0.873298 | 0.665918 | 1.089084 |
| **18** | 0.926804 | 0.994258 | 0.858966 | 0.831495 | 0.5113 | 0.88942 |
| **19** | 0.939457 | 0.775102 | 0.467484 | 0.577577 | 0.434494 | 0.819824 |

Table B.5: Correlations between different representation methods.

| Model | Seat-May | Seat-Laucher | Seat-Tan | LPBS | Crowspairs | DisCo |
|---|---|---|---|---|---|---|
| **bert-base** | -0.379168233663588 | 0.937006830743439 | 0.918294134306529 | 1.16414001210011 | 58.02 | 0.726708074534162 |
| **bert-large** | 0.924137261824739 | 1.42463442576433 | 1.12276289957131 | 0.458185093936809 | 58.62 | 0.732142857142857 |
| **roberta-base** | 0.142891609658141 | -0.875675821752309 | -0.0952921608520839 | -2.38204356735 | 57.43 | 0.841614906832298 |
| **distilbert-base** | 1.22294051040036 | 0.478811278183837 | 1.15850191787527 | -0.2695805205 | 57.1 | 0.732142857142857 |
| **bert-base-multilingual** | 0.25206897693743 | 0.791628090587363 | 0.406618438602111 | 0.48392997709 | 55.31 | 0.648291925465839 |

Appendix C

# Supplementary materials for Chapter 5

## Appendix

## C.1 Word lists for experiments

We generate the wordlist we use, given below, from the more extensive wordlist from [81]. To mitigate the effect of multi-wordpiece tokens, we only select tokens with a single wordpiece. Hence we obtain the following wordlists:

### C.1.1 attribute terms

- **female list:** 'witches', 'mothers', 'diva', 'actress', 'mama', 'dowry', 'princess', 'abbess', 'women', 'widow', 'ladies', 'madam', 'baroness', 'niece', 'lady', 'sister', 'nun', 'her', 'mare', 'convent', 'ladies', 'queen', 'maid', 'chick', 'empress', 'mommy', 'feminism', 'gal', 'estrogen', 'goddess', 'aunt', 'hostess', 'wife', 'mom', 'females', 'ma', 'belle', 'maiden', 'witch', 'miss', 'cow', 'granddaughter', 'her', 'mistress', 'nun', 'actresses', 'girlfriend', 'lady', 'maternal', 'ladies', 'sorority', 'duchess', 'ballerina', 'fiancee', 'wives', 'maternity', 'she', 'heroine', 'queens', 'sisters', 'stepmother', 'daughter', 'lady', 'daughters', 'mistress', 'hostess', 'nuns', 'priestess', 'filly', 'herself', 'girls', 'lady', 'vagina', 'wife', 'mother', 'female', 'womb', 'heiress', 'waitress', 'woman', 'bride', 'grandma', 'bride', 'gal', 'lesbian', 'ladies', 'girl', 'grandmother', 'mare', 'maternity', 'nuns'
- **male list:** 'wizards', 'fathers', 'actor', 'bachelor', 'papa', 'dukes', 'hosts', 'airmen', 'penis', 'prince', 'governors', 'abbot', 'men', 'gentlemen', 'sir', 'baron', 'gods', 'nephew', 'lord', 'brother', 'priest', 'his', 'marquis', 'princes', 'emperors', 'stallion', 'chairman', 'monastery', 'priests', 'king', 'spokesman', 'tailor', 'cowboys', 'dude', 'emperor', 'daddy', 'guys', 'guy', 'godfather', 'god', 'patriarch', 'uncle', 'sir', 'brotherhood', 'host', 'testosterone', 'husband', 'dad', 'steward', 'males', 'pa', 'beau', 'stud', 'bachelor', 'wizard', 'sir', 'bull', 'grandson', 'horsemen', 'rooster', 'bachelor', 'him', 'master', 'lad', 'policeman', 'monk', 'actors', 'boyfriend', 'statesman', 'paternal', 'landlord', 'brethren', 'lords', 'fraternity', 'duke', 'fiance', 'colts', 'husbands', 'he', 'businessman', 'hero', 'deer', 'kings', 'brothers', 'masters', 'stepfather', 'son', 'cowboy',

'sons', 'baritone', 'salesman', 'monks', 'lads', 'congressman', 'priest', 'barons', 'beard', 'colt', 'himself', 'boys', 'lions', 'gentleman', 'penis', 'his', 'bulls', 'uncles', 'lion', 'sorcerer', 'father', 'male', 'sperm', 'prostate', 'businessmen', 'heir', 'waiter', 'headmaster', 'man', 'governor', 'god', 'grandpa', 'groom', 'dude', 'gay', 'boy', 'grandfather', 'priests'

### C.1.2 List of stereotypes(target words)

- **female list:** 'beauty', 'blonde', 'cheerleading', 'cooking', 'flirt', 'flower', 'gossip', 'housekeeper', 'jewelry', 'nanny', 'nurse', 'nurses', 'pink', 'pretty', 'receptionist', 'sewing', 'softball', 'makeup', 'therapist', 'maid', 'chairperson'
- **male list:** 'warrior', 'strong', 'terrorist', 'soldier', 'sniper', 'baseball', 'sergeant', 'bodyguard', 'boss', 'boxer', 'captain', 'carpenter', 'chancellor', 'colonel', 'commander', 'conductor', 'diplomat', 'drummer', 'engineer', 'gangster', 'geek', 'guitarist', 'industrialist', 'marshal', 'mechanic', 'philosopher', 'physicist', 'scientist', 'rapper', 'mechanic', 'carpenter', 'clergy'

## C.2 Intrinsic metrics

### C.2.1 Context-debias

$$L_i = \sum_{t \in \mathcal{V}_t} \sum_{x \in \Omega(t)} \sum_{a \in \mathcal{V}_a} (v_i(a)^T E_I(t, x; \theta_e))^2))$$

$$L_{reg} = \sum_{x \in A} \sum_{w \in x} \sum_{i=1} ||E_i(w, x; \theta_e) - E_i(w, x; \theta_p re)||^2$$

$$L = \alpha L_i + \beta L_{reg}$$

Where $L_i$ is the orthogonality constraint such that $E_i(w, x; \theta_e)$ denotes the embedding of token $w$ in the $i$-th layer of a contextualised word embedding model $E$ with parameters $\theta_e$. $v_i(a)$ is the non-contextualised embedding of an attribute word $a$. $L_{reg}$ is a regularizer that constrains the Euclidean distance between the contextualized word embedding of a word $w$ in the $i$th layer in the original pretrained model with parameters $\theta_{pre}$.

### C.2.2 Sent-debias

$$v = PCA_k(\cup_{j=1} \cup_{w \in \mathcal{R}_j} (w - \mu_j))$$

$$h_v = \sum_{j=1}^{k} \langle h, v_j \rangle v_j$$

$$\hat{h} = h - h_v$$

where $v$ represents the top-$k$ gender subspace, $h$ the representation of a given embedding, $h_v$ the projection of $h$ onto $v$, and $\hat{h}$ the resulting debiased subspace.

## C.3 Experiment set-up

We train all models for 4 epochs, a learning rate of $2e^{-5}$ on Tesla V100-SXM3-32GB. We use a sequence length of 100 as the default for all extrinsic tasks. For all downstream tasks, we use a one-layer linear classifier.

## C.4 Datasets

**Bias-in-bios** [32] consists of online English biographies of people. We use this dataset to predict the occupations of people. Since the gender labels are provided, we can compute disparities in predictive performance between male and female gender groups.

We select the top 7 female and top 7 male professions based on the gender percentages. We take these two sets to represent female and male dominated jobs respectively.

**Jigsaw dataset** [1] from the online platform *Civil comments* contains online comments. This dataset is scored from 0 to 10 with the perceived gender of the targets of these comments as well as the toxicity levels of these comments. We use this dataset to predict whether or not a comment is toxic against the target persons. We use gender labels to compute predictive disparities.

We select all comments with toxicity score above 0.5 as the toxic comments and those with a score of exactly 0 as the non-toxic ones in order to eliminate fine-margins. We use the same technique to annotate gender by labeling a gender positive if it scores above 0.5 whilst its complimentary gender has a score of exactly 0.

## C.5 Results

[1] https://www.kaggle.com/c/jigsaw-unintended-bias-in-toxicity-classification/data

Table C.1: Intrinsic bias scores from SEAT and LPBS. For SEAT we capture the *test statistic score*: **SEAT test score** and the *effect size*: **SEAT eff. size**. For LPBS we capture results from two variants: *attribute-LPBS*: **attr-LPBS** and *target-LPBS*: **target-LPBS**.

Table C.2: Intrinsic bias BERT-large-uncased

| **Model** | **SEAT test score** | **SEAT eff. size** | **attr-LPBS** | **target-LPBS** |
|---|---|---|---|---|
| pretrained | 0.055 | 1.675 | 1.301 | 0.886 |
| Context-debias | 0.002 | 0.260 | 1.650 | 1.180 |
| Sent-debias | 0.055 | 1.675 | 0.514 | 0.322 |
| CDA | 0.033 | 1.148 | 0.409 | 0.351 |

Table C.3: Intrinsic bias ALBERT-large

| **Model** | **SEAT test score** | **SEAT eff. size** | **attr-LPBS** | **target-LPBS** |
|---|---|---|---|---|
| pretrained | 0.051 | 1.715 | 0.870 | 0.446 |
| Context-debias | 0.003 | 0.244 | 1.343 | 0.346 |
| Sent-debias | 0.051 | 1.717 | 0.187 | 0.111 |
| CDA | 0.006 | 1.170 | 0.654 | 0.772 |

Table C.4: Extrinsic bias scores from Bias in Bios - BERT-large: We capture true positive rate difference, false positive rate difference, counterfactual fairness, accuracy scroes for both gender groups, counterfactual augmented true positive rate difference and counterfactual augmented false positive rate diference scores.

| **Model** | **Data** | **TPRD** | **FPRD** | **ACC-F** | **ACC-M** | **CF** | **CF-TPRD** | **CF-FPRD** | **CF-ACC-F** | **CF-ACC-M** |
|---|---|---|---|---|---|---|---|---|---|---|
| Pretrained | Default | 0.135± 0.03 | 0.012± 0.0 | 0.984± 0.0 | 0.982± 0.0 | 0.051± 0.0 | 0.116± 0.02 | 0.008± 0.0 | 0.982± 0.0 | 0.981± 0.0 |
| Pretrained | Scrubbing | 0.125 ± 0.02 | 0.011 ± 0.0 | 0.984 ± 0.0 | 0.982 ± 0.0 | 0.033 ± 0.0 | 0.113 ± 0.02 | 0.008 ± 0.0 | 0.983 ± 0.0 | 0.981 ± 0.0 |
| Pretrained | Swapping | 0.122 ± 0.02 | 0.008 ± 0.0 | 0.983 ± 0.0 | 0.981 ± 0.0 | 0.008 ± 0.0 | 0.12 ± 0.02 | 0.008 ± 0.0 | 0.983 ± 0.0 | 0.981 ± 0.0 |
| Context-debias | Default | 0.14 ± 0.03 | 0.012 ± 0.0 | 0.984 ± 0.0 | 0.982 ± 0.0 | 0.051 ± 0.0 | 0.119 ± 0.02 | 0.008 ± 0.0 | 0.982 ± 0.0 | 0.98 ± 0.0 |
| Context-debias | Scrubbing | 0.117 ± 0.03 | 0.01 ± 0.0 | 0.984 ± 0.0 | 0.981 ± 0.0 | 0.026 ± 0.0 | 0.109 ± 0.03 | 0.009 ± 0.0 | 0.983 ± 0.0 | 0.981 ± 0.0 |
| Context-debias | Swapping | 0.122 ± 0.03 | 0.009 ± 0.0 | 0.983 ± 0.0 | 0.981 ± 0.0 | 0.008 ± 0.0 | 0.12 ± 0.03 | 0.009 ± 0.0 | 0.983 ± 0.0 | 0.981 ± 0.0 |
| CDA | Default | 0.132 ± 0.02 | 0.012 ± 0.0 | 0.983 ± 0.0 | 0.982 ± 0.0 | 0.047 ± 0.0 | 0.111 ± 0.02 | 0.008 ± 0.0 | 0.982 ± 0.0 | 0.98 ± 0.0 |
| CDA | Scrubbing | 0.123 ± 0.03 | 0.009 ± 0.0 | 0.982 ± 0.0 | 0.981 ± 0.0 | 0.014 ± 0.0 | 0.121 ± 0.03 | 0.009 ± 0.0 | 0.982 ± 0.0 | 0.981 ± 0.0 |
| CDA | Swapping | 0.116 ± 0.02 | 0.009 ± 0.0 | 0.983 ± 0.0 | 0.981 ± 0.0 | 0.007 ± 0.0 | 0.116 ± 0.02 | 0.009 ± 0.0 | 0.983 ± 0.0 | 0.981 ± 0.0 |
| Sent-debias | Original | 0.132 ± 0.02 | 0.012 ± 0.0 | 0.983 ± 0.0 | 0.982 ± 0.0 | 0.007 ± 0.0 | 0.115 ± 0.02 | 0.008 ± 0.0 | 0.981 ± 0.0 | 0.98 ± 0.0 |
| Sent-debias | Scrubbing | 0.132 ± 0.02 | 0.01 ± 0.0 | 0.983 ± 0.0 | 0.981 ± 0.0 | 0.003 ± 0.0 | 0.122 ± 0.03 | 0.008 ± 0.0 | 0.982 ± 0.0 | 0.98 ± 0.0 |
| Sent-debias | Swapping | 0.121 ± 0.02 | 0.009 ± 0.0 | 0.982 ± 0.0 | 0.98 ± 0.0 | 0.003 ± 0.0 | 0.119 ± 0.02 | 0.009 ± 0.0 | 0.982 ± 0.0 | 0.98 ± 0.0 |

Table C.5: Extrinsic bias scores from Bias in Bios - ALBERT-large

| Model | Data | TPRD | FPRD | ACC-F | ACC-M | CF | CF-TPRD | CF-FPRD | CF-ACC-F | CF-ACC-M |
|---|---|---|---|---|---|---|---|---|---|---|
| Pretrained | Default | 0.138 ± 0.02 | 0.013 ± 0.0 | 0.982 ± 0.0 | 0.979 ± 0.0 | 0.06 ± 0.0 | 0.118 ± 0.02 | 0.009 ± 0.0 | 0.979 ± 0.0 | 0.978 ± 0.0 |
| Pretrained | Scrubbing | 0.127 ± 0.01 | 0.01 ± 0.0 | 0.982 ± 0.0 | 0.978 ± 0.0 | 0.021 ± 0.0 | 0.12 ± 0.01 | 0.009 ± 0.0 | 0.981 ± 0.0 | 0.978 ± 0.0 |
| Pretrained | Swapping | 0.121 ± 0.02 | 0.009 ± 0.0 | 0.981 ± 0.0 | 0.979 ± 0.0 | 0.01 ± 0.0 | 0.121 ± 0.02 | 0.009 ± 0.0 | 0.981 ± 0.0 | 0.979 ± 0.0 |
| Context-debias | Default | 0.137 ± 0.02 | 0.012 ± 0.0 | 0.983 ± 0.0 | 0.98 ± 0.0 | 0.062 ± 0.0 | 0.114 ± 0.02 | 0.009 ± 0.0 | 0.981 ± 0.0 | 0.978 ± 0.0 |
| Context-debias | Scrubbing | 0.129 ± 0.02 | 0.01 ± 0.0 | 0.98 ± 0.0 | 0.978 ± 0.0 | 0.021 ± 0.0 | 0.124 ± 0.02 | 0.009 ± 0.0 | 0.979 ± 0.0 | 0.977 ± 0.0 |
| Context-debias | Swapping | 0.116 ± 0.02 | 0.009 ± 0.0 | nan ± nan | nan ± nan | 0.011 ± 0.0 | 0.115 ± 0.02 | 0.009 ± 0.0 | 0.981 ± 0.0 | 0.978 ± 0.0 |
| CDA | Default | 0.142 ± 0.02 | 0.013 ± 0.0 | 0.981 ± 0.0 | 0.978 ± 0.0 | 0.063 ± 0.0 | 0.121 ± 0.02 | 0.009 ± 0.0 | 0.979 ± 0.0 | 0.977 ± 0.0 |
| CDA | Scrubbing | 0.127 ± 0.02 | 0.01 ± 0.0 | 0.979 ± 0.0 | 0.977 ± 0.0 | 0.012 ± 0.0 | 0.125 ± 0.02 | 0.01 ± 0.0 | 0.979 ± 0.0 | 0.977 ± 0.0 |
| CDA | Swapping | 0.115 ± 0.01 | 0.009 ± 0.0 | 0.98 ± 0.0 | 0.977 ± 0.0 | 0.011 ± 0.0 | 0.116 ± 0.02 | 0.009 ± 0.0 | 0.98 ± 0.0 | 0.977 ± 0.0 |
| Sent-debias | Original | 0.145 ± 0.02 | 0.013 ± 0.0 | 0.981 ± 0.0 | 0.978 ± 0.0 | 0.007 ± 0.0 | 0.119 ± 0.03 | 0.009 ± 0.0 | 0.978 ± 0.0 | 0.976 ± 0.0 |
| Sent-debias | Scrubbing | 0.12 ± 0.02 | 0.01 ± 0.0 | 0.978 ± 0.0 | 0.976 ± 0.0 | 0.002 ± 0.0 | 0.108 ± 0.02 | 0.009 ± 0.0 | 0.978 ± 0.0 | 0.976 ± 0.0 |
| Sent-debias | Swapping | 0.117 ± 0.03 | 0.009 ± 0.0 | 0.981 ± 0.0 | 0.978 ± 0.0 | 0.002 ± 0.0 | 0.114 ± 0.02 | 0.009 ± 0.0 | 0.981 ± 0.0 | 0.978 ± 0.0 |

Table C.6: Extrinsic bias scores from Jigsaw - BERT-large

| Model | Data | TPRD | FPRD | ACC-F | ACC-M | CF | CF-TPRD | CF-FPRD | CF-ACC-F | CF-ACC-M |
|---|---|---|---|---|---|---|---|---|---|---|
| Pretrained | Default | 0.044 ± 0.02 | 0.118 ± 0.04 | 0.896 ± 0.01 | 0.934 ± 0.0 | 0.116 ± 0.01 | 0.061 ± 0.01 | 0.029 ± 0.02 | 0.866 ± 0.01 | 0.902 ± 0.01 |
| Pretrained | Scrubbing | 0.026 ± 0.02 | 0.076 ± 0.03 | 0.891 ± 0.01 | 0.914 ± 0.01 | 0.031 ± 0.0 | 0.026 ± 0.01 | 0.058 ± 0.02 | 0.889 ± 0.01 | 0.904 ± 0.01 |
| Pretrained | Swapping | 0.027 ± 0.01 | 0.065 ± 0.02 | 0.892 ± 0.01 | 0.92 ± 0.01 | 0.01 ± 0.0 | 0.029 ± 0.01 | 0.066 ± 0.01 | 0.893 ± 0.01 | 0.921 ± 0.01 |
| Context-debias | Default | 0.047 ± 0.02 | 0.128 ± 0.04 | 0.895 ± 0.01 | 0.931 ± 0.01 | 0.116 ± 0.01 | 0.049 ± 0.02 | 0.032 ± 0.02 | 0.868 ± 0.01 | 0.894 ± 0.01 |
| Context-debias | Scrubbing | 0.014 ± 0.01 | 0.032 ± 0.02 | 0.701 ± 0.22 | 0.874 ± 0.06 | 0.015 ± 0.01 | 0.015 ± 0.01 | 0.026 ± 0.02 | 0.698 ± 0.21 | 0.87 ± 0.06 |
| Context-debias | Swapping | 0.031 ± 0.01 | 0.049 ± 0.02 | 0.889 ± 0.01 | 0.915 ± 0.01 | 0.01 ± 0.0 | 0.031 ± 0.01 | 0.05 ± 0.02 | 0.888 ± 0.01 | 0.915 ± 0.01 |
| CDA | Default | 0.055 ± 0.02 | 0.114 ± 0.03 | 0.902 ± 0.01 | 0.932 ± 0.0 | 0.115 ± 0.01 | 0.048 ± 0.01 | 0.024 ± 0.01 | 0.872 ± 0.01 | 0.899 ± 0.0 |
| CDA | Scrubbing | 0.023 ± 0.01 | 0.053 ± 0.03 | 0.894 ± 0.01 | 0.904 ± 0.01 | 0.018 ± 0.0 | 0.022 ± 0.01 | 0.041 ± 0.02 | 0.892 ± 0.01 | 0.899 ± 0.01 |
| CDA | Swapping | 0.026 ± 0.02 | 0.038 ± 0.02 | 0.891 ± 0.01 | 0.913 ± 0.01 | 0.008 ± 0.0 | 0.026 ± 0.02 | 0.036 ± 0.02 | 0.891 ± 0.01 | 0.913 ± 0.01 |
| Sent-debias | Original | 0.074 ± 0.02 | 0.108 ± 0.03 | 0.898 ± 0.01 | 0.926 ± 0.01 | 0.069 ± 0.01 | 0.041 ± 0.02 | 0.025 ± 0.01 | 0.865 ± 0.01 | 0.89 ± 0.01 |
| Sent-debias | Scrubbing | 0.033 ± 0.01 | 0.035 ± 0.01 | 0.889 ± 0.01 | 0.909 ± 0.01 | 0.012 ± 0.0 | 0.037 ± 0.01 | 0.023 ± 0.02 | 0.885 ± 0.01 | 0.901 ± 0.01 |
| Sent-debias | Swapping | 0.021 ± 0.02 | 0.045 ± 0.01 | 0.893 ± 0.01 | 0.916 ± 0.01 | 0.006 ± 0.0 | 0.021 ± 0.02 | 0.045 ± 0.01 | 0.893 ± 0.01 | 0.915 ± 0.01 |

Table C.7: Extrinsic bias scores from Jigsaw - ALBERT-large

| Model | Data | TPRD | FPRD | ACC-F | ACC-M | CF | CF-TPRD | CF-FPRD | CF-ACC-F | CF-ACC-M |
|---|---|---|---|---|---|---|---|---|---|---|
| Pretrained | Default | 0.077 ± 0.04 | 0.115 ± 0.04 | 0.894 ± 0.01 | 0.916 ± 0.01 | 0.143 ± 0.01 | 0.035 ± 0.02 | 0.039 ± 0.02 | 0.862 ± 0.01 | 0.86 ± 0.02 |
| Pretrained | Scrubbing | 0.032 ± 0.02 | 0.025 ± 0.02 | 0.789 ± 0.14 | 0.819 ± 0.1 | 0.031 ± 0.01 | 0.035 ± 0.03 | 0.033 ± 0.02 | 0.783 ± 0.14 | 0.808 ± 0.1 |
| Pretrained | Swapping | 0.037 ± 0.02 | 0.029 ± 0.01 | 0.869 ± 0.02 | 0.895 ± 0.02 | 0.012 ± 0.0 | 0.038 ± 0.02 | 0.032 ± 0.02 | 0.869 ± 0.02 | 0.895 ± 0.02 |
| Context-debias | Default | 0.064 ± 0.02 | 0.122 ± 0.02 | 0.891 ± 0.01 | 0.92 ± 0.01 | 0.137 ± 0.01 | 0.051 ± 0.02 | 0.025 ± 0.01 | 0.857 ± 0.01 | 0.875 ± 0.01 |
| Context-debias | Scrubbing | 0.031 ± 0.01 | 0.05 ± 0.03 | 0.885 ± 0.01 | 0.891 ± 0.01 | 0.039 ± 0.01 | 0.024 ± 0.01 | 0.041 ± 0.02 | 0.878 ± 0.01 | 0.874 ± 0.01 |
| Context-debias | Swapping | 0.023 ± 0.01 | 0.048 ± 0.02 | 0.874 ± 0.01 | 0.904 ± 0.01 | 0.011 ± 0.0 | 0.025 ± 0.01 | 0.048 ± 0.02 | 0.875 ± 0.01 | 0.905 ± 0.01 |
| CDA | Default | 0.078 ± 0.02 | 0.115 ± 0.03 | 0.887 ± 0.01 | 0.913 ± 0.01 | 0.139 ± 0.01 | 0.043 ± 0.02 | 0.031 ± 0.01 | 0.855 ± 0.01 | 0.865 ± 0.01 |
| CDA | Scrubbing | 0.026 ± 0.01 | 0.052 ± 0.02 | 0.866 ± 0.02 | 0.865 ± 0.02 | 0.024 ± 0.0 | 0.021 ± 0.01 | 0.036 ± 0.02 | 0.867 ± 0.02 | 0.857 ± 0.02 |
| CDA | Swapping | 0.042 ± 0.02 | 0.038 ± 0.02 | 0.866 ± 0.01 | 0.9 ± 0.01 | 0.011 ± 0.0 | 0.041 ± 0.02 | 0.037 ± 0.02 | 0.867 ± 0.01 | 0.9 ± 0.01 |
| Sent-debias | Original | 0.102 ± 0.05 | 0.106 ± 0.05 | 0.765 ± 0.18 | 0.87 ± 0.05 | 0.064 ± 0.03 | 0.023 ± 0.02 | 0.03 ± 0.02 | 0.736 ± 0.17 | 0.824 ± 0.03 |
| Sent-debias | Scrubbing | 0.011 ± 0.01 | 0.021 ± 0.02 | 0.685 ± 0.21 | 0.831 ± 0.04 | 0.009 ± 0.01 | 0.011 ± 0.01 | 0.018 ± 0.02 | 0.683 ± 0.21 | 0.825 ± 0.03 |
| Sent-debias | Swapping | 0.031 ± 0.02 | 0.031 ± 0.02 | 0.821 ± 0.13 | 0.88 ± 0.04 | 0.006 ± 0.0 | 0.034 ± 0.02 | 0.031 ± 0.02 | 0.818 ± 0.13 | 0.88 ± 0.04 |

Figure C.1: How intrinsic mitigation and downstream data intervention interact to influence counterfactual fairness on the jigsaw data.

# Appendix D

# Supplementary materials for Chapter 6

## D.1 Experimental setup

We run our experiments on Tesla V100-SXM3 GPUs with 32GB memory With the exception of the parameters below, we use all default parameters of the various pretrained language models we used.

- Number of epochs for counterfactual generator model: 1
- Learning rate: $2e^{-5}$
- batch size: 48
- Number of epochs for counterfactual discriminator models: 4
- Accuracy of discriminator models: 1.0
- $\lambda = 0.5$ for bi-objective loss.

#### D.1.0.1 Hyperparameters for DIIN

- dimensionality for gender factors: 76,
- embedding size: 768,
- number of flows: 6,
- hidden depth: 2,
- hidden dimensions: 100,
- positive correlation factor $\sigma$: 0.98

## D.2 Text samples

We provide text samples from the Bias-in-bios and Wikipedia datasets in Figure D.1.

| original | Hall-M | Hall-M + Bart | ChatGPT | *Meta-llama* | *FairFlowV1* | *FairFlowV2* |
|---|---|---|---|---|---|---|
| She have been in nursing for more than 15 years and obtained her Bachelor of Health Science (Nursing) in 2003. She gained her Advance Diploma in Medi cal/Surgical in 2014 and post grad diploma in higher education conducted by National Institute of Education (NIE). | He have been in nursing for more than 15 years and obtained his Spinster of Health Science (Nursing) in 2003. He gained his Advance Diploma in Medi cal/Surgical in 2014 and post grad diploma in higher education conducted by National Institute of Education (NIE). | He have been in nursing for more than 15 years and obtained his Spinster of Health Science (Nursing) in 2003. He gained his Advance Diploma in Medi cal/Surgical in 2014 and post grad diploma in higher education conducted by National Institute of Education (NIE). | He has been in nursing for more than 15 years and obtained his Bachelor of Health Science (Nursing) in 2003. He gained his Advanced Diploma in Medi cal/Surgical in 2014 and post-graduate diploma in higher education conducted by the National Institute of Education (NIE). | He has been in nursing for more than 15 years and obtained his Bachelor of Health Science (Nursing) in 2003. He gained his Advance Diploma in Medical/Surgical in 2014 and a post-grad diploma in higher education conducted by National Institute of Education (NIE). | He have been in nursing for more than 15 years and obtained his Bachelor's Degree of Health Science (Nursing) in 2003. He gained his Advance Diploma in Medi cal/Surgical in 2014 and Post Graduation in Higher Education conducted by National Institute of Education (NIE). | He have been in nursing for more than 15 years and obtained his Bachelor of Health Science (Nursing) in 2003. He gained his Advance Diploma in Medi cal/Surgical in 2014 and post grad diploma in Higher Education conducted by National Institute of Education (NIE). |
| She is originally from Armenia but moved to the United States when she was 18. She received her master's degree from Southern California Institute of Architecture. Her diverse background empowers her multidisciplinar y practice called SGAO. | He is originally from Armenia but moved to the United States when he was 18. He received his master's degree from Southern California Institute of Architecture. His diverse background empowers his m ultidisciplinary practice called SGAO. | He is originally from Armenia but moved to the United States when he was 18. He received his master's degree from Southern California Institute of Architecture. His diverse background empowers his m ultidisciplinary practice called SGAO. | He is originally from Armenia but moved to the United States when he was 18. He received his master's degree from Southern California Institute of Architecture. His diverse background empowers his m ultidisciplinary practice called SGAO. | He is originally from Armenia but moved to the United States when he was 18. He received his master's degree from Southern California Institute of Architecture. His diverse background empowers his multidisciplinary practice called SGAO. | He is originally from Armenia but moved to the United States when he was 18. He received his master's degree from Southern California Institute of Architecture. His diverse background empowers his m ultidisciplinary practice called SGAO. | He is originally from Armenia but moved to the United States when he was 18. He received his master's degree from Southern California Institute of Architecture. His diverse background empowers his m ultidisciplinary practice called SGAO |
| His office is dedicated to helping people obtain attentive medical care for their feet. As a podiatrist and foot surgeon, Dr. Siegel is highly effective in treating and caring for the elderly, diabetics and people with other foot injuries or foot conditions. | Her office is dedicated to helping people obtain attentive medical care for their feet. As a podiatrist and foot surgeon, Dr. Siegel is highly effective in treating and caring for the elderly, diabetics and people with other foot injuries or foot conditions. | Her office is dedicated to helping people obtain attentive medical care for their feet. As a podiatrist and foot surgeon, Dr. Siegel is highly effective in treating and caring for the elderly, diabetics and people with other foot injuries or foot conditions. | Her office is dedicated to helping people obtain attentive medical care for their feet. As a podiatrist and foot surgeon, Dr. Siegel is highly effective in treating and caring for the elderly, diabetics, and people with other foot injuries or foot conditions. | Her office is dedicated to helping people obtain attentive medical care for their feet. As a podiatrist and foot surgeon, Dr. Siegel is highly effective in treating and caring for the elderly, diabetics and people with other foot injuries or foot conditions. | Her office is dedicated to helping people obtain attentive medical care for their feet. As a podiatrist and foot surgeon, Dr. Siegel is highly effective in treating and caring for the elderly, diabetics and people with other foot injuries or foot conditions. | Her office is dedicated to helping people obtain attentive medical care for their feet. As a podiatrist and foot surgeon, Dr. Siegel is highly effective in treating and caring for the elderly, diabetics and people with other foot injuries or foot conditions |
| She is Honorary Researcher in Psychology at Lancaster University and is co-author, with Paul Martin, of Extreme: Why Some People Thrive at the Limits. | He is Honorary Researcher in Psychology at Lancaster University and is co-author, with Emily Ella, of Extreme: Why Some People Thrive at the Limits. | He is Honorary Researcher in Psychology at Lancaster University and is co-author, with Emily Ella, of Extreme: Why Some People Thrive at the Limits. | He is an Honorary Researcher in Psychology at Lancaster University and is co-author, with Paul Martin, of Extreme: Why Some People Thrive at the Limits. | He is Honorary Researcher in Psychology at Lancaster University and is co-author, with Paul Martin, of Extreme: Why Some People Thrive at the Limits. | He is Honorary Researcher in Psychology at Lancaster University and is co-author, with Emily Ella, of Extreme: Why Some People Thrive at the Limits. | He is honorary researcher in psychology at Lancaster University and is co-author, with Emily Ella, of Extreme: Why Some People Thrive at the Limits. |
| His accredited day surgery in Merivale, Christchurch offers a range of plastic and reconstructive surgeries, tailored for his patients, with a focus on best possible care and outcomes. | Her accredited day surgery in Merivale, Christchurch offers a range of plastic and reconstructive surgeries, tailored for her patients, with a focus on best possible care and outcomes. | Her accredited day surgery in Merivale, Christchurch offers a range of plastic and reconstructive surgeries, tailored for her patients, with a focus on best possible care and outcomes. | Her accredited day surgery in Merivale, Christchurch offers a range of plastic and reconstructive surgeries, tailored for her patients, with a focus on best possible care and outcomes. | Her accredited day surgery in Merivale, Christchurch offers a range of plastic and reconstructive surgeries, tailored for her patients, with a focus on best possible care and outcomes. | Her accredited day surgery in Merivale, Christchurch offers a range of plastic and reconstructive surgeries, tailored for her patients, with a focus on best possible care and outcomes. | Her accredited day surgery in Merivale, Christchurch offers a range of plastic and reconstructive surgeries, tailored for her patients, with a focus on best possible care and outcomes. |

Figure D.1: Text samples from Bias-in-bios and Wikipedia demonstrate that *FairFlow* and *ChatGPT-4* generate more robust counterfactual texts. Compared to *ChatGPT-4, Meta-llama-3-8B-Instruct* generates more inaccurate counterfactuals.